

Memories

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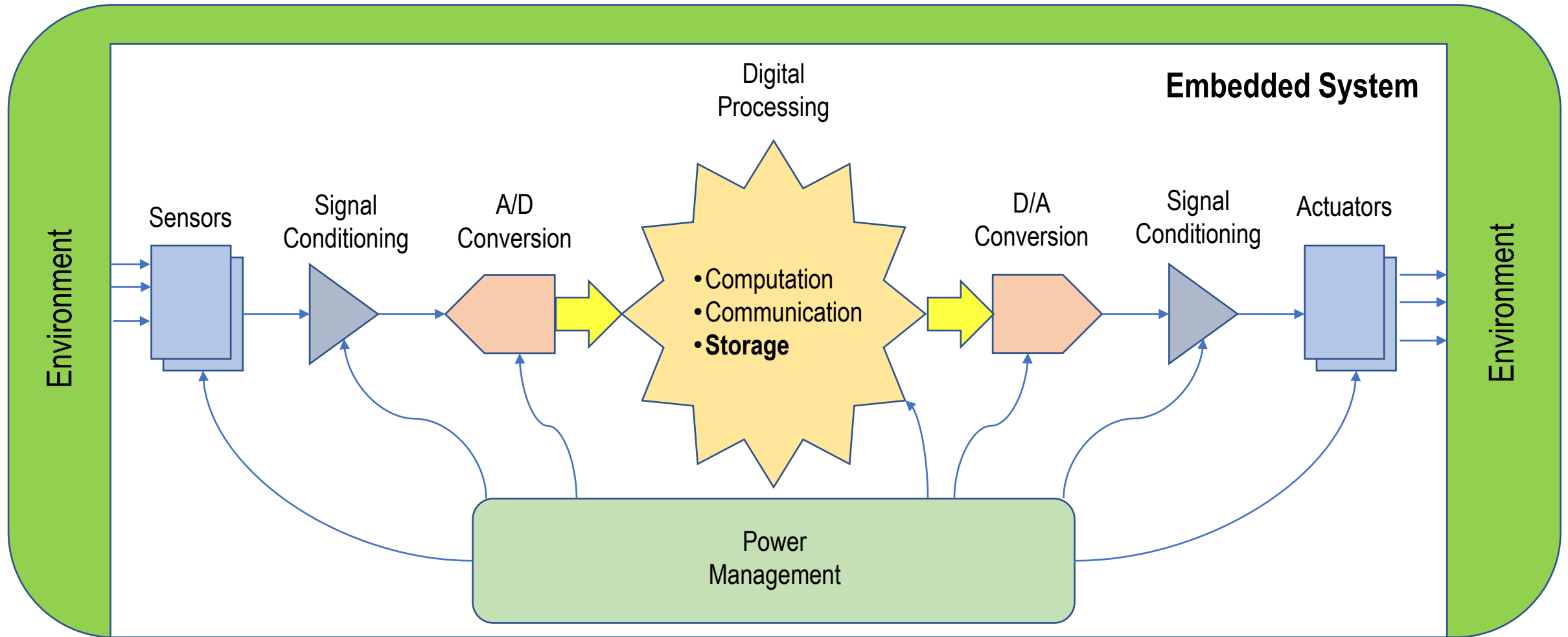
(Electronics for Embedded Systems)

Outline

- Introduction – Memories in general
 - Characteristics, types, operations, interfaces, timing diagrams, composition of memories
 - Memory performance parameters
- Read Only Memories (ROM)
 - Mask Programmable ROMs, OTP-ROMs
 - EPROM, E²PROM, NOR Flash, NAND Flash
 - FAMOS and FLOTOX transistors
- Random Access Memories (RAM)
 - Static RAMs
 - Dynamic RAMs, SDRAM, DDR
- Cache Memories
 - CAM, Cache
- Memory controllers

Introduction

- Typical architecture of an Embedded System

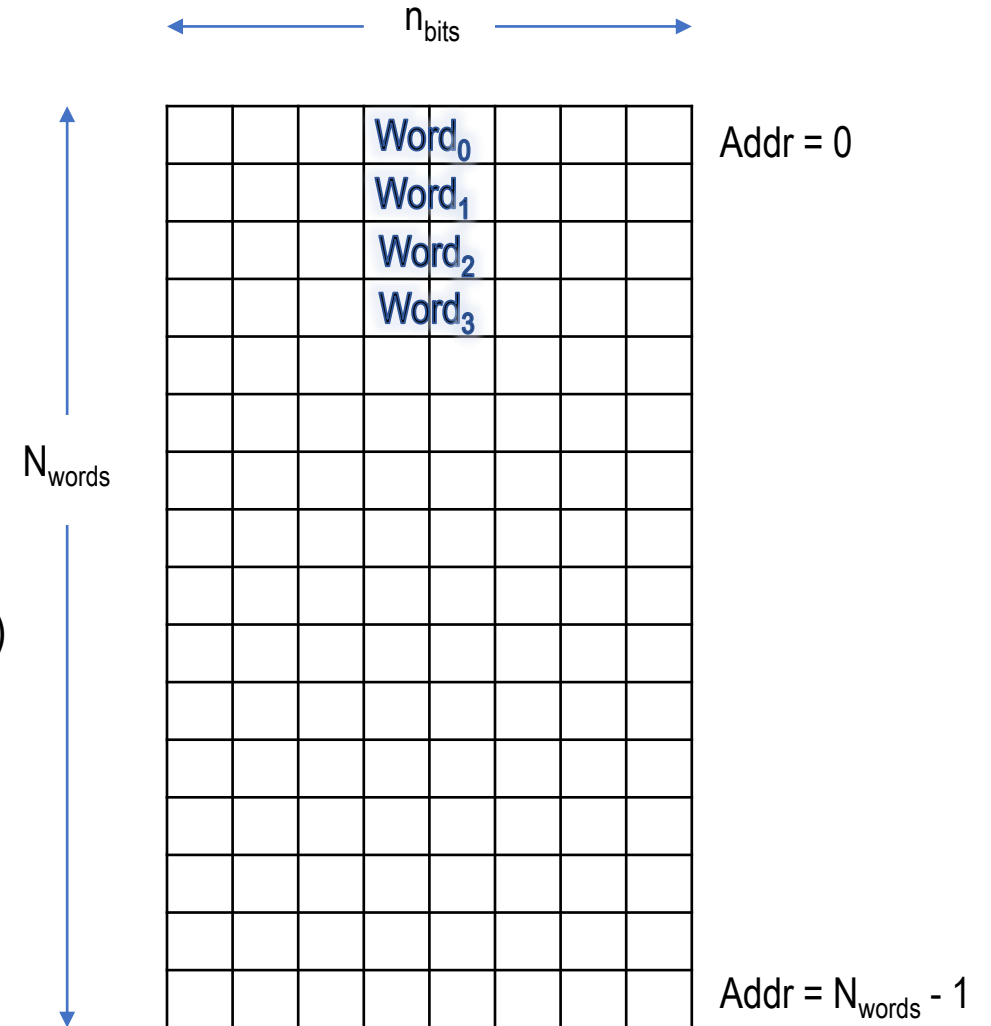


Introduction

- What is a memory?
 - A device that keeps some information for a certain amount of time
- Memory kinds
 - Your brain
 - Chemical and electrical effects
 - Very efficient
 - Books
 - Long lasting (thousands of years)
 - Electronic devices
 - Capacitors (analog memory)
 - Hard disks and tapes (magnetic)
 - CDs, DVDs, Blue Rays (optical)
 - Solid state circuits
 - Used in Embedded Systems

Introduction

- Digital Memory (Chapter 2)
 - An ordered collection of data
 - Not all data is available or accessed concurrently
 - Affects memory organization
 - Data is made of words
 - Words are made of bits
 - The memory is a collection of binary storage elements
 - Flip-Flops are binary storage elements (not used in memories)
 - All words have the same size
 - Word *width*, in number of bits
 - Words are contiguous in memory
 - Each word is identified by its index, or *address*
 - Addresses start from 0
 - The last address is $A_{last} = N_{words} - 1$



Introduction

- Digital Memory

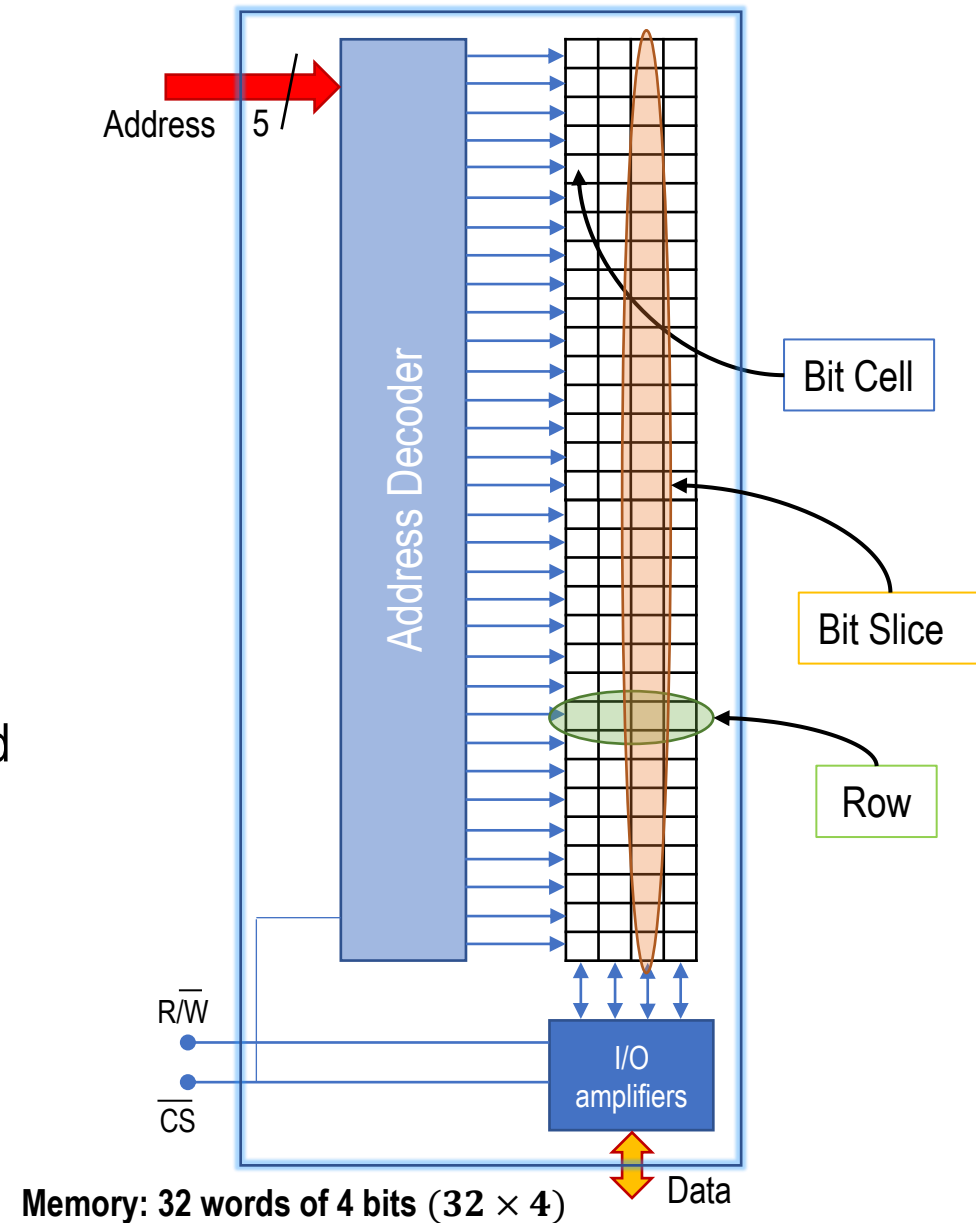
- The number of words is always a power of 2
 - Addresses can be expressed as binary numbers on $k = \log_2(N_{words})$ bits
- The number of bits in a word may or may not be a power of 2
 - But often it is a power of 2
 - Examples: 1, 2, 4, 8, 9, 12, 16, 18, 24, 32, 36, ...
- Main parameter: *SIZE*
 - Expressed as the total number of bits in the memory
 - $SIZE = N_{words} \times n_{bits}$
 - Examples:
 - 256 x 8 (2048 bits, 256 bytes)
 - 1G x 1 (1073741824 bits, 134217728 bytes, 131075 Kbytes, 128 Mbytes)
 - 16K x 16 (262144 bits, 32768 bytes, 32 Kbytes)
 - 4K x 9 (36864 bits, 4608 bytes, 4.5 Kbytes)

Introduction

- Digital Memory
 - Two memories can have the same size but different organization
 - Same total number of bits, but different word size and number of words
 - Examples: 16384 bits (2 Kbytes)
 - 16K x 1, 8K x 2, 4K x 4
 - 2K x 8, 1K x 16, 512 x 32
 - The organization affects the memory interface
 - Number and kind of pins
 - External access also affects the memory interface
 - Parallel, Serial
 - Random, Sequential
 - Special access modes

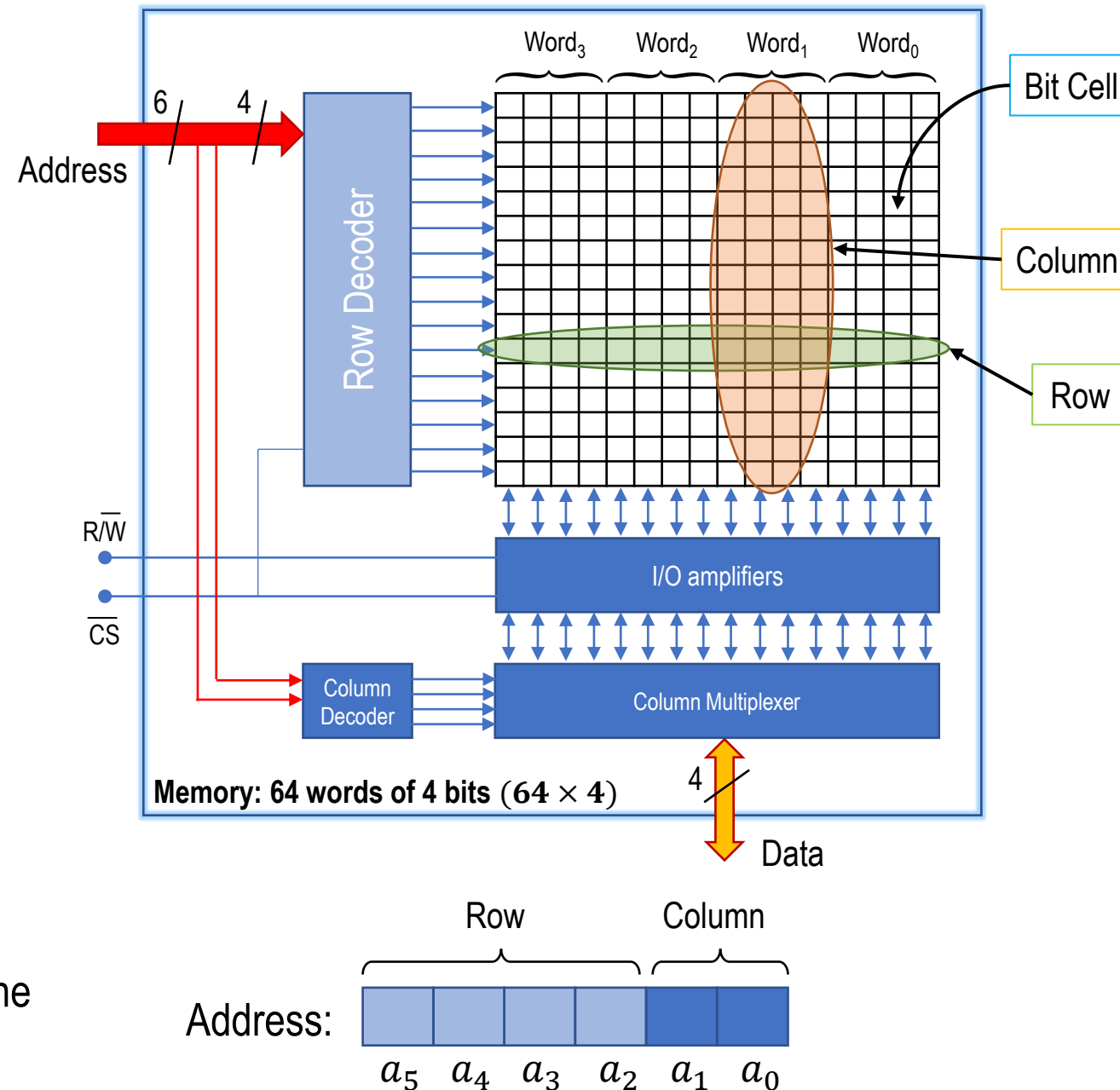
Introduction

- Digital memory
 - Logical Structure
 - Array of bit cells
 - One row for each word
 - One bit slice for each bit in the word
 - Address decoder to select a single row
 - From k address lines to 2^k word lines
 - A single word line is active at any given time
 - Sense amplifier and I/O circuit to interface to the external world
 - Not the best choice
 - Aspect ratio
 - Size of address decoder
 - Exponential on the number of address bits
 - Slow
 - Large parasitic capacitance on bit lines



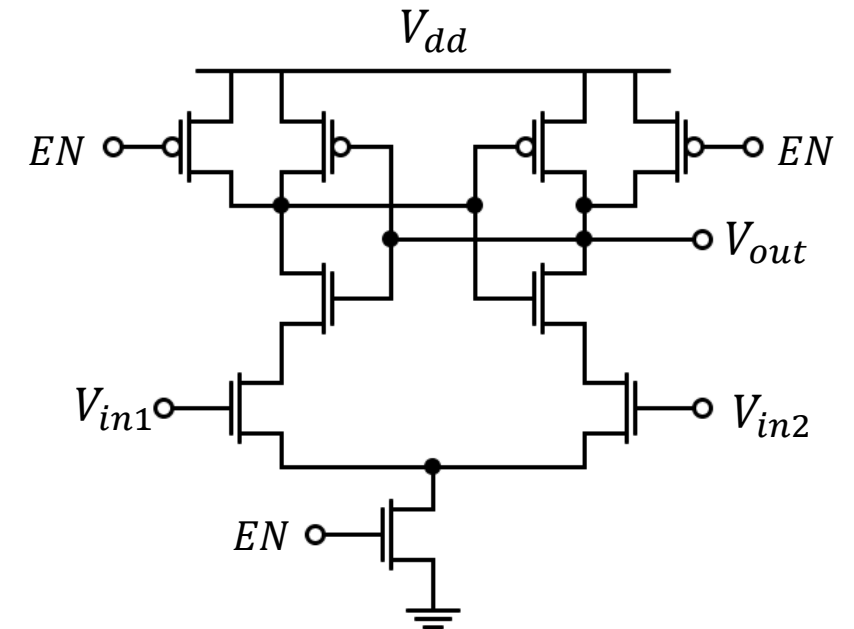
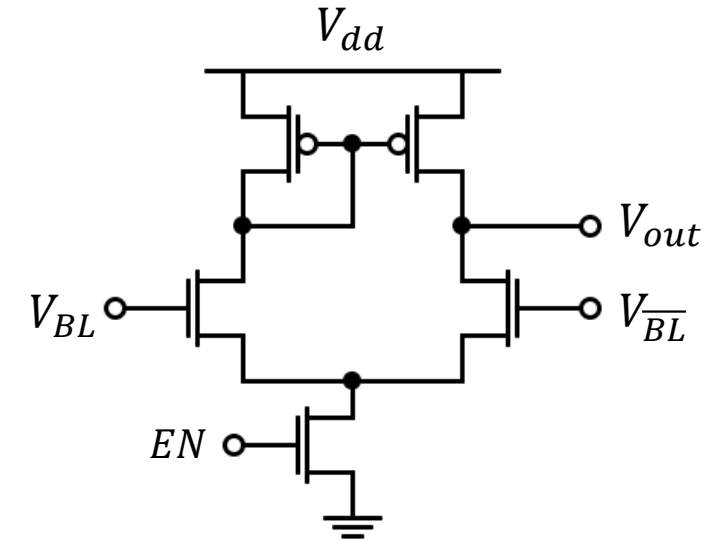
Introduction

- Digital memory
 - Real Structure
 - Array of bit cells
 - Square array (optimal aspect ratio)
 - More than one word for each row
 - One bit slice for each bit in the words
 - One column for each word
 - Address decoder to select a single row
 - It becomes a *row decoder*
 - Uses a subset of the address lines
 - A single row line is active at any given time
 - Column decoder to select a word
 - It drives a multiplexer
 - Sense amplifier and I/O circuit to interface to the external world



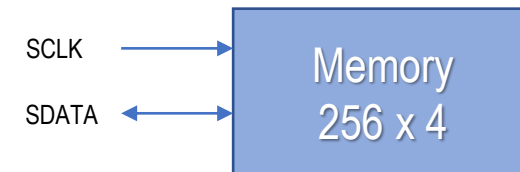
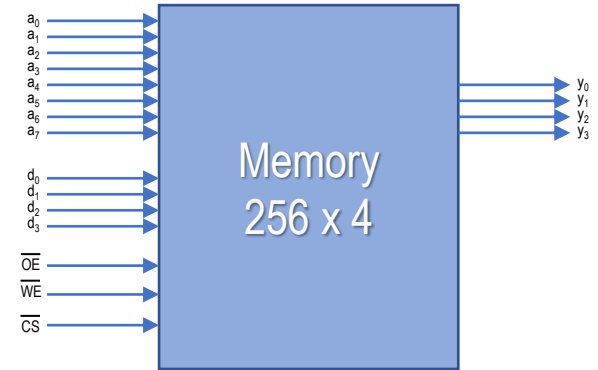
Introduction

- Digital memory
 - Sense Amplifier
 - Many designs
 - Used to detect slight voltage differences
 - A few hundred millivolts
 - Compares two lines
 - A bit line
 - A reference line
 - The reference line can be another bit line
 - Performance
 - Strong positive feedback to achieve a fast decision
 - Large impact on the memory speed
 - Impact on memory power consumption



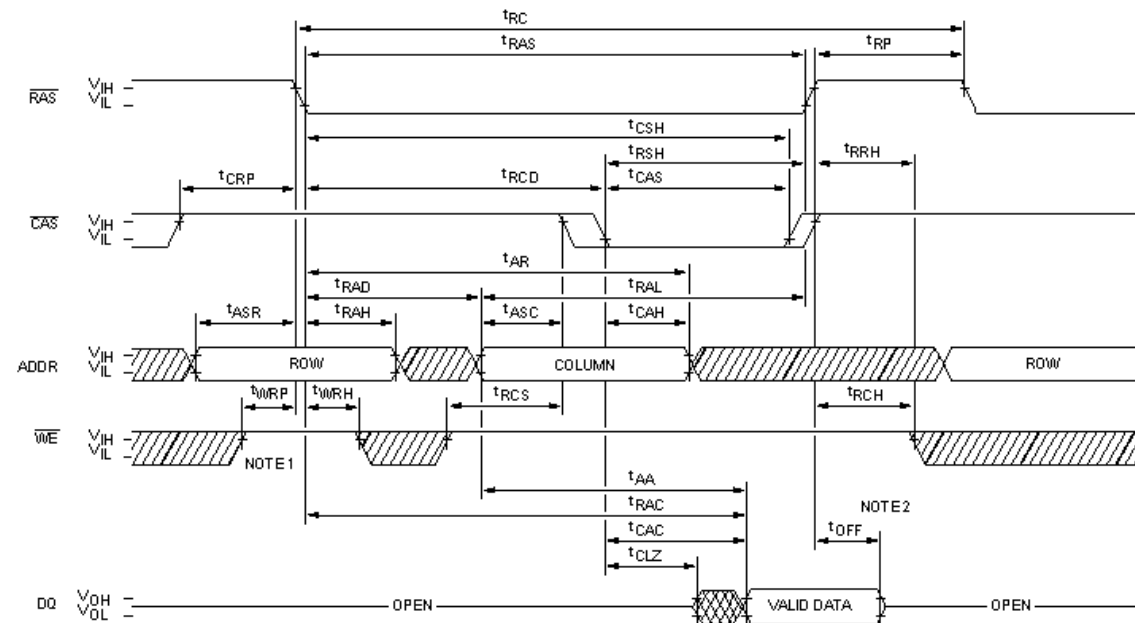
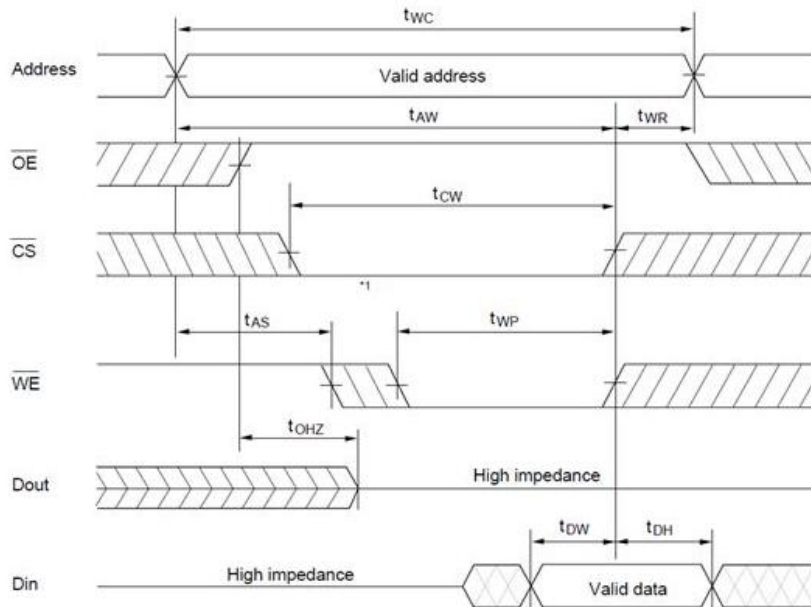
Introduction

- Digital Memory External Interface
 - Device pins used to access the memory content
 - Usually grouped depending on their function
 - Address Bus
 - Identifies the memory location where an operation is performed
 - A straightforward address bus is made of k pins
 - Data Bus
 - Provides the data to/from the memory location
 - Bidirectional, or split in monodirectional input and output busses
 - A straightforward data bus is made of n_{bits} pins
 - Control Bus
 - Defines the kind of operation to perform
 - Clock, Reset, Chips Select, Output Enable, Write Enable, $R/\overline{W}$, ...
 - ...



Introduction

- Operations on memories
 - Data is accessed using a protocol
 - The protocol allows organizing the memory to have better performance and area size
 - It defines the sequences of signal activations to correctly perform different operations
 - It defines the timing constraints
 - Often described as a timing diagram in the device datasheet



Introduction

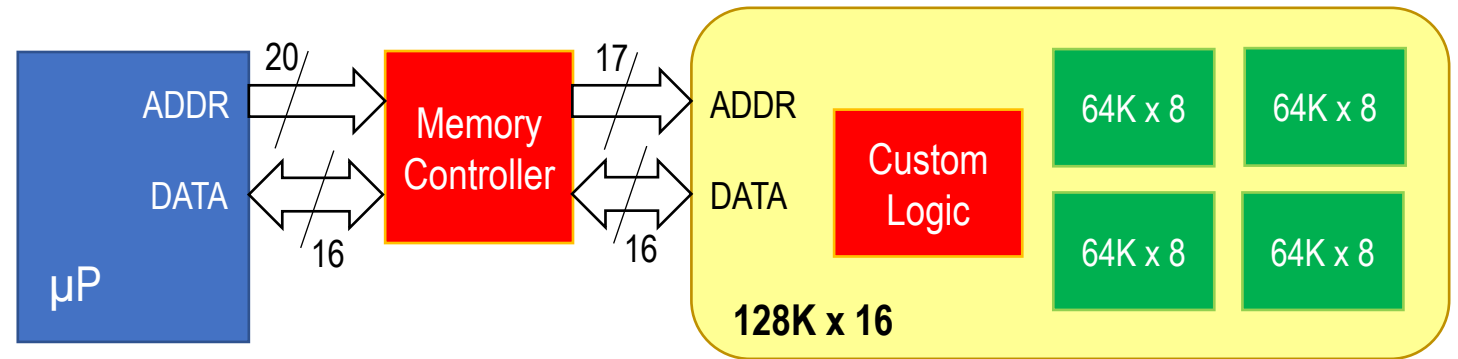
- Operations on memories
 - Read
 - Given an address, returns the word contained in memory at that address
 - Common to all memories
 - Write
 - Given a word and an address, stores the word in memory at that address
 - Common to all memories, but may be limited
 - Match
 - Given a word, searches the memory for that particular word content and returns the address
 - Specific of Content Addressable Memories (CAM, used in caches, for instance)
 - Erase
 - Initializes the memory to a pre-defined content
 - Typical of some memories
 - Burst operation, Partial operation, Precharge, Refresh, ...

Memory interfacing

- Interfacing a memory to a digital circuit
 - The digital circuit shall connect to all pins of the memory
 - Drive the inputs according to the specifications in the datasheet
 - Read the outputs
 - Comply with the timing diagrams
 - In a custom digital circuit, the interface is designed specifically for the given memory
 - In a standard digital circuit, there is a generic memory interface
 - It may require external components to correctly drive a certain memory
 - Example: an embedded processor
 - Example: an On-Chip Bus or Network
 - The number of address pins may be different
 - The number of data pins may be different
 - The function and number of control pins may be different

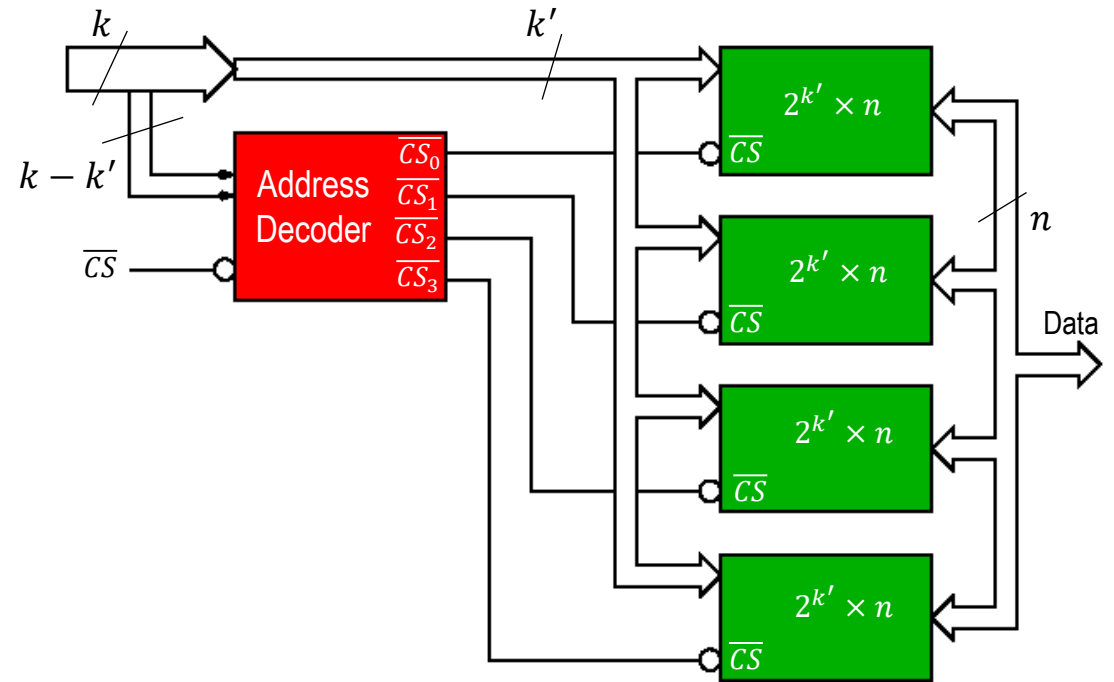
Memory interfacing

- Interfacing to a digital circuit
 - Example
 - A microprocessor has 20 address pins and 16 data pins
 - It can address $2^{20} = 1048576$ locations (1 Mwords), where each word is of 2 bytes
 - Total addressable memory is 2 Mbytes (1M x 16)
 - The embedded system only requires 256 Kbytes of memory (128 Kwords x 16 bits = 2 Mbits)
 - $\log_2(131072) = 17$ address lines would be sufficient
 - The available memory chips are 64 Kwords x 8 bits = 512 Kbits
 - 16 address lines
 - 8 data lines
 - Need 4 memory chips



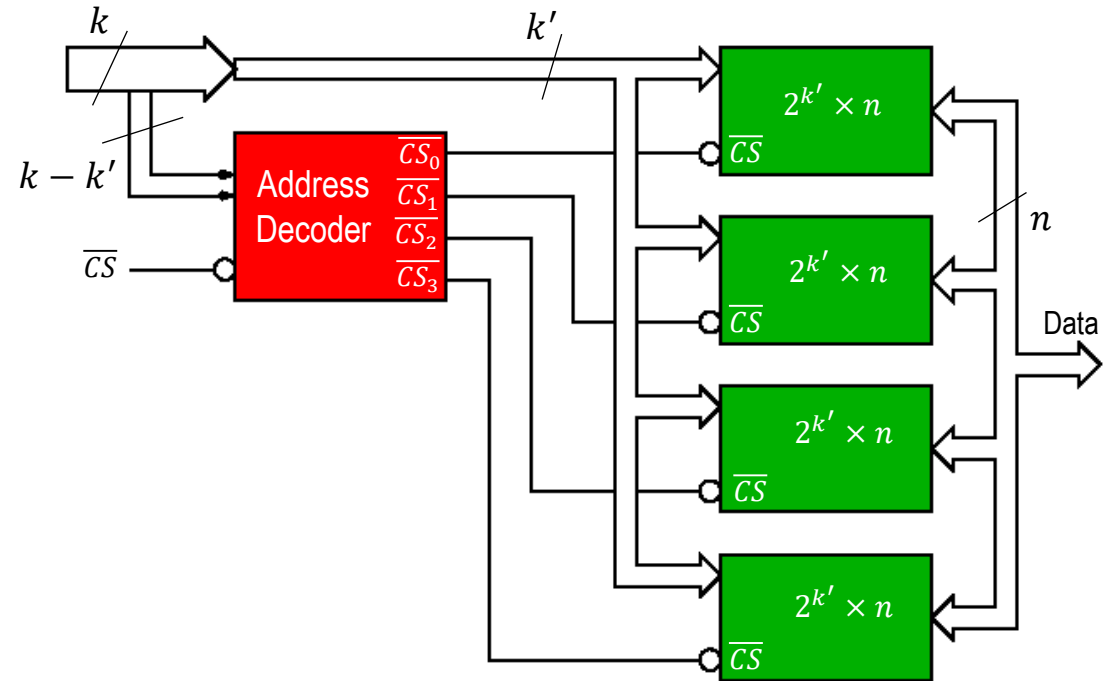
Memory interfacing

- Composition of memories
 - Required memory needs more words than those available in a single memory chip
 - Assume word width is the same
 - k required address lines
 - k' address lines in a memory chip
 - $k' < k$
 - Need more than one memory chip
 - $n_{chips} = \frac{2^k}{2^{k'}} = 2^{k-k'}$
 - Need a custom logic to drive the $\overline{CS}$ signal of the memory chips
 - The custom logic is a decoder
 - From $k - k'$ additional address lines to $2^{k-k'}$ chip select lines



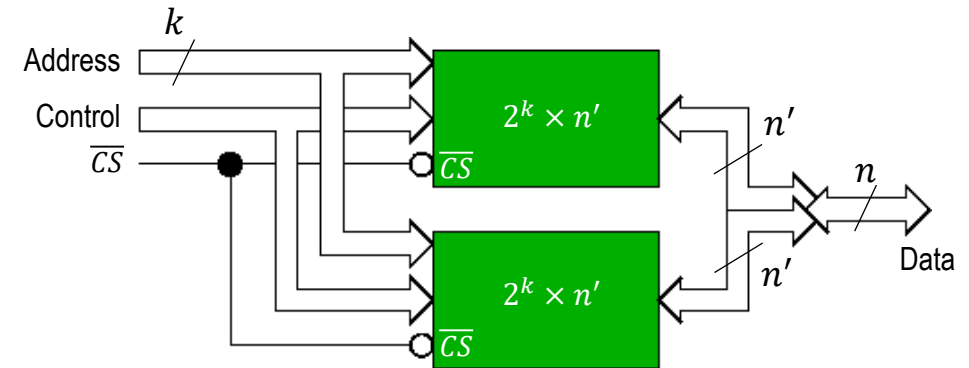
Memory interfacing

- Composition of memories
 - Required memory needs more words than those available in a single memory chip
 - The data bus is shared among all memory chips
 - May lead to potential conflicts, but...
 - No risk of short circuits, because a single memory chip is active at any given time
 - ... if the address decoder works correctly
 - Part of the address bus is shared among all memory chips
 - You can decide which lines go to the decoder
 - The most significant bits of the address
 - The least significant bits of the address
 - All other lines in the Control Bus are shared among all memory chips



Memory interfacing

- Composition of memories
 - Required memory needs more bits in a word than those available in a single memory chip
 - Assume number of words is the same
 - n required data lines
 - n' data lines in a memory chip
 - $n' < n$
 - Need more than one memory chip
 - $n_{chips} = \left\lceil \frac{n}{n'} \right\rceil$
 - All memory chips share the same address
 - All memory chips share the same Chip Select
 - Their data busses are combined (concatenated) to form a larger data bus

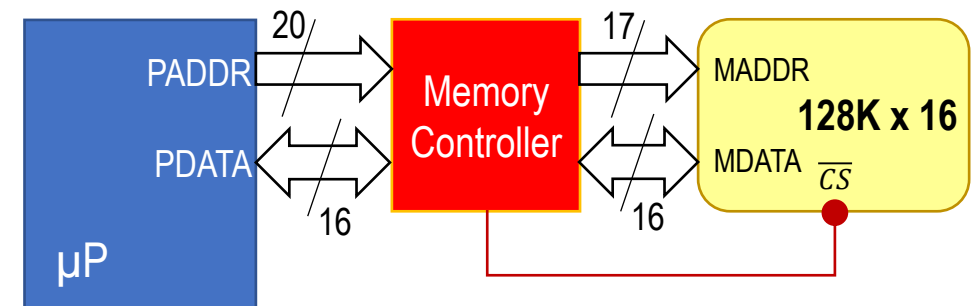


Memory interfacing

- Composition of memories
 - Required memory needs more words than those available in a single memory chip and needs more bits in a word than those available in a single memory chip
 - Combination of the two previous cases
 - k required address lines, but k' address lines in a memory chip, with $k' < k$
 - n required data lines, but n' data lines in a memory chip, with $n' < n$
 - Need more than one memory chip
 - $n_{chips} = \left\lceil \frac{n}{n'} \right\rceil \cdot 2^{k-k'}$
- Composition
 - First, create a structure with n data lines using $\lceil n/n' \rceil$ chips
 - Second case of the previous slides
 - Then replicate the structure $2^{k-k'}$ times, with an address decoder
 - First case of the previous slides

Memory interfacing

- Composition of memories
 - Without a memory controller
 - The microprocessor accesses the same location in memory with multiple addresses
 - $a_{16} \div a_0$ of the μP connected to the memory, $a_{19} \div a_{17}$ not connected (floating)
 - Example: $A_0 = 000\ 0\ 0000\ 0000\ 0000\ 0000 \rightarrow$ First location in memory
 - Example: $A_1 = 001\ 0\ 0000\ 0000\ 0000\ 0000 \rightarrow$ First location in memory
 - A memory controller is used to translate addresses and activate the memory
 - The memory is mapped to a certain address range in the microprocessor addressable space
 - Not necessarily starting from address 0, use $a_{19} \div a_{17}$ in the memory controller
 - The address range is usually aligned to a power of 2 to simplify the logic
 - The compiler for the microprocessor should know the chosen address range
 - It generates code to correctly access the memory
 - The memory controller may be inside the microprocessor

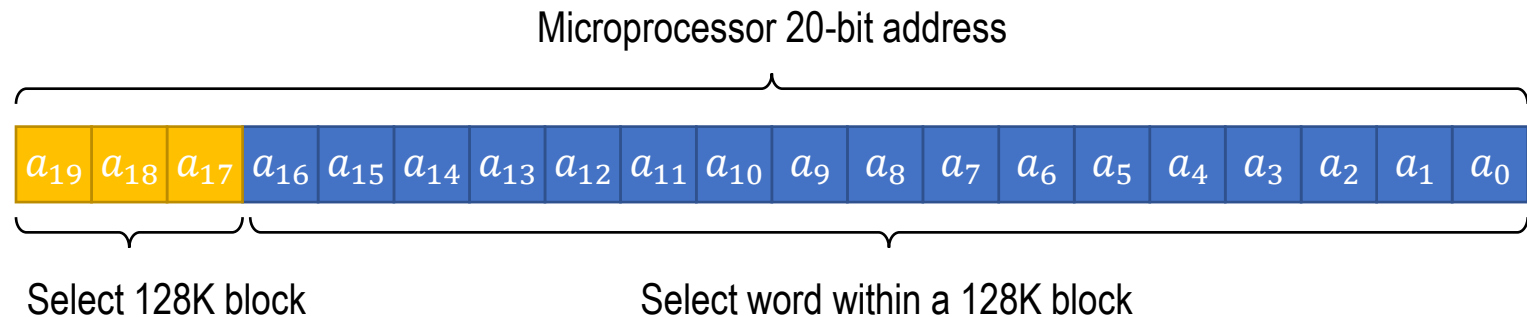


Memory interfacing

- Composition of memories
 - Microprocessor with 20-bit Address Bus, 8-bit Data Bus
 - Can address up to 1 Mbyte of memory
 - Memory is $128K \times 8$
 - Mapped in the memory space starting at 0x40000 (256K)
 - What is the output of the following C program?
 - With and without the memory controller

```
int main(void) {
    int *p = 0x410C4;
    *p = 10;
    int a = *p;
    p = 0x810C4;
    int b = *p;
    p = 0x010C4;
    int c = *p;
    printf("a=%d, b=%d, c=%d\n", a, b, c);
}
```

			a_{19}	a_{18}	a_{17}
0xFFFFF		1M	1	1	1
0xE0000		896K	1	1	0
0xC0000		768K	1	0	1
0xA0000		640K	1	0	0
0x80000		512K	0	1	1
0x60000	Memory	384K	0	1	0
0x40000		256K	0	0	1
0x20000		128K	0	0	0
0x00000		0	0	0	0



Memory interfacing

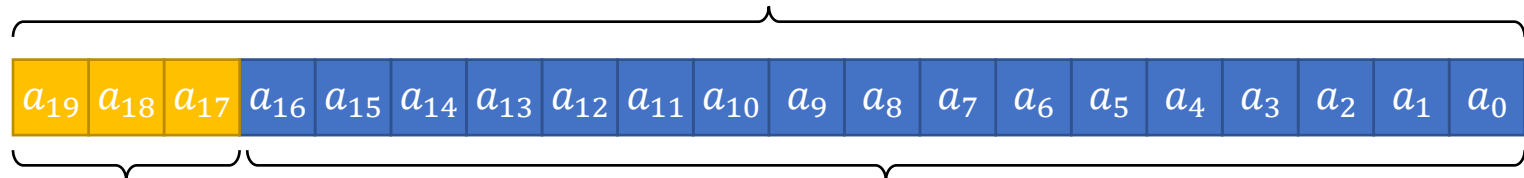
- Composition of memories
 - Microprocessor with 20-bit Address Bus, 8-bit Data Bus
 - Can address up to 1 Mbyte of memory
 - Memory is $128K \times 8$
 - Mapped in the memory space starting at 0x40000 (256K)
 - VHDL implementation of the simple memory controller

```

1  entity simplememcontroller is
2      port (
3          PADDR: in std_logic_vector(19 downto 0);
4          PDATA: out std_logic_vector(15 downto 0);
5          MADDR: out std_logic_vector(16 downto 0);
6          MDATA: in std_logic_vector(15 downto 0);
7          M_CSn: out std_logic
8      );
9  end simplememcontroller;
10
11 architecture rtl of simplememcontroller is
12 begin
13     M_CSn <= '0' when PADDR(19 downto 17) = "010"
14           else '1';
15     PDATA <= MDATA;
16     MADDR <= PADDR(16 downto 0);
17 end rtl;
    
```

			a_{19}	a_{18}	a_{17}
0xFFFFF		1M	1	1	1
0xE0000		896K	1	1	0
0xC0000		768K	1	0	1
0xA0000		640K	1	0	0
0x80000		512K	0	1	1
0x60000	Memory	384K	0	1	0
0x40000		256K	0	0	1
0x20000		128K	0	0	0
0x00000		0	0	0	0

Microprocessor 20-bit address

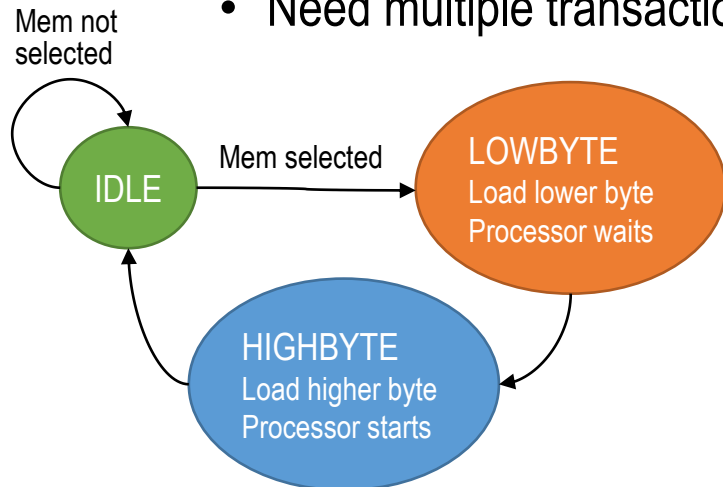


Select 128K block

Select word within a 128K block

Memory interfacing

- Composition of memories
 - Microprocessor with 20-bit Address Bus, 16-bit Data Bus
 - Can address up to 2 Mbyte of memory
 - Memory is 128K × 8
 - Mapped in the memory space starting at 0x40000 (256K)
 - More complex memory controller
 - The Data Bus is not compatible
 - Need multiple transactions



```
1  entity memcontroller is
2      port (
3          clk, rst: in std_logic;
4          -- Processor interface
5          PWAIT: out std_logic;
6          PADDR: in std_logic_vector(19 downto 0);
7          PDATA: out std_logic_vector(15 downto 0);
8          -- Memory interface
9          MADDR: out std_logic_vector(16 downto 0);
10         MDATA: in std_logic_vector(7 downto 0);
11         M_CSn: out std_logic
12     );
13 end memcontroller;
```

```
15 architecture behav of memcontroller is
16     type status_type is (IDLE, LOWBYTE, HIGHBYTE);
17     signal status: status_type;
18 begin
19     MADDR(16 downto 1) <= PADDR(15 downto 0);
20     M_CSn <= '0';
21
22     FSM: process (clk, rst)
23     begin
24         if rst = '0' then
25             status <= IDLE;
26         elsif clk'event and clk = '0' then
27             case status is
28                 when IDLE =>
29                     if PADDR(19 downto 16) = "0100" then
30                         status <= LOWBYTE;
31                     end if;
32                 when LOWBYTE =>
33                     status <= HIGHBYTE;
34                 when HIGHBYTE =>
35                     status <= IDLE;
36             end case;
37         end if;
38     end process;
39
40     OUTPUTS: process (status)
41     begin
42         case status is
43             when IDLE =>
44                 PWAIT <= '0';
45                 MADDR(0) <= '0';
46             when LOWBYTE =>
47                 PWAIT <= '1';
48                 PDATA(7 downto 0) <= MDATA(7 downto 0);
49                 MADDR(0) <= '1';
50             when HIGHBYTE =>
51                 PWAIT <= '0';
52                 PDATA(15 downto 8) <= MDATA(7 downto 0);
53             end case;
54         end process;
55
56 end behav;
```

Memory Parameters

- Size
 - Expressed in bits, bytes or multiples
 - K (1024), k (1000)
 - M (1048576)
 - G (1073741824)
 - Requirements based on the application specifications
 - Strong impact on costs
 - Try to reduce memory requirements
 - Small Embedded Systems use only some Kbytes of memory

Memory Parameters

- Speed
 - Can be defined in several different ways
 - Example: asynchronous memory read
 - From the time instant when the address is stable ...
 - ... to the time instant when the output data is guaranteed to be stable
 - Known as *access time*, measured in seconds
 - Example: asynchronous memory write
 - From the time instant when the address and the input data are stable ...
 - ... to the time instant when the data is guaranteed to be stored
 - Known as *writing time*, measured in seconds
- Throughput
 - Maximum number of operations per second
- Synchronous Memories: time measured in clock cycles given the clock frequency

Memory Parameters

- Area
 - Size (in m^2 , or better μm^2) of the bit cell
 - Total array area size
 - Total number of bits multiplied by the size of a single bit cell
 - Total area includes also decoders, multiplexers and I/O circuits
 - Different kind of memories have different bit cells
 - Different bit cells provide different features
 - Very strong impact on the total area
 - Very strong impact on the total cost
 - Memories are at the cutting edge of technology
 - First to adopt smaller gate lengths

Memory Parameters

- Cost
 - Related to
 - Size of the bit cell
 - Size of the memory (in number of bits)
 - Production volume
 - Distinguish between
 - Unit cost
 - NRE cost
- Power consumption
 - Depends on the bit cell structure and technology
 - Very important for portable systems

Memory Parameters

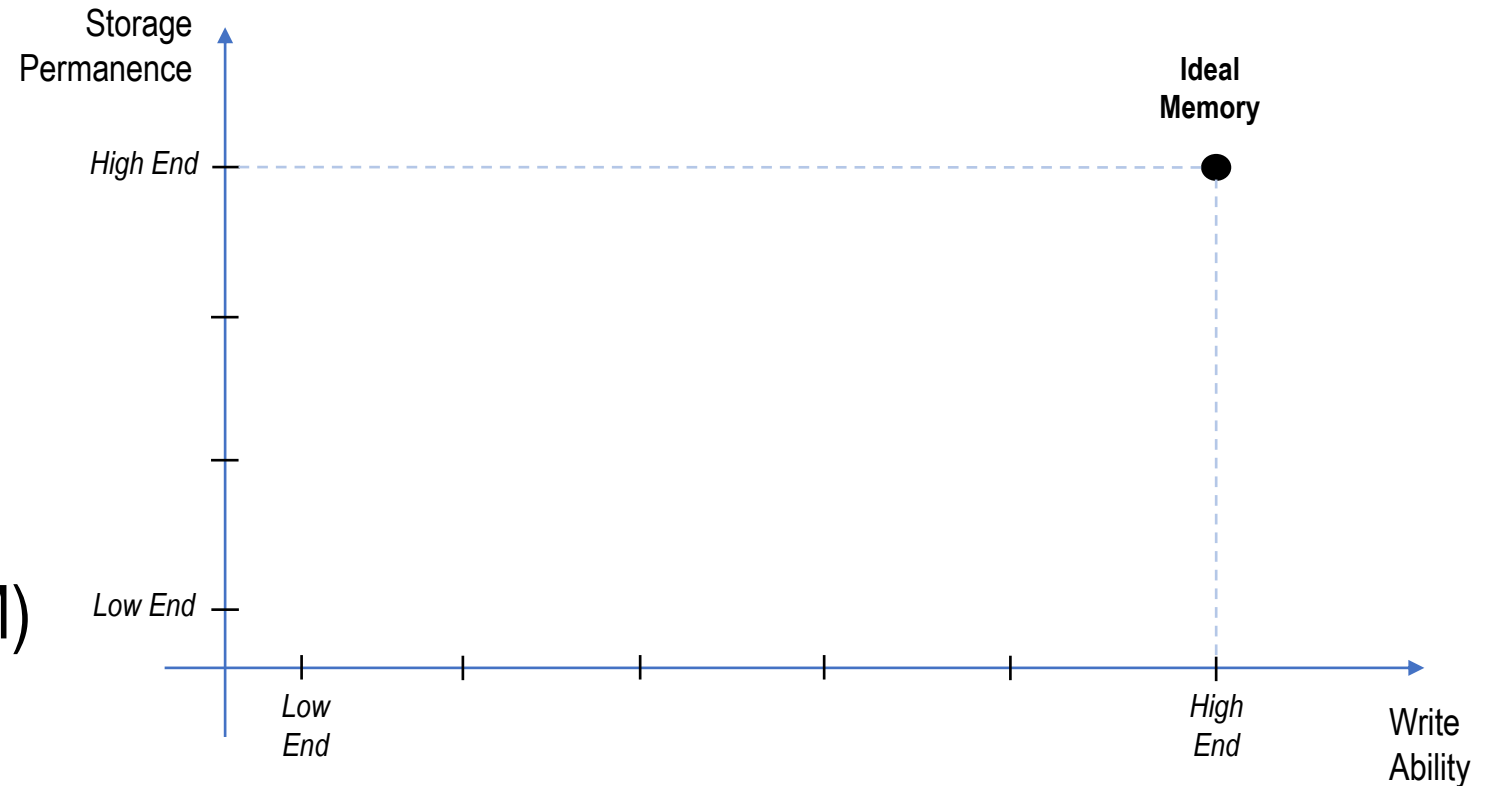
- Write ability
 - Manner and speed a memory can be written
 - High End
 - Processor writes to memory simply and quickly
 - In-system, unlimited cycles
 - Middle range
 - Processor writes to memory, but slower than the high end
 - In-system, thousands or tens of thousands of cycles
 - Lower range
 - Special equipment to be used to program the memory (external programmer)
 - Thousands of cycles
 - Low End
 - Written only during fabrication

Memory Parameters

- Storage permanence
 - Also known as *volatility*
 - Ability of the memory to hold stored bits after they are written
 - High End
 - Bits are never lost, even without power
 - Lifetime of the device
 - Middle range
 - Bits are held for years, even without power
 - Sometimes more than the lifetime of the embedded system
 - Lower range
 - As long as power is supplied to the memory
 - Low End
 - Memory content is lost after a certain amount of time, even if power is supplied
 - Often after a few tens of milliseconds

Memory Parameters

- Graph of Write Ability and Storage Permanence
 - Different kind of memories classified based on these two parameters
 - Ideal memory
 - High storage permanence
 - High write ability
 - Real memories
 - Tradeoffs
 - Based on requirements
 - Must consider costs
- Read Only Memories (ROM)
- Random Access Memories (RAM)



Read Only Memory (ROM)

- Mask Programmable Read Only Memories

- Written at fabrication time

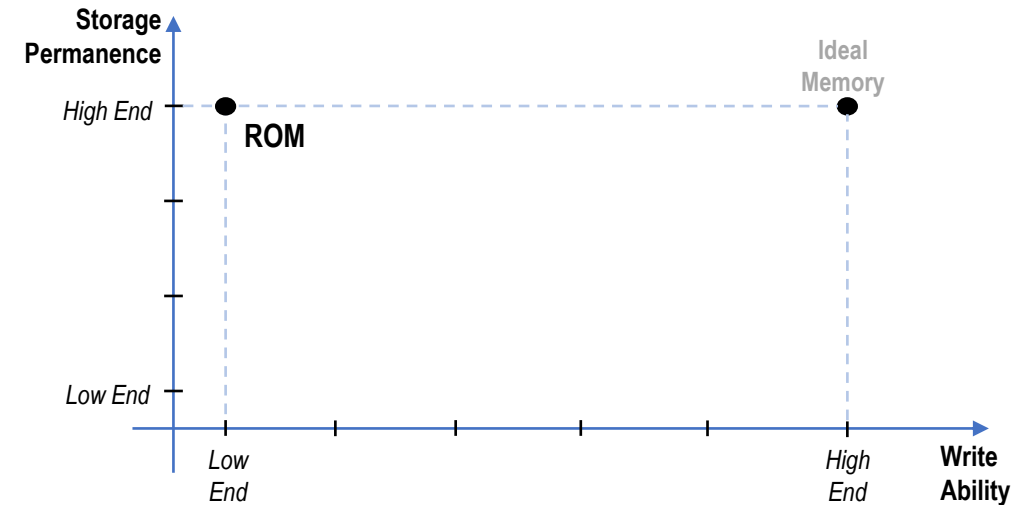
- Lowest write ability (only once)
 - Highest storage permanence (for ever)
 - Can only be read and not written to

- Costs

- High NRE costs
 - Need to develop a dedicated set of masks given the memory content
 - Some masks may be shared for memories of the same size and organization
 - Low unit cost
 - Good for very large volume productions

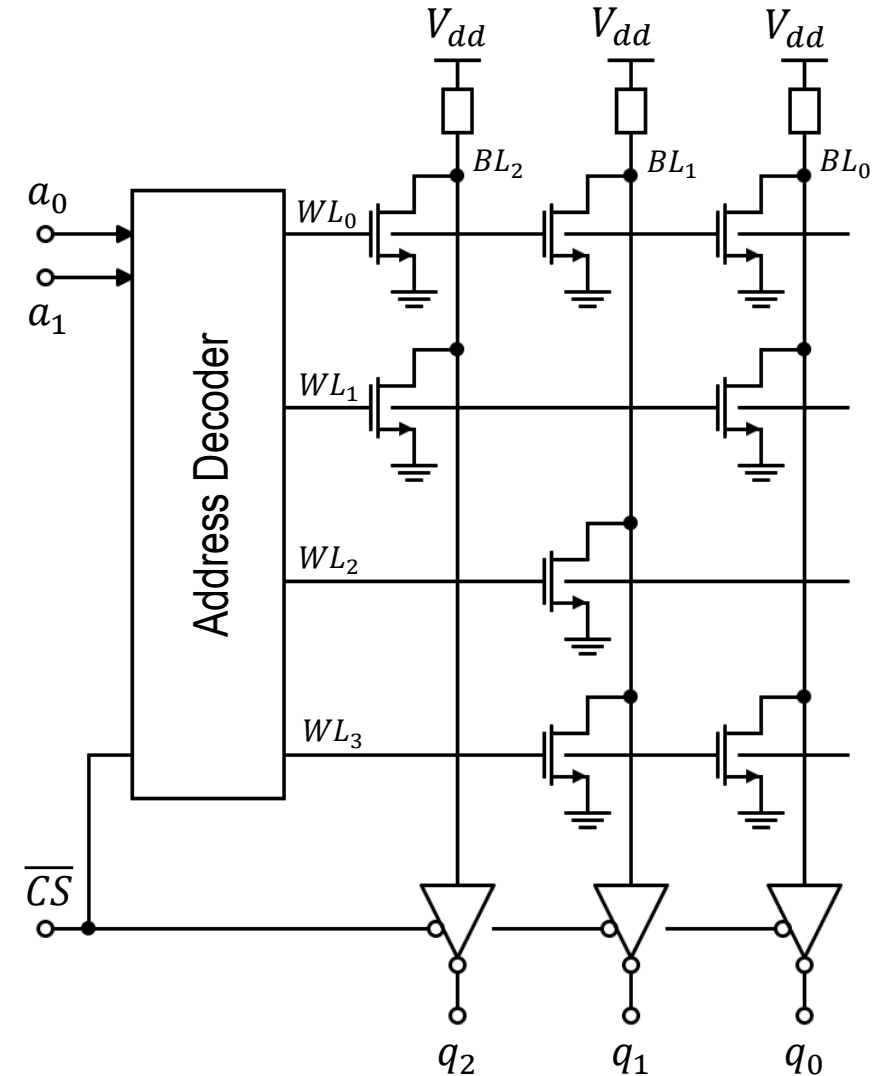
- Examples

- Permanent code and data for an embedded system
 - Reliable constant memory in safety critical applications



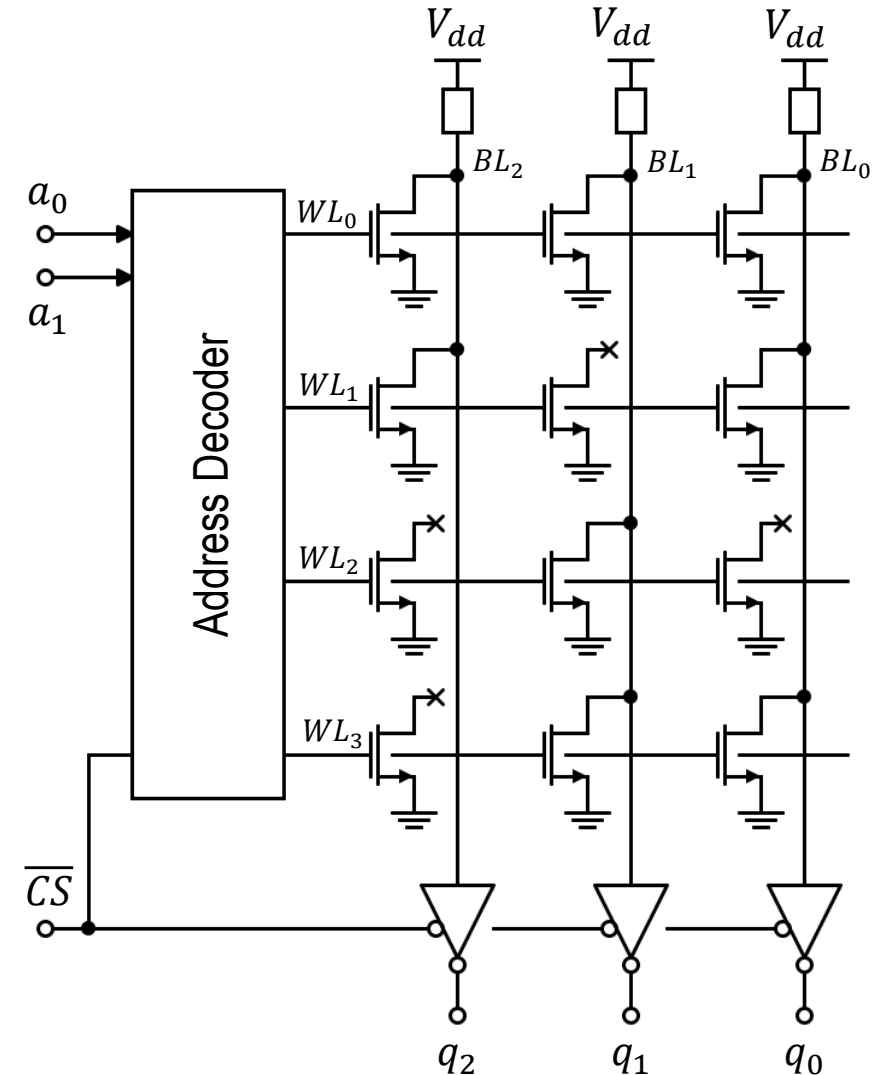
Read Only Memory (ROM)

- Mask Programmable Read Only Memories
 - NOR structure
 - All Bit Lines are normally pulled-up to V_{dd}
 - The output is logic 0 because of the inverter
 - A single Word Line is active (high) at any given time
 - Transistors in the corresponding word are conducting
 - Transistors in all other words are in cut-off
 - A conducting transistor lowers a Bit Line to 0
 - Presence of a transistor in the bit cell
 - Logic 1 at the output
 - Absence of a transistor in the bit cell
 - Logic 0 at the output
 - Transistor defined at the mask level
 - Diffusions, Thin Oxide, Polysilicon, Metal



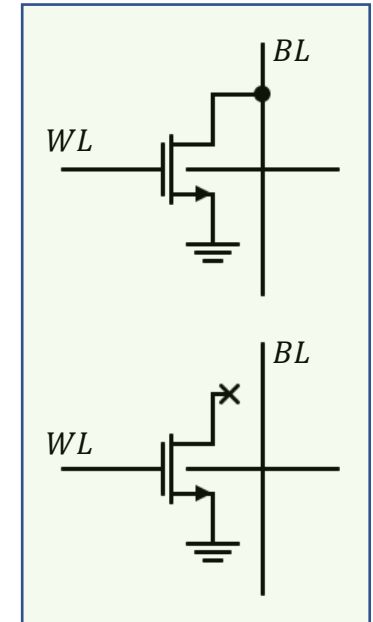
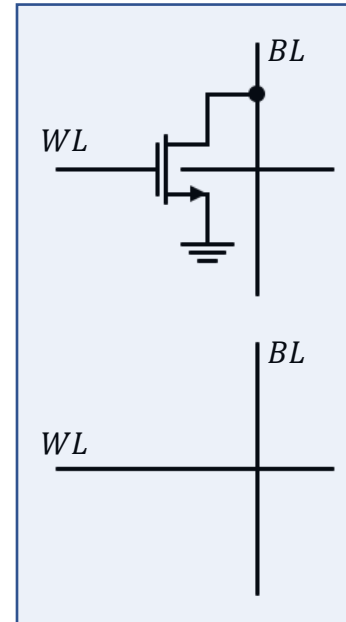
Read Only Memory (ROM)

- Mask Programmable Read Only Memories
 - NOR structure
 - All Bit Lines are normally pulled-up to V_{dd}
 - The output is logic 0 because of the inverter
 - A single Word Line is active (high) at any given time
 - Transistors in the corresponding word are conducting
 - Transistors in all other words are in cut-off
 - A connected conducting transistor lowers a Bit Line to 0
 - Connection of a transistor to the Bit Line
 - Logic 1 at the output
 - No connection of a transistor to the Bit Line
 - Logic 0 at the output
 - Connections defined at the mask level
 - Metal1 and Metal2



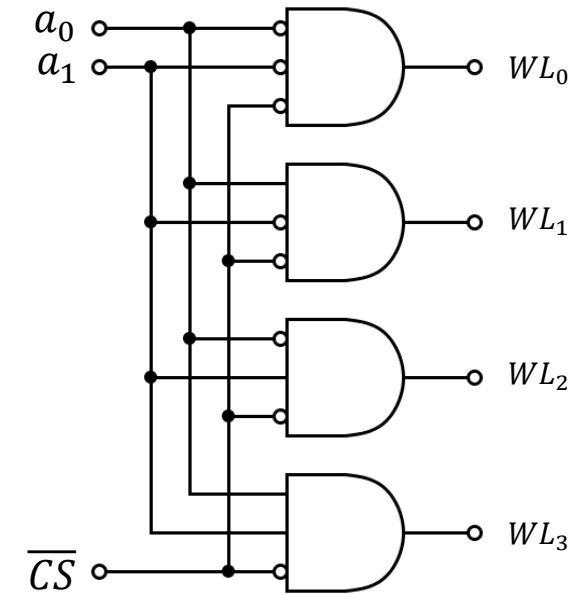
Read Only Memory (ROM)

- Mask Programmable Read Only Memories
 - NOR structure
 - A bit cell may or may not contain a transistor
 - A bit cell always has a transistor, but it may or may not be connected to the Bit Line
 - A bit cell may or may not contain a diode
 - Not shown in the slides
 - Memory content set during manufacturing
 - Cannot be changed afterwards
 - Memories manufactured with the same size but different content may share a subset of the masks
 - Save costs



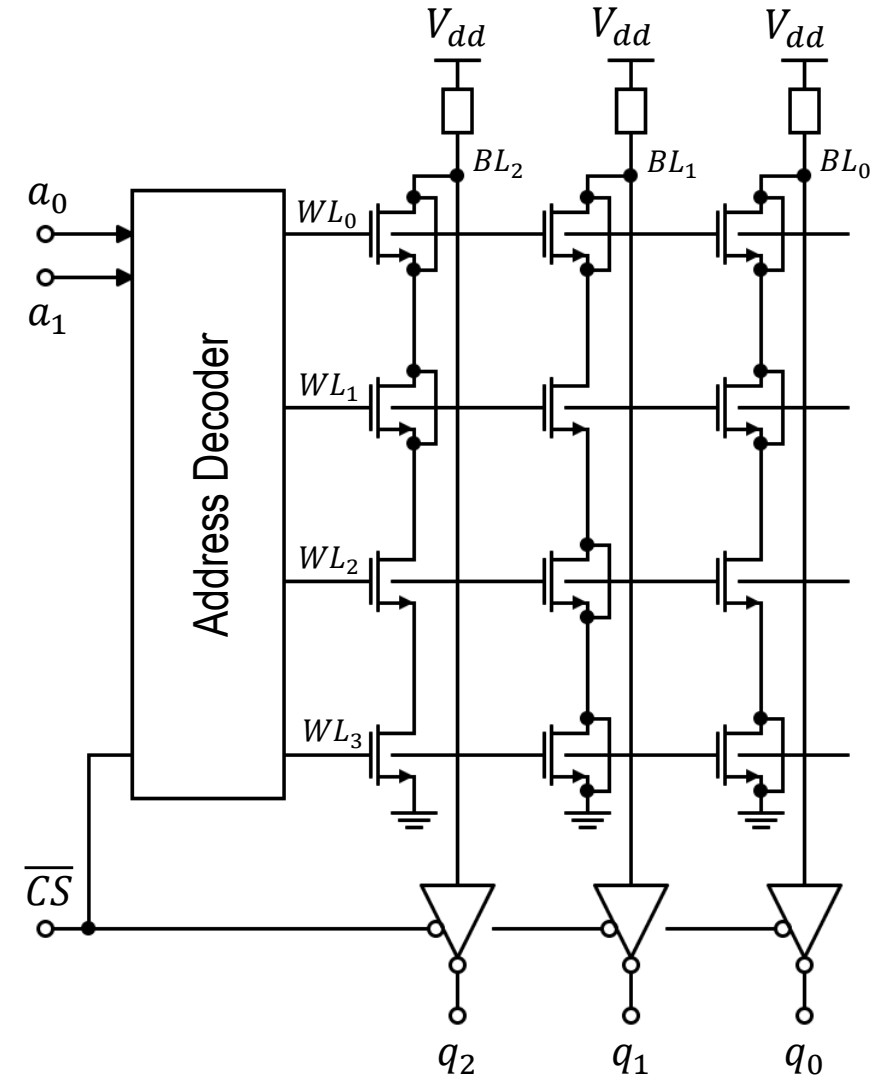
Read Only Memory (ROM)

- Mask Programmable Read Only Memories
 - NOR structure
 - Address Decoder
 - From k lines to 2^k
 - A single output line is active at any given time
 - If the memory is not enable, all outputs are inactive
 - The number of AND gate is exponential on k
 - Use row and column decoders to minimize area
 - $2^{k/2} + 2^{k/2} = 2 \cdot 2^{k/2} = 2^{1+k/2} \ll 2^k$
 - Example: $k = 16$, $2^{1+k/2} = 512$, $2^k = 65536$
 - The number of inputs for each AND gate is linear on k
 - AND gates with many inputs are slow
 - Multiple logic levels using cascaded AND gates with less inputs



Read Only Memory (ROM)

- Mask Programmable Read Only Memories
 - NAND structure
 - All Bit Lines are normally pulled-up to V_{dd}
 - The output is logic 0 because of the inverter
 - A single Word Line is active (low) at any given time
 - Transistors in the corresponding word are in cut-off
 - Transistors in all other words are conducting
 - A shorted transistor lowers a Bit Line to 0
 - Short circuit of a transistor
 - Logic 1 at the output
 - No short circuit of a transistor
 - Logic 0 at the output
 - Connections defined at the mask level
 - Metal1 and Metal2



Read Only Memory (ROM)

- Mask Programmable Read Only Memories

- Performance

- Delay of the Address Decoder

- Delay of the address lines inverters (inv_1)

- Driving half of the NAND gates

- Large parasitic capacitance

- Delay of the NAND gates

- Large parasitic resistance when all pull-down transistors are ON

- Delay of the Word Line inverters (inv_2)

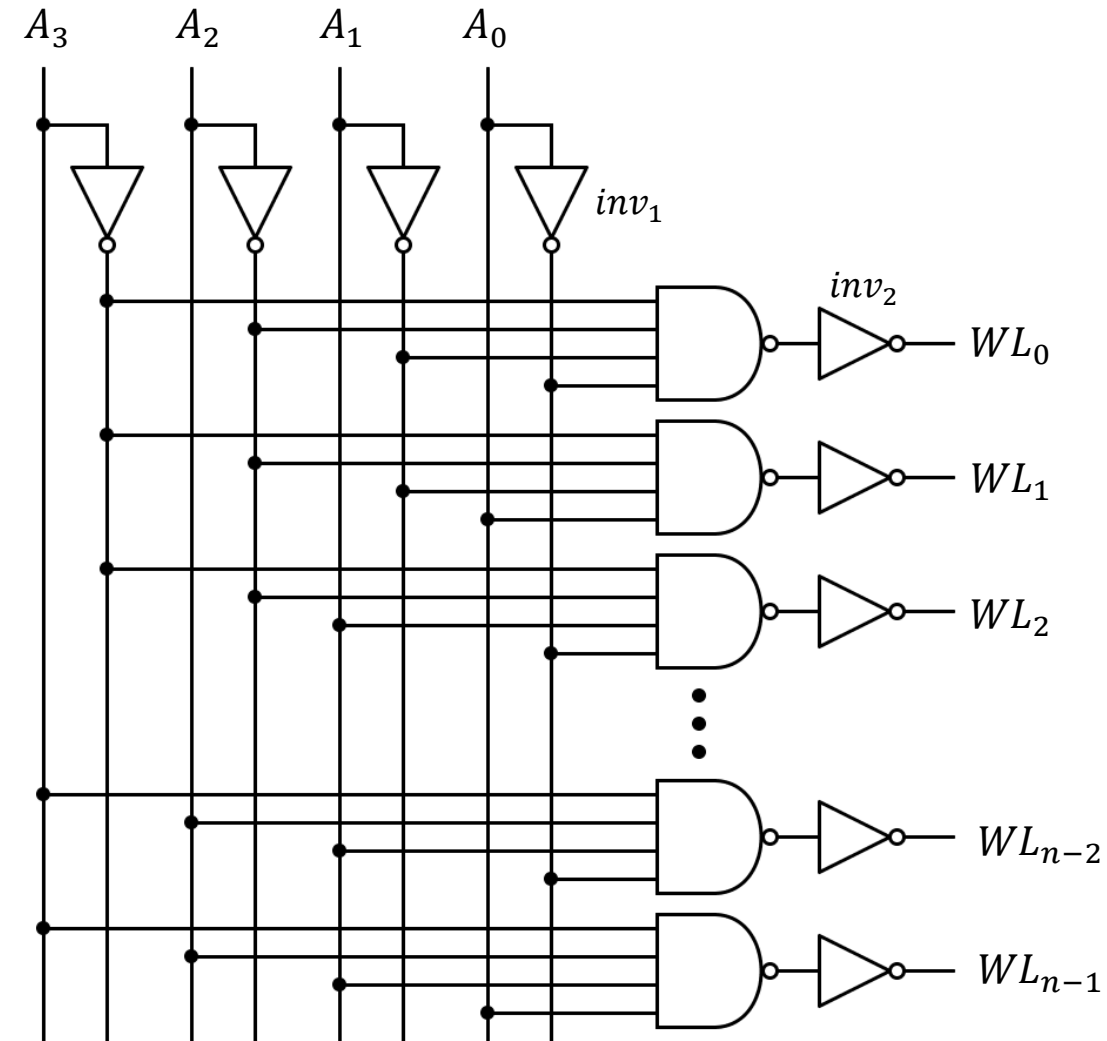
- Driving all transistors in the bit cells in a row

- Large parasitic capacitance

- Word Line is long

- Large parasitic resistance

- $t_{delay} = t_{inv_1} + t_{NAND} + t_{inv_2}$



Read Only Memory (ROM)

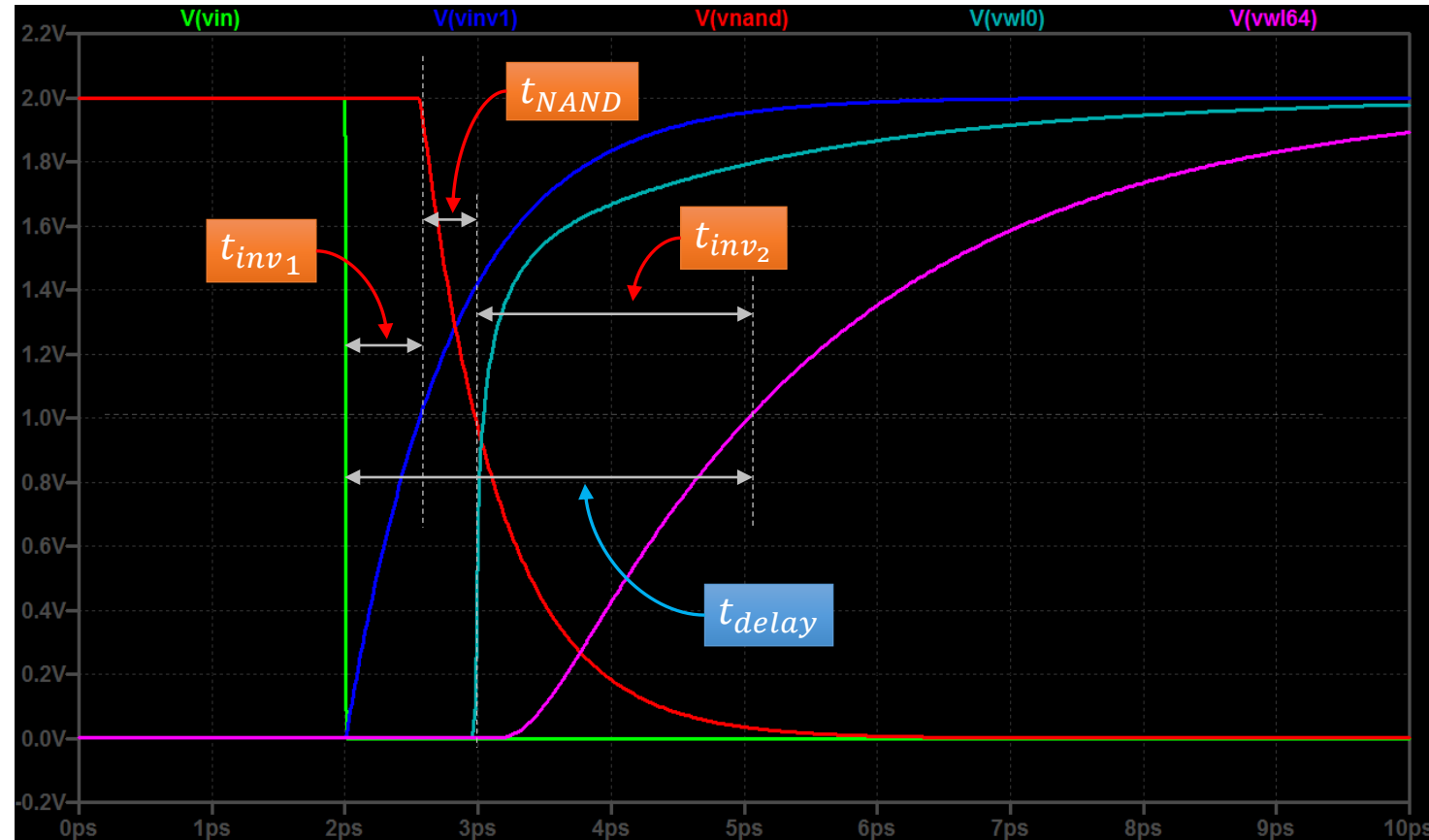
- Mask Programmable Read Only Memories

- Performance

- Delay of the Address Decoder
 - $t_{delay} = t_{inv_1} + t_{NAND} + t_{inv_2}$
 - After t_{delay} transistors in the bit cells turn ON
 - But discharging the Bit Lines takes time
 - Different for NOR and NAND topologies

Example with 64 bit cells in a row

- The delay for the first bit cell is much lower than the delay of the last bit cell
 - Must always consider the worst case

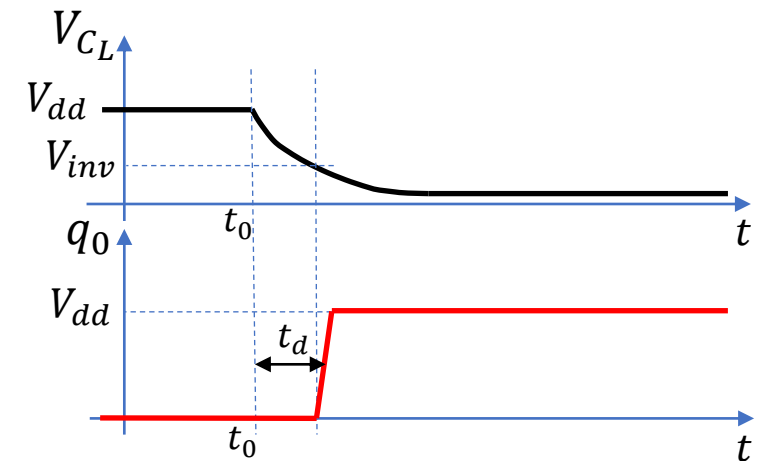
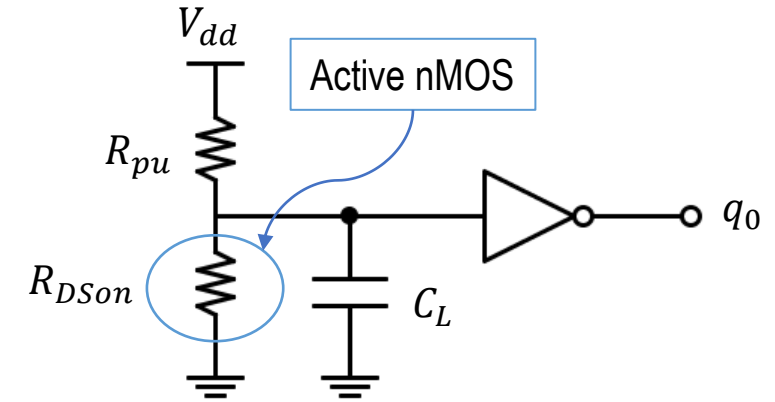


Read Only Memory (ROM)

- Mask Programmable Read Only Memories

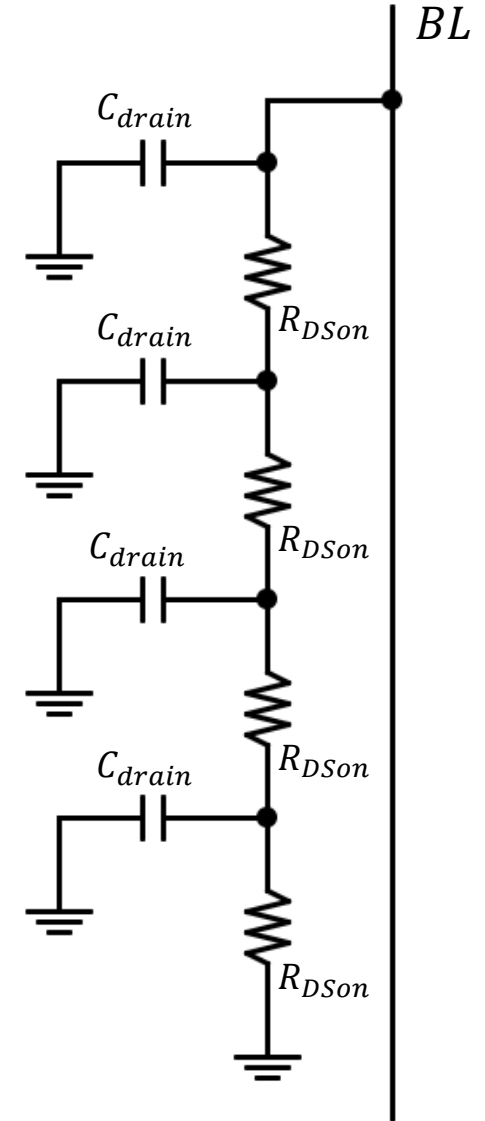
- Performance of NOR topology

- The Bit Line has a parasitic capacitance C_L
 - When the memory is idle, the Bit Line is pulled-up to V_{dd}
 - The parasitic capacitance C_L is charged
 - Afterwards, there is no current flowing in the pull-up
 - When an nMOS transistor is activated at time t_0
 - The Bit Line is pulled-down towards ground
 - There is a transient with time constant $\tau = C_L(R_{DSon} \parallel R_{pu})$
 - The final value is the result of a voltage divider
 - Close to 0V because $R_{DSon} \ll R_{pu}$
 - The output inverter or buffer detects the change earlier
 - When the voltage V_{CL} on the Bit Line crosses the inverter threshold V_{inv}
 - There is a final delay t_d



Read Only Memory (ROM)

- Mask Programmable Read Only Memories
 - Performance of NOR topology
 - In NOR topology the parasitic capacitance dominates
 - The resistance of the bit line is small (metal), all capacitances are in parallel
 - $\tau = R_{DSon} \cdot C_L = R_{DSon} \cdot C_{drain} \cdot n_{tr}$
 - Performance of NAND topology
 - In NAND topology the parasitic resistance dominates
 - It is the channel resistance of transistors in the series
 - The structure can be studied using the Elmore delay
 - $\tau = \frac{RC}{2}$
 - Where R and C are the total resistance and capacitance
 - $R = n_{tr} \cdot R_{DSon}$, $C = n_{tr} \cdot C_{drain}$
 - $\tau = \frac{1}{2} \cdot R_{DSon} \cdot C_{drain} \cdot n_{tr}^2$
 - The delay has a quadratic dependency on the number of transistors

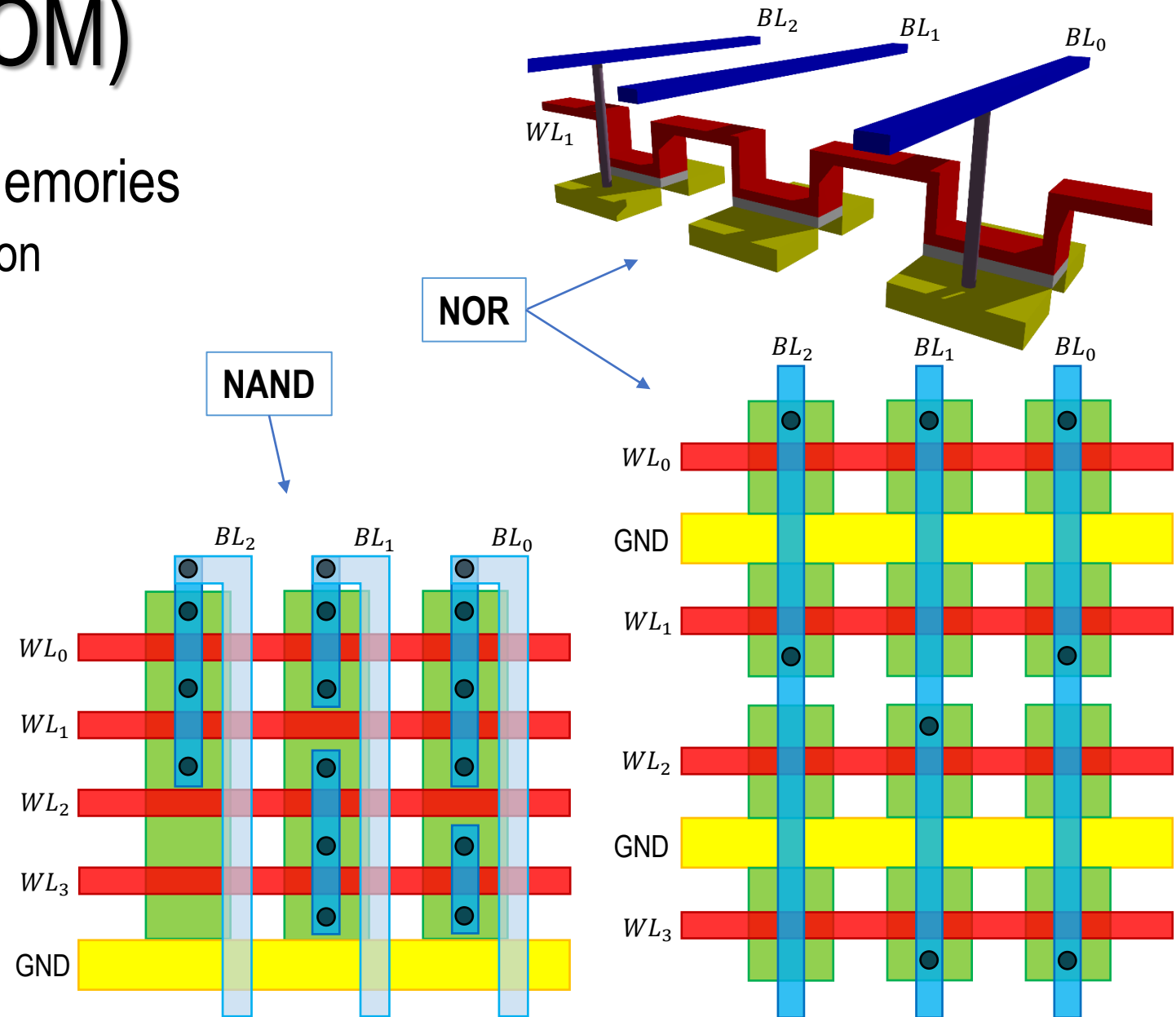


Read Only Memory (ROM)

- Mask Programmable Read Only Memories

- NOR and NAND structure comparison

- NAND structure is denser
 - Given the same memory size
 - NAND is cheaper
 - Given the same area (cost)
 - NAND has more bit cells
 - Performance
 - NAND has higher parasitic parameters on the Bit Line
 - Quadratic vs. linear on the number of transistors
 - NAND is slower
 - Trade-off between costs and performance



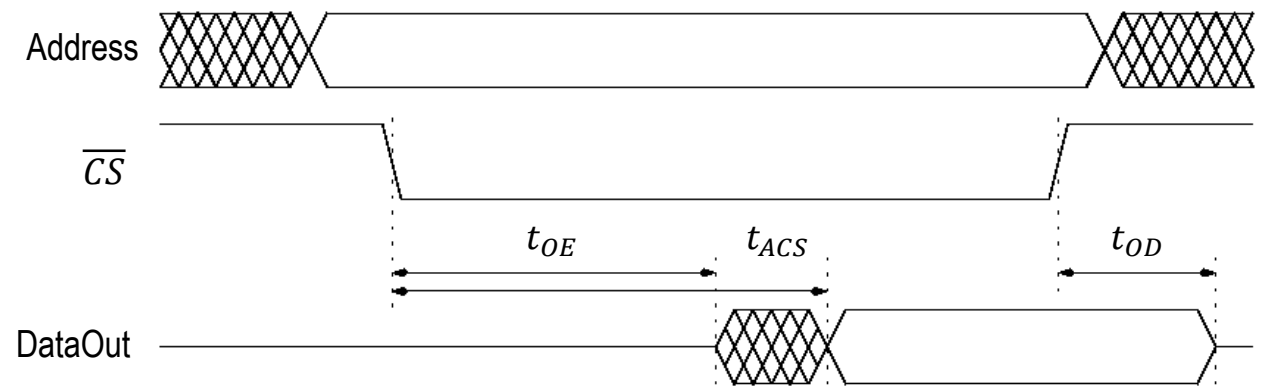
Read Only Memory (ROM)

- Mask Programmable Read Only Memories

- Timing diagram (read operation, no write possible)

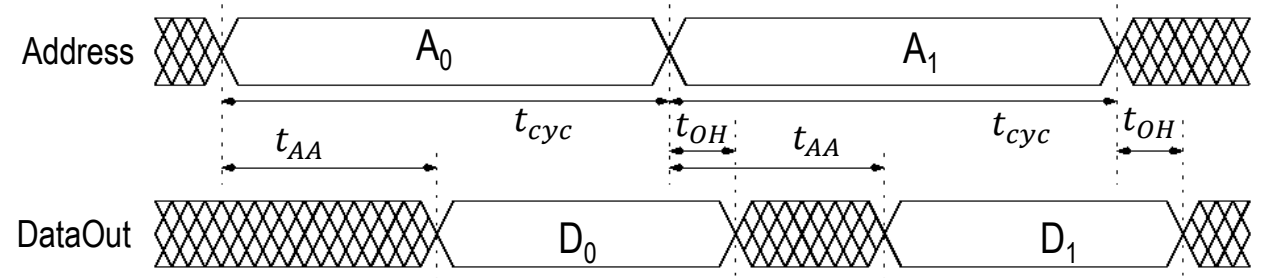
- Address is stable, Chip Select $\overline{CS}$ is not active (high), DataOut is in high impedance (tri-state)
 - Chip Select $\overline{CS}$ is activated (low)
 - After at least t_{OE} (minimum Output Enable time) the output takes some value, but it is not yet stable
 - The value on DataOut might be wrong, but DataOut is not in high impedance anymore
 - After at most t_{ACS} (maximum Chip Select Access time) the output is stable
 - Chip Select $\overline{CS}$ is deactivated (high)
 - After at most t_{OD} (maximum Output Disable time) the output returns to high impedance (tri-state)
 - While the memory is not selected, the address might change

- How many clock cycles?



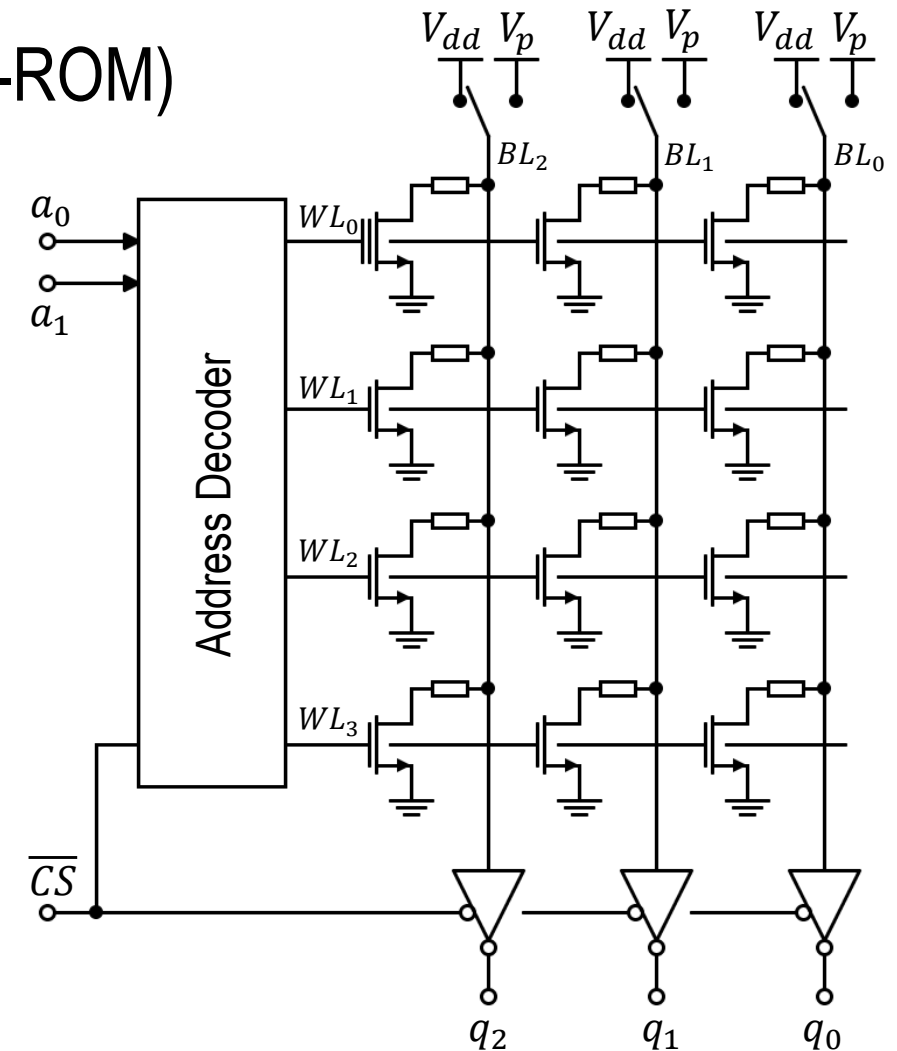
Read Only Memory (ROM)

- Mask Programmable Read Only Memories
 - Timing diagram (read operation, no write possible)
 - Chip Select $\overline{CS}$ is active (low, not shown) and stable, Address changes
 - Since $\overline{CS}$ is active, DataOut is not in high impedance
 - Address becomes stable with value A_0
 - After at most t_{AA} (maximum Address Access time) DataOut takes the value D_0
 - Address changes to value A_1
 - For at least t_{OH} (minimum Output Hold time) DataOut remains at value D_0
 - After at most t_{AA} (maximum Address Access time) DataOut takes the value D_1
 - Address changes to some other value
 - For at least t_{OH} (minimum Output Hold time) DataOut remains at value D_1
 - Addresses should remain stable for at least t_{cyc} (minimum cycle time)



Read Only Memory (ROM)

- One Time Programmable Read Only Memories (OTP-ROM)
 - Based on fuses or anti-fuses
 - All bit cells have a transistor
 - The drain of each transistor is connected to the Bit Line through a fuse
 - All fuses are short circuit after fabrication
 - All transistors are connected to the Bit Lines
 - The initial content of the memory is all 1s
 - Blowing a fuse turns a logic 1 into a logic 0
 - The corresponding transistor is disconnected from the Bit Line
 - The fuse cannot be restored
 - The operation is One Time
 - Cannot turn a logic 0 back to a logic 1



Read Only Memory (ROM)

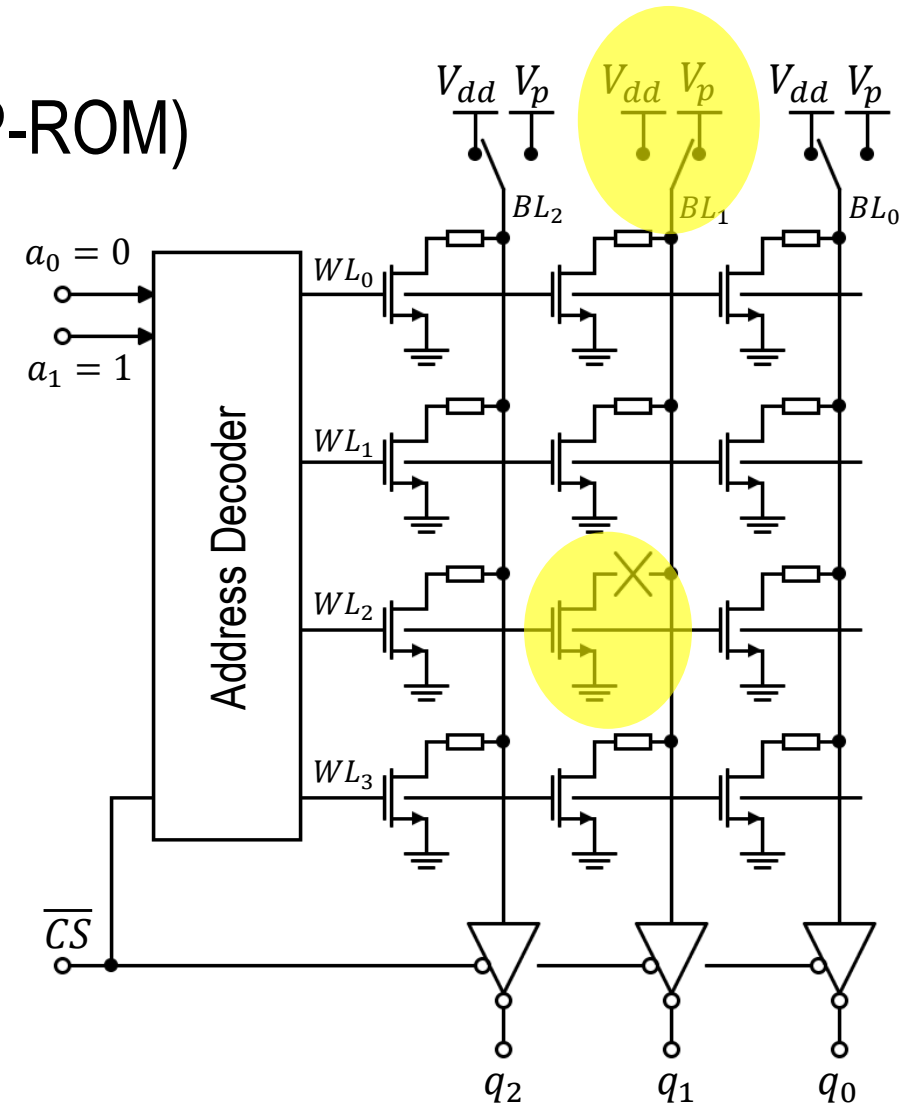
- One Time Programmable Read Only Memories (OTP-ROM)

- How to blow a fuse

- Select the word to be changed by providing its address
 - Example: $A = 10_2 = 2_{10}$
 - The desired Bit Line is set to a programming voltage $V_p \gg V_{dd}$
 - Example: $V_p = 12V$ for BL_1
 - The higher drain current blows the corresponding fuse
 - The transistor is now disconnected and corresponds to a logic 0
 - Repeat the operation for all bits that need to become logic 0

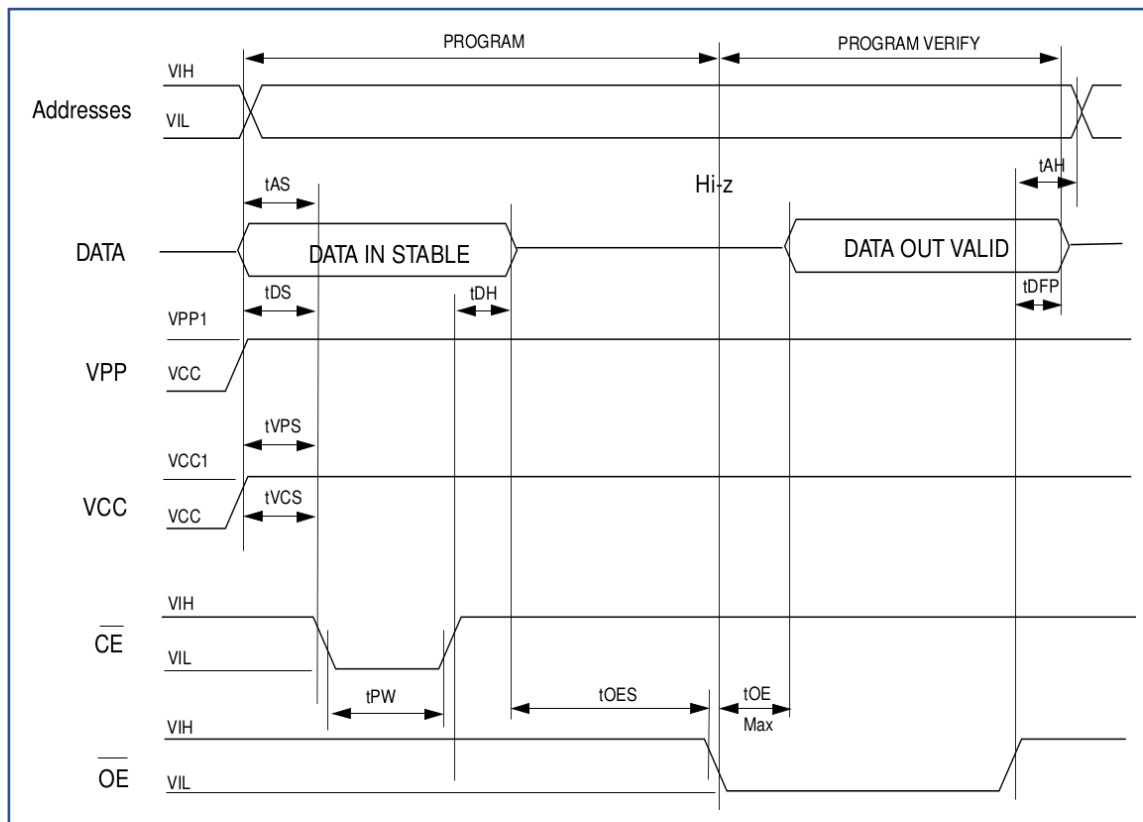
- External interface

- During the operation the data bus is an input
 - Used to specify the Bit Lines to connect to V_p
 - Provide V_p on a dedicated pin of the memory



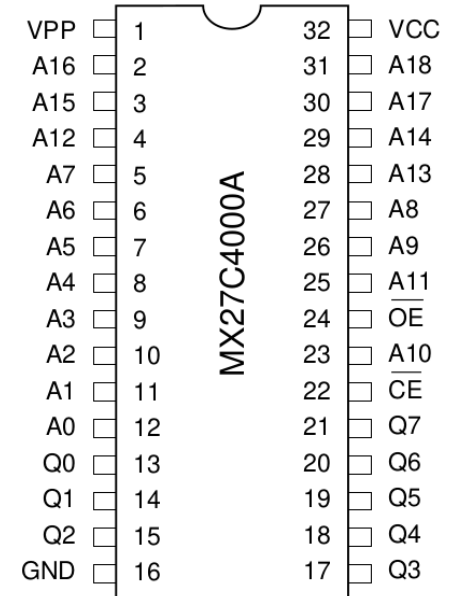
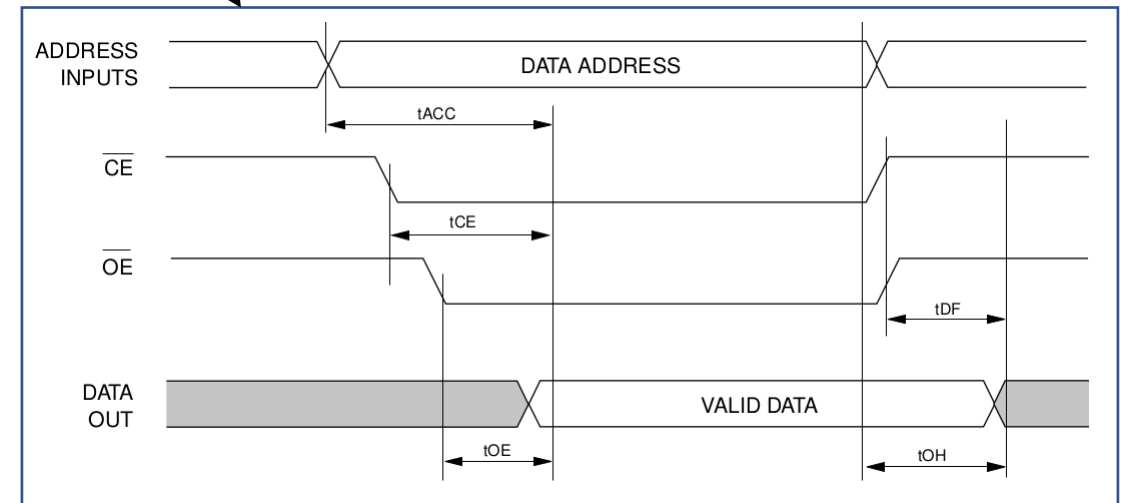
Read Only Memory (ROM)

- One Time Programmable Read Only Memories (OTP-ROM)
 - Timing diagrams (MX27C4000A – 4Mbit – 512K x 8)



Programming Cycle

Read Cycle



Read Only Memory (ROM)

- One Time Programmable Read Only Memories (OTP-ROM)

- Timing diagrams
(MX27C4000A – 4Mbit – 512K x 8)

- $t_{ACC} = t_{CE} = 150 \text{ ns}$
- $t_{PW} = 10 \mu\text{s}$
 - Almost 2 orders of magnitude higher!

AC CHARACTERISTICS

		27C4000A-90		27C4000A-10		27C4000A-12		27C4000A-15		
Symbol	PARAMETER	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	Unit
t _{ACC}	Address to Output Delay		90		100		120		150	ns
t _{CE}	Chip Enable to Output Delay		90		100		120		150	ns
t _{OE}	Output Enable to Output Delay		40		45		50		65	ns
t _{DF}	$\overline{\text{OE}}$ High to Output Float, or $\overline{\text{CE}}$ High to Output Float	0	30	0	30	0	35	0	50	ns
t _{OH}	Output Hold from Address, $\overline{\text{CE}}$ or $\overline{\text{OE}}$ which ever occurred first	0		0		0		0		ns

DC PROGRAMMING CHARACTERISTICS TA = 25°C ± 5°C

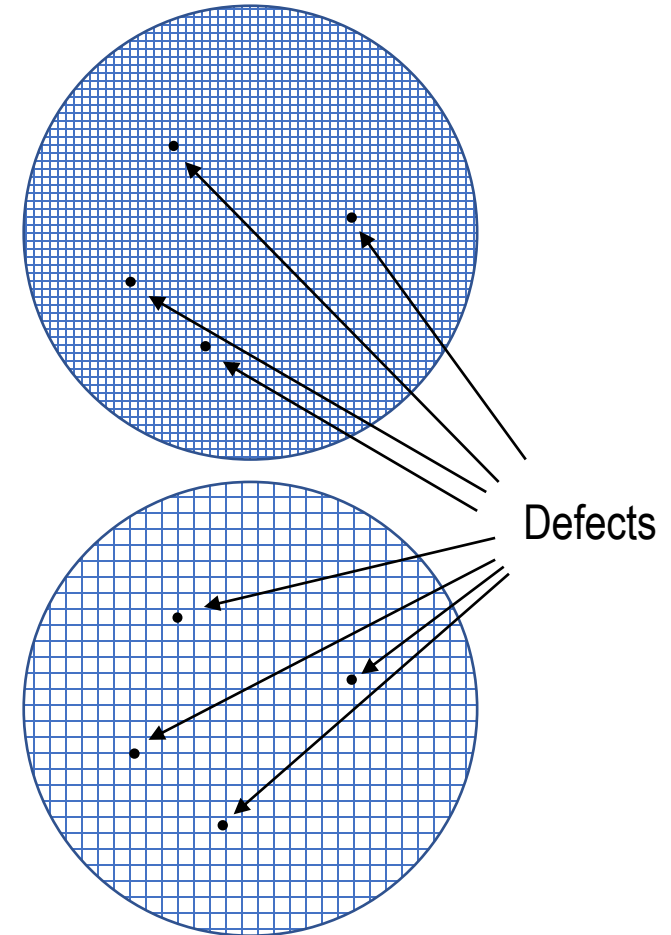
SYMBOL	PARAMETER	MIN.	MAX.	UNIT
VOH	Output High Voltage	2.4		V
VOL	Output Low Voltage		0.4	V
VIH	Input High Voltage	2.0	VCC + 0.5	V
VIL	Input Low Voltage	-0.3	0.8	V
ILI	Input Leakage Current	-10	10	uA
VH	A9 Auto Select Voltage	11.5	12.5	V
ICC3	VCC Supply Current (Program & Verify)		50	mA
IPP2	VPP Supply Current(Program)		30	mA
VCC1	Fast Programming Supply Voltage	6.00	6.50	V
VPP1	Fast Programming Voltage	12.5	13.0	V

AC PROGRAMMING CHARACTERISTICS TA = 25°C ± 5°C

SYMBOL	PARAMETER	MIN.	MAX.	UNIT
t _{AS}	Address Setup Time	2.0		us
t _{OES}	$\overline{\text{OE}}$ Setup Time	2.0		us
t _{DS}	Data Setup Time	2.0		us
t _{AH}	Address Hold Time	0		us
t _{DH}	Data Hold Time	2.0		us
t _{DFP}	Output Enable to Output Float Delay	0	130	ns
t _{VPS}	VPP Setup Time	2.0		us
t _{PW}	PGM Program Pulse Width	10	50	us
t _{VCS}	VCC Setup Time	2.0		us
t _{OE}	Data valid from $\overline{\text{OE}}$		150	ns

Read Only Memory (ROM)

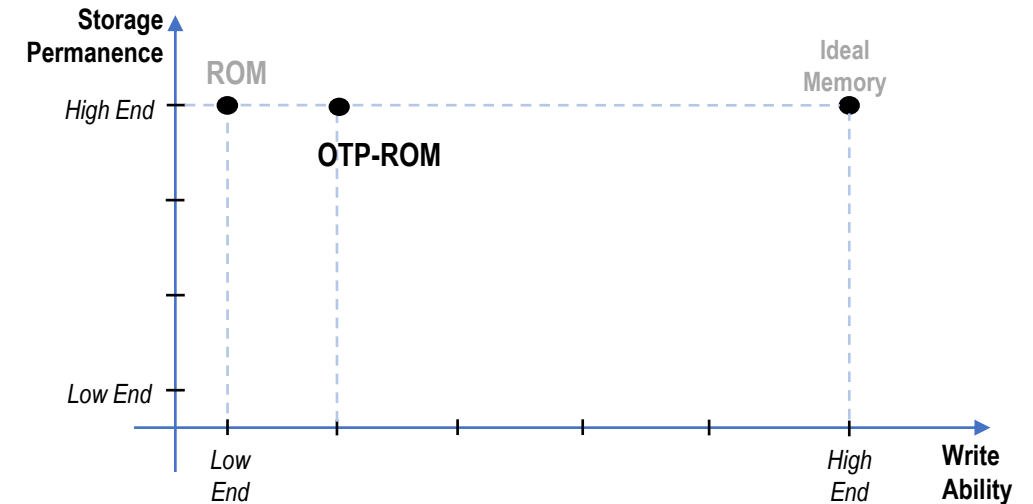
- One Time Programmable Read Only Memories (OTP-ROM)
 - Read operation
 - Identical to Mask Programmable ROMs
 - Disadvantages
 - High resistance
 - Fuse compared to short circuit
 - Slower compared to a Mask Programmable ROM
 - Untestable
 - Small size to have a good yield
 - The number of defective ICs in a wafer does not depend on the IC size
 - There are more small ICs in a wafer than large ICs
 - Yield $y = \frac{n_{IC_{ok}}}{n_{IC_{tot}}} = 1 - \frac{n_{IC_{defective}}}{n_{IC_{tot}}}$
 - Anti-fuses
 - Initially they are open circuits
 - When blown, they become short circuits



Read Only Memory (ROM)

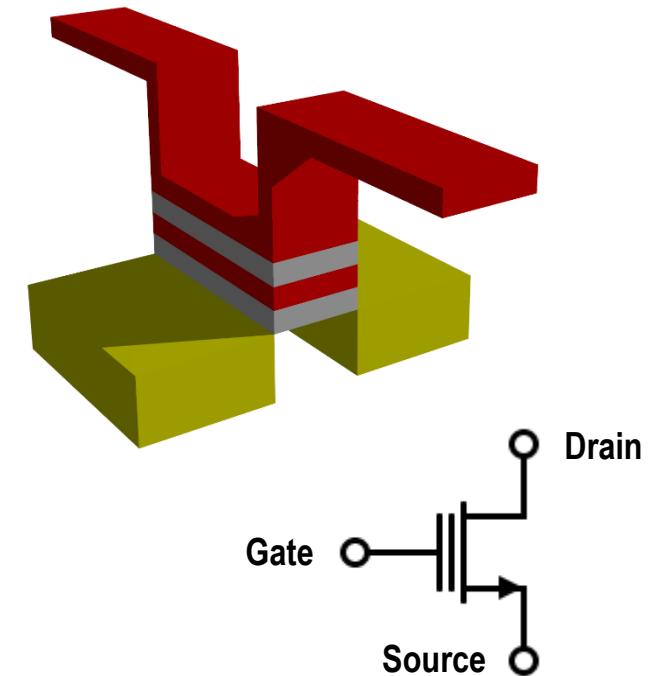
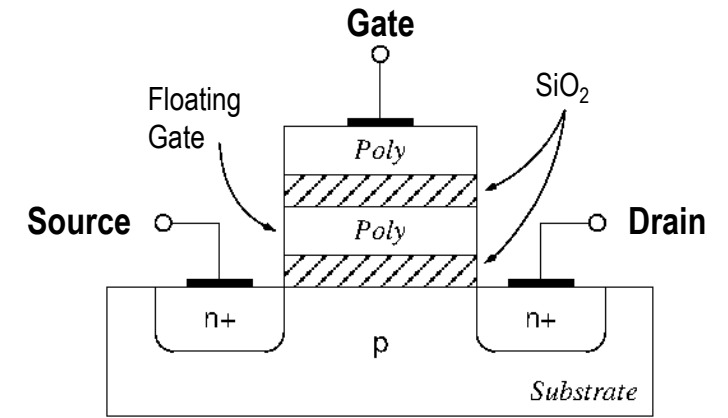
- One Time Programmable Read Only Memories (OTP-ROM)

- Low write ability
 - Can be written only once after fabrication
 - Programmed out of circuit
 - Special hardware programmer
 - Better than Mask Programmable ROM
- High storage permanence
 - For ever
- Costs
 - Low NRE costs
 - No need of dedicated masks
 - Low per unit costs
 - Good for small volume productions
 - Prototyping



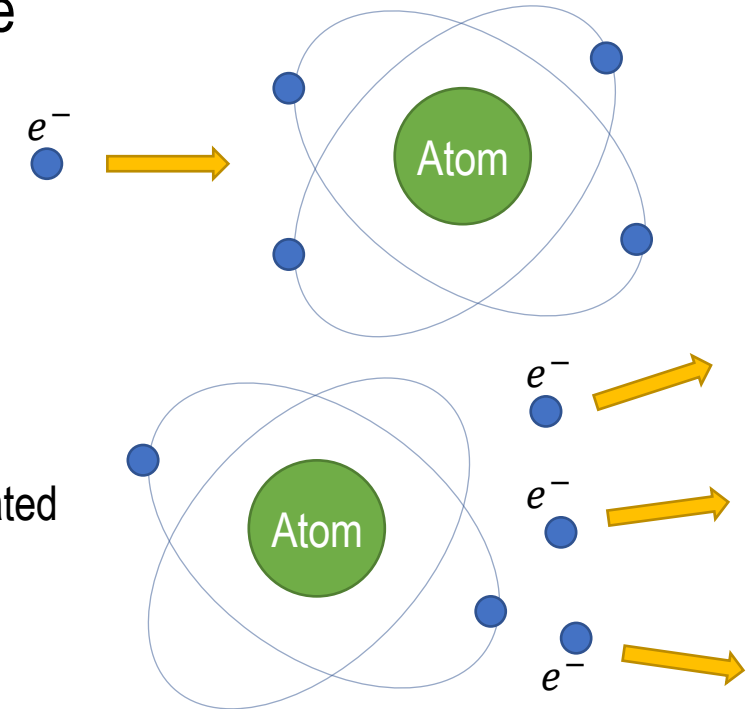
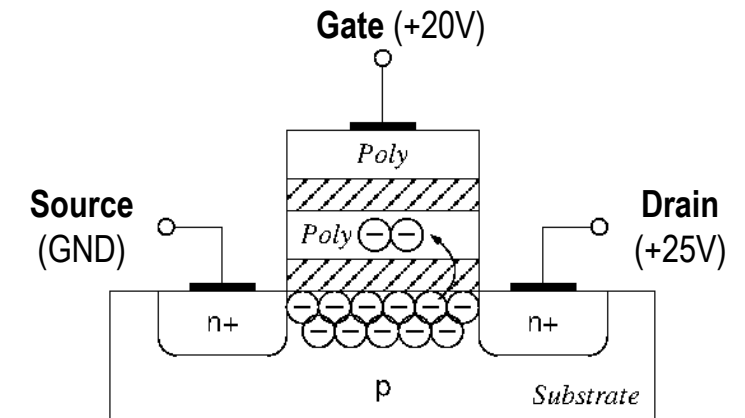
Programmable ROM

- Electrically change the behavior of a transistor
 - Change the threshold voltage V_{th}
 - Increase the threshold voltage so that $V_{th} > V_{dd}$
 - The transistor is always an open circuit with normal gate voltages
- FAMOS transistor
 - Floating gate Avalanche effect MOS
 - Similar to a MOS transistor
 - Additional polysilicon layer between two thin SiO_2 layers below the gate
 - Floating Gate
 - Charges trapped in the floating gate change the threshold voltage V_{th}
 - Negative charges increase the threshold
 - Positive charges decrease the threshold
 - The floating gate is completely isolated
 - How to transfer charges to the floating gate?



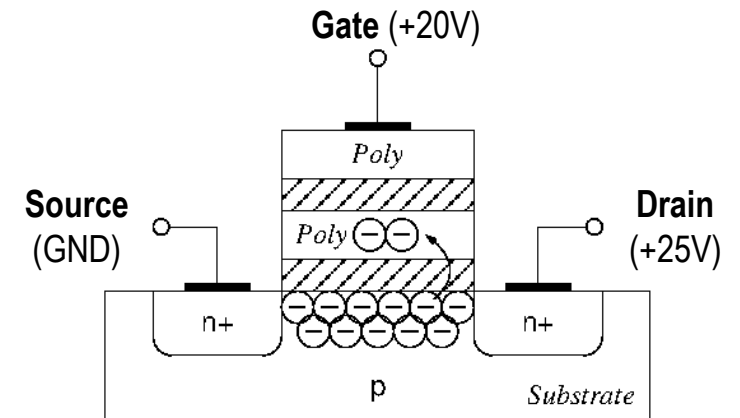
Programmable ROM

- Programming a FAMOS transistor
 - Process known as HOT ELECTRON PROGRAMMING
 - Apply a high programming voltage V_p to the Gate
 - A rich channel forms
 - Apply a high programming voltage V_p' between Drain and Source
 - Electrons in the channel are subject to strong electrical fields
 - Strong horizontal electrical field between Drain and Source
 - There is a large current flowing in the transistor channel
 - Electrons in the channel are accelerated to a high speed
 - High speed electrons collide with other atoms in the lattice
 - They set some electrons free
 - The new electrons contribute to increase the current and are accelerated
 - An avalanche effect is started
 - Electrons with a very high energy become abundant



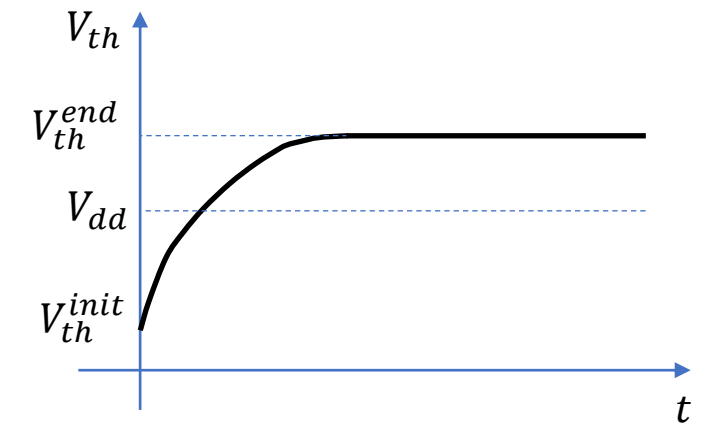
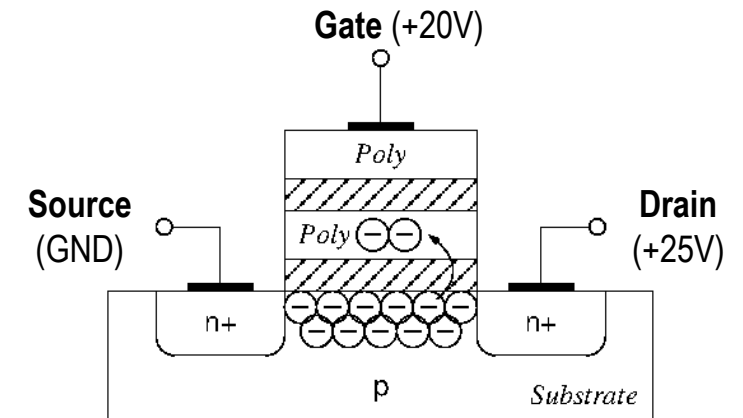
Programmable ROM

- Programming a FAMOS transistor
 - Process known as HOT ELECTRON PROGRAMMING
 - Apply a high programming voltage V_p to the Gate
 - A rich channel forms
 - Apply a high programming voltage V_p' between Drain and Source
 - Electrons in the channel are subject to strong electrical fields
 - Strong vertical electrical field from the Gate
 - The high energy electrons may overcome the SiO_2 barrier
 - Electrons reach the floating gate
 - Electrons loose energy, there is no current flowing in the floating gate
 - The floating gate is isolated
 - There is no horizontal electrical field
 - They are trapped in the floating gate
 - The threshold voltage V_{th} increases



Programmable ROM

- Programming a FAMOS transistor
 - As the threshold voltage V_{th} increases
 - The channel is less rich
 - The vertical electrical field is less strong
 - The number of electrons with enough energy to jump in the floating gate decreases
 - The process is SELF REGULATED
 - The final threshold voltage is quite consistent among all transistors
 - Regardless of the length of the process
 - Design doping levels, densities, geometries
 - V_{th} grows above V_{dd} by a certain margin



Programmable ROM

- Programming a FAMOS transistor
 - Disadvantages
 - Process takes time and requires high voltages
 - How to generate those voltages
 - Provided externally using pins
 - Generated internally from the standard voltage supply
 - Damages to the lattice
 - Cannot apply programming too many times
 - 1000 – 10000 programming cycles
 - Leakage current removes electrons from the floating gate
 - The information is lost over time
 - Several tens of years

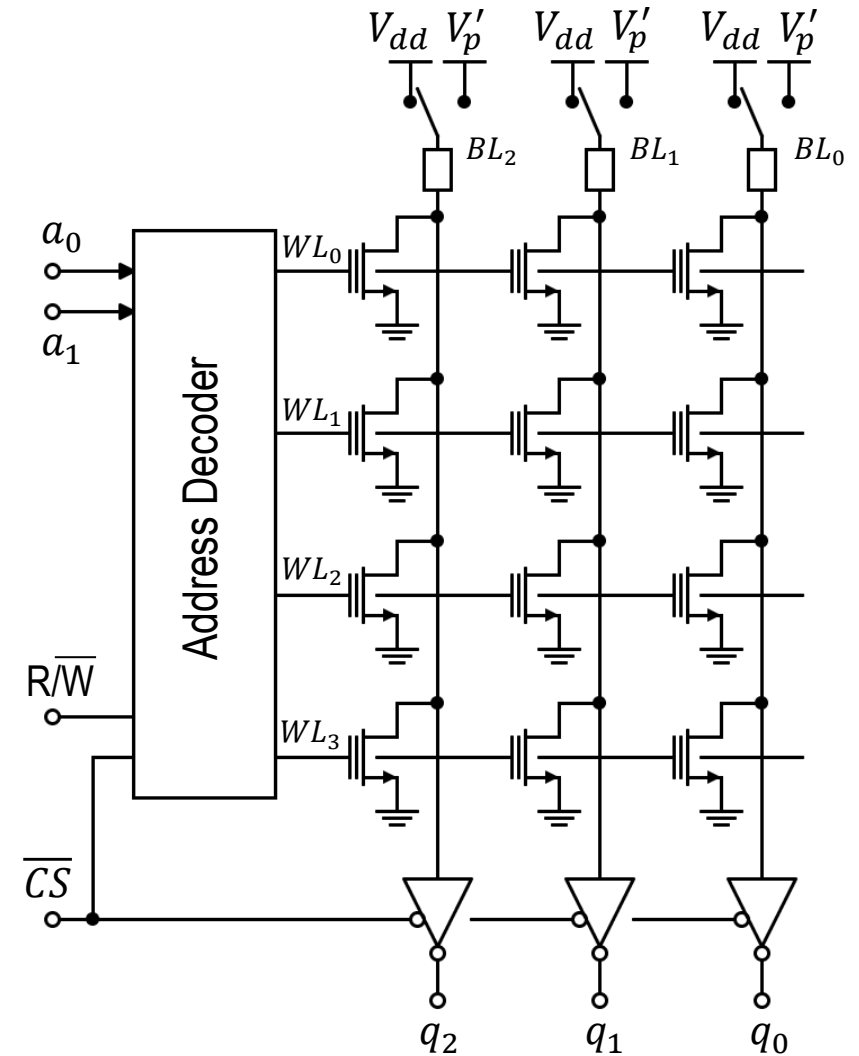
Programmable ROM

- Erasing a FAMOS transistor
 - Electrons cannot be removed electrically from the floating gate
 - Provide energy to the electrons with an UV lamp
 - High energy photons hit electrons
 - The electrons jump back to the substrate and diffuse
 - After a sufficiently long amount of time (minutes) all electrons are removed
 - Need a window on the integrated circuit
 - Shield the window with a sticker when not erasing
 - Cannot target a single transistor
 - All transistors are erased at the same time
 - Out of circuit erasing



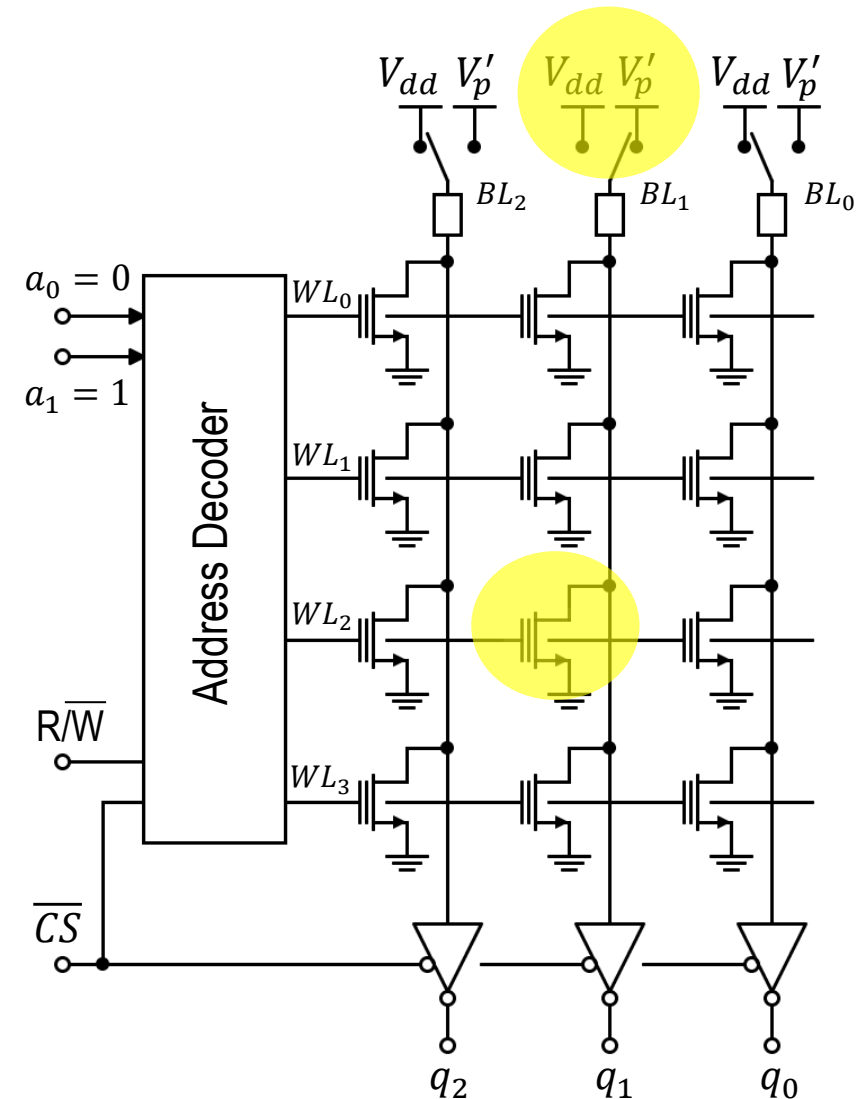
Programmable ROM

- Electrically Programmable ROM (EPROM)
 - Based on FAMOS transistors
 - All bit cells have a FAMOS transistor
 - The drain of each transistor is connected to the Bit Line
 - All FAMOS transistors are not programmed after fabrication
 - All transistors behave normally
 - The initial content of the memory is all 1s
 - Programming a transistor turns a logic 1 into a logic 0
 - The programmed transistor is always OFF at normal voltages
 - The memory can be reset to the initial state
 - Shine UV rays through a window
 - All programmed transistors are erased
 - All non programmed transistors are not affected
 - The memory content returns to all 1s
 - Cannot erase individual bits



Programmable ROM

- Electrically Programmable ROM (EPROM)
 - How to program a transistor
 - Select the word to be changed by providing its address
 - Example: $A = 10_2 = 2_{10}$
 - The selected Word Line is set to a high programming voltage
 - All other Word Lines are set to GND
 - The desired Bit Line is set to a programming voltage $V_p' \gg V_{dd}$
 - Example: $V_p' = 25V$ for BL_1
 - The corresponding transistor is programmed
 - Repeat the operation for all bits that need to become logic 0
 - In parallel if they belong to the same word
 - Avoid affecting other transistors
 - Transistors in other words (rows) have the Gate grounded
 - Other transistors in the same word have the Drain at V_{dd}



Programmable ROM

- Electrically Programmable ROM (EPROM)
 - Read operation
 - Identical to Mask Programmable ROM and OTP-ROM
 - Write operation
 - Programming
 - Move negative charges to the floating gate
 - Makes the transistor a permanent open circuit
 - Turns a 1 into a 0 (if the Bit Line is inverted)
 - Turns a 0 into a 1 (if the Bit Line is affirmed)
 - It works at the word level
 - Erasing
 - Remove the negative charges from the floating gate
 - Restores the normal transistor behavior
 - It applies to the entire memory at once

Given a certain memory, it is either the first case or the second case. The reverse operation is *erasing*.



- Programming is not writing
- Erasing is not writing
- Writing requires both erasing and programming

Programmable ROM

- Electrically Programmable ROM (EPROM)

- Write operation

- Bit Lines with inverters at the bottom
- Write 1100_2 at location 0x05

Addr	d_3	d_2	d_1	d_0
0x00	0	0	1	0
0x01	1	0	0	1
0x02	1	0	1	1
0x03	0	0	1	0
0x04	0	0	0	0
0x05	1	0	1	0
0x06	1	1	1	1
0x07	0	1	1	0

2 *Real steps*

But erasing applies to the entire memory at once

Program d_1
Erase d_2

1

Desired behavior

Addr	d_3	d_2	d_1	d_0
0x00	1	1	1	1
0x01	1	1	1	1
0x02	1	1	1	1
0x03	1	1	1	1
0x04	1	1	1	1
0x05	1	1	1	1
0x06	1	1	1	1
0x07	1	1	1	1

3 *Real steps*

Rewrite the full content by programming all the bits that should be equal to 0

New value

4

Final result

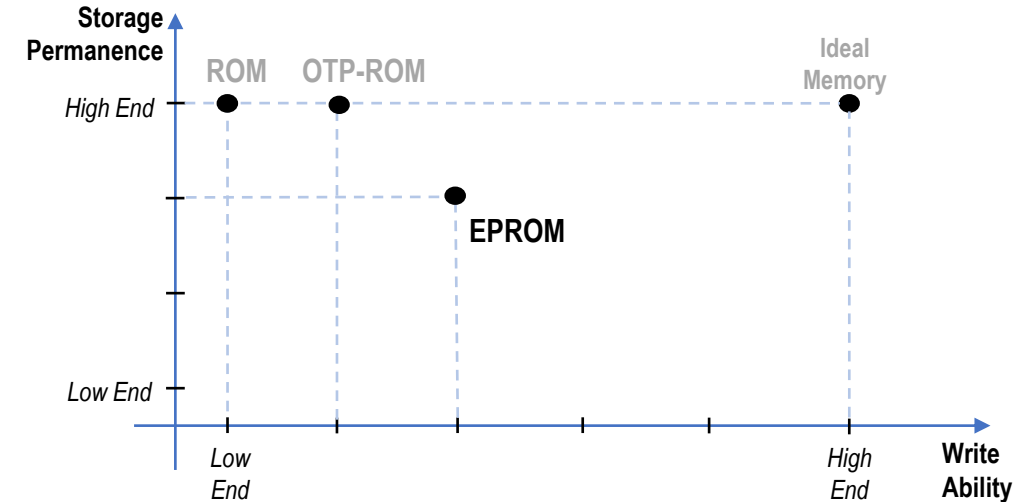
Addr	d_3	d_2	d_1	d_0
0x00	0	0	1	0
0x01	1	0	0	1
0x02	1	0	1	1
0x03	0	0	1	0
0x04	0	0	0	0
0x05	1	1	0	0
0x06	1	1	1	1
0x07	0	1	1	0

Programmable ROM

- Electrically Programmable ROM (EPROM)
 - External interface
 - Provide programming voltages
 - Signal to select programming
 - When programming voltage is applied, the memory goes in programming mode
 - The Data Bus becomes an input to the memory
 - Almost completely superseded OTP-ROMs
 - Much more flexible
 - Cheaper during prototyping
 - Can be reused
 - Not suitable for data that is frequently changed

Programmable ROM

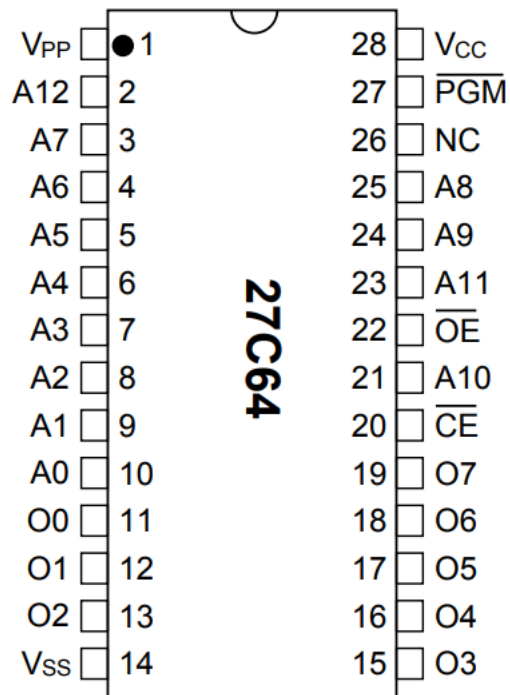
- Electrically Programmable ROM (EPROM)
 - Good write ability
 - Can be written thousands of time
 - Programmed out of circuit
 - Special hardware programmer
 - Erased out of circuit
 - Very long, always erase entire memory
 - High storage permanence
 - Tens of years
 - Leakage currents
 - Can be rewritten to restore the content before it is lost
 - Costs
 - Low NRE costs
 - Low per unit costs



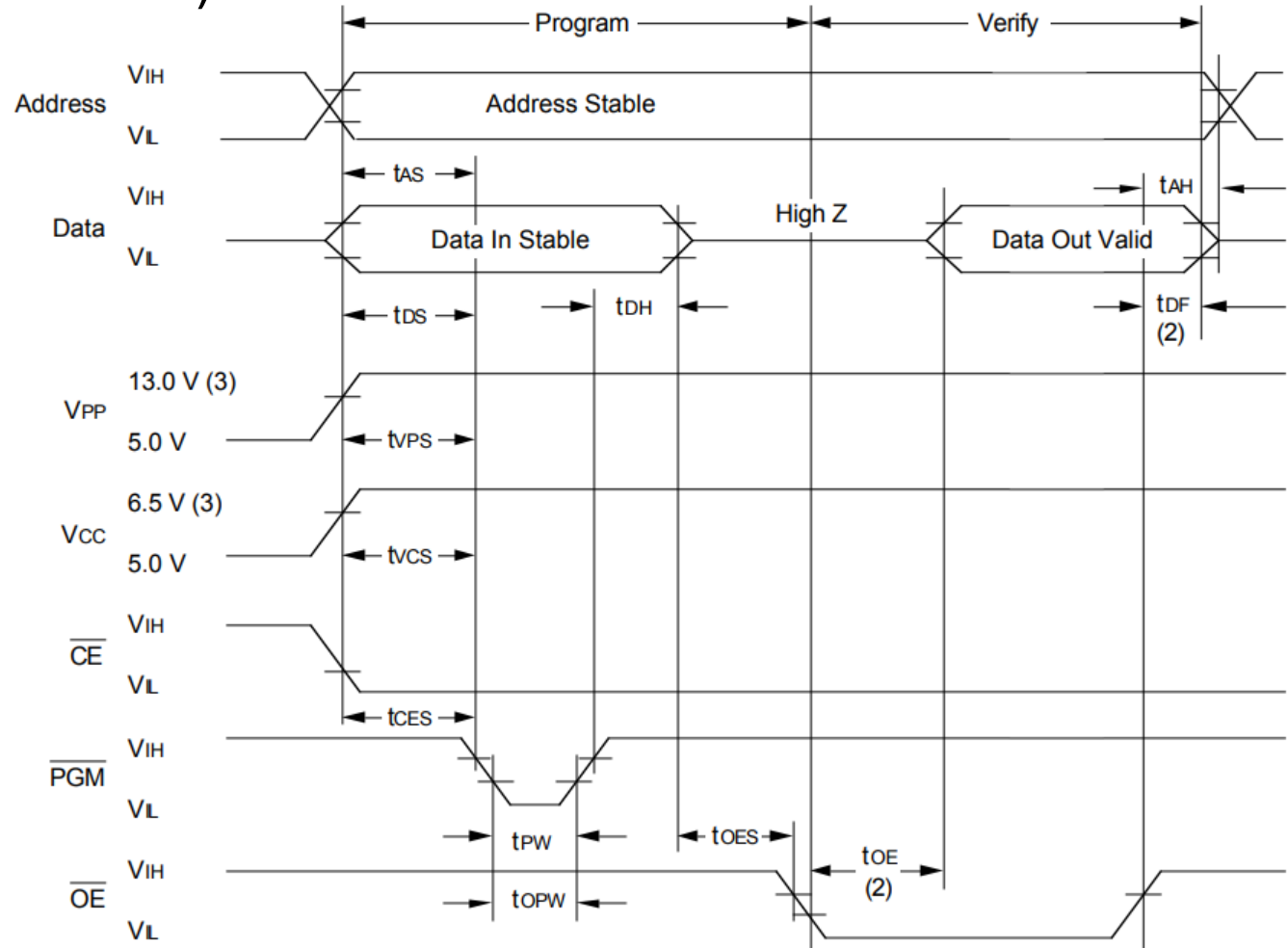
Programmable ROM

- Electrically Programmable ROM (EPROM)

- Datasheet of Microchip 27C64
 - 64K (8K x 8) CMOS EPROM
 - $t_{PW} = 100 \mu s$, $t_{ACC} = 120 ns$

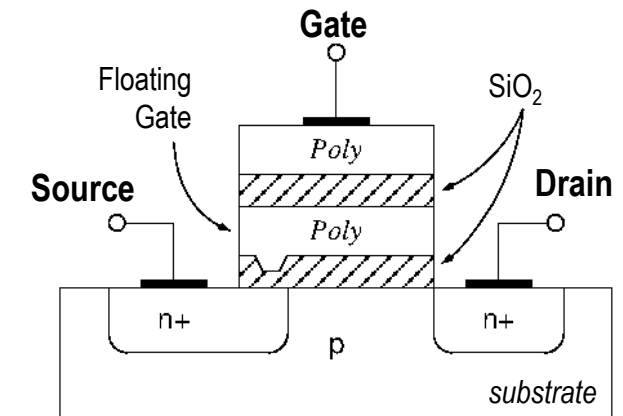
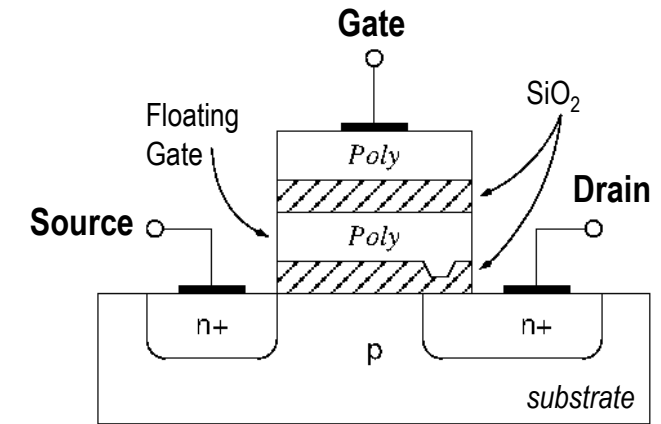


Param	Time
t_{PW}	100 μs
t_{CES}	2 μs
t_{OES}	2 μs
t_{OE}	150 ns



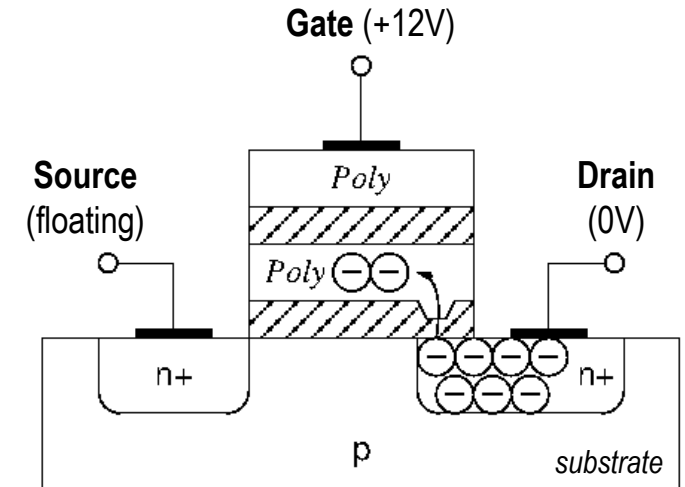
Programmable ROM

- FLOTOX transistor
 - **F**loating gate **T**unneling **O**xide transistor
 - Similarly to FAMOS, it has a floating gate
 - Different programming mechanism compared to FAMOS
 - Electrically erasable
 - Tunneling effect
 - Quantum mechanical phenomenon
 - A particle (electron) can pass a potential barrier even if it has insufficient energy
 - Thinner oxide layer on the Drain (Source) side
 - Makes tunneling easier
 - Electrons come from the Drain (Source) $n+$ region rather than from the substrate or channel
 - High electrical field necessary
 - Also known as Fowler-Nordheim effect in floating gate transistors



Programmable ROM

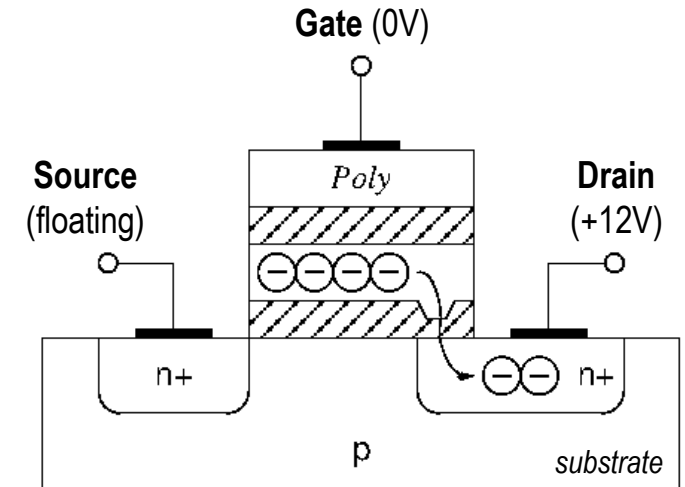
- Programming a FLOTOX transistor
 - Apply a high programming voltage V_p to the Gate
 - A rich channel forms
 - But the channel is **NOT** used for programming
 - Leave the Source floating (or grounded)
 - Connect the Drain to ground
 - There is no current flowing in the transistor between Drain and Source
 - The Source is floating or $V_{DS} = 0$
 - Electrons in the Drain $n+$ region are subject to strong electrical fields
 - Electrons jump to the floating gate by tunneling through the thin oxide
 - The threshold voltage V_{th} increases
 - Process is not Self Regulated
 - The threshold voltage V_{th} increases as long as the voltages are applied



There is NOT a hole in the oxide
Tunneling is a probabilistic effect due to the Heisenberg uncertainty principle. A thin oxide makes the probability of electrons passing through the barrier higher than with a thicker oxide.

Programmable ROM

- Erasing a FLOTOX transistor
 - Connect the Gate to ground
 - No channel forms
 - Leave the Source floating (or grounded)
 - Apply a high programming voltage V_p to the Drain
 - There is no current flowing in the transistor between Drain and Source
 - The Source is floating or there is no channel
 - Electrons in the Floating Gate are subject to strong electrical fields
 - Electrons jump to the Drain $n+$ region by tunneling through the thin oxide
 - The threshold voltage V_{th} decreases
 - After a sufficiently long amount of time, all electrons are removed from the floating gate
 - The threshold voltage V_{th} returns to the initial value

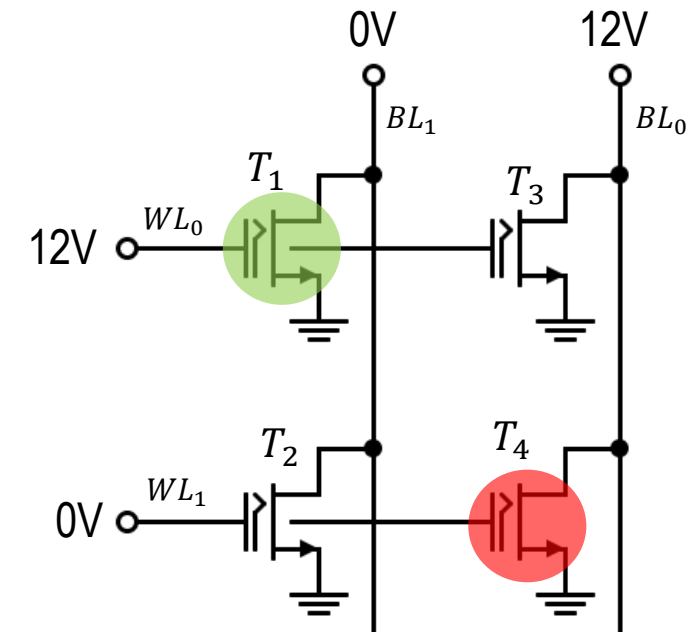


Programmable ROM

- Comparison of programming mechanisms
 - Hot Electron Programming is faster than Tunneling
 - May increase performance
 - But performance may be limited by other issues
 - Tunneling causes less damages to the lattice than Hot Electron Programming
 - Higher life-time of the device
 - Many more programming cycles
 - Hot Electron Programming is Self Regulated, Tunneling is not
 - Tunneling requires careful timing to achieve good consistency
 - The resulting distribution of V_{th} after programming has a larger standard deviation
 - Requires larger margins during operations
 - FLOTOX transistors is larger and harder to fabricate
 - Lower yield, higher costs

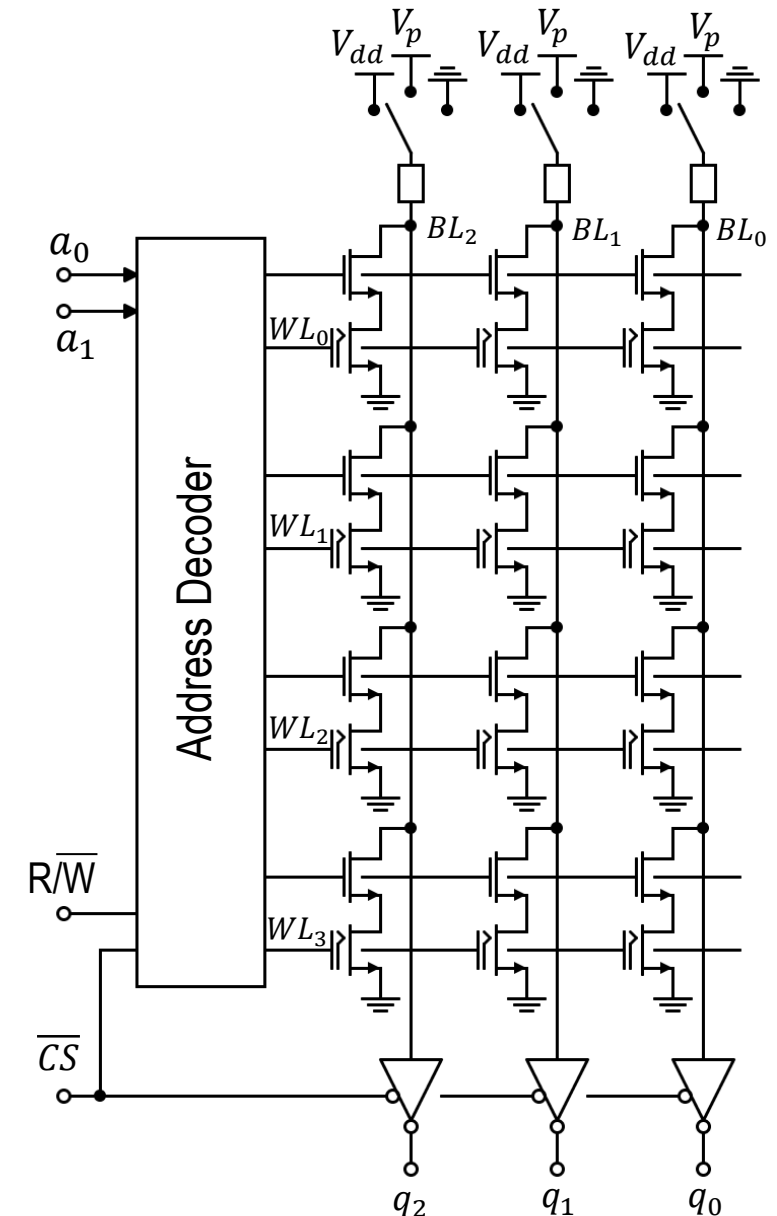
Programmable ROM

- Electrically Erasable Programmable ROM (E²PROM)
 - Based on FLOTOX transistors
 - All bit cells have a FLOTOX transistor
 - Thin oxide on the Drain side
 - Straightforward implementation (but it doesn't work!)
 - The Drain of each transistor is connected to the Bit Line
 - The Source of each transistor is grounded
 - Problem: programming a transistor erases another one
 - To program $T_1 \rightarrow$ Gate: $WL_0 = 12V$, Drain: $BL_1 = 0V$
 - To avoid programming $T_2 \rightarrow$ Gate: $WL_1 = 0V$
 - To avoid programming $T_3 \rightarrow$ Drain: $BL_0 = 12V$, such that $V_{GD} = 0$
 - But T_4 is erased $\rightarrow$ Gate: $WL_1 = 0V$, Drain: $BL_0 = 12V$
 - Solution
 - Use an access transistor for each bit cell



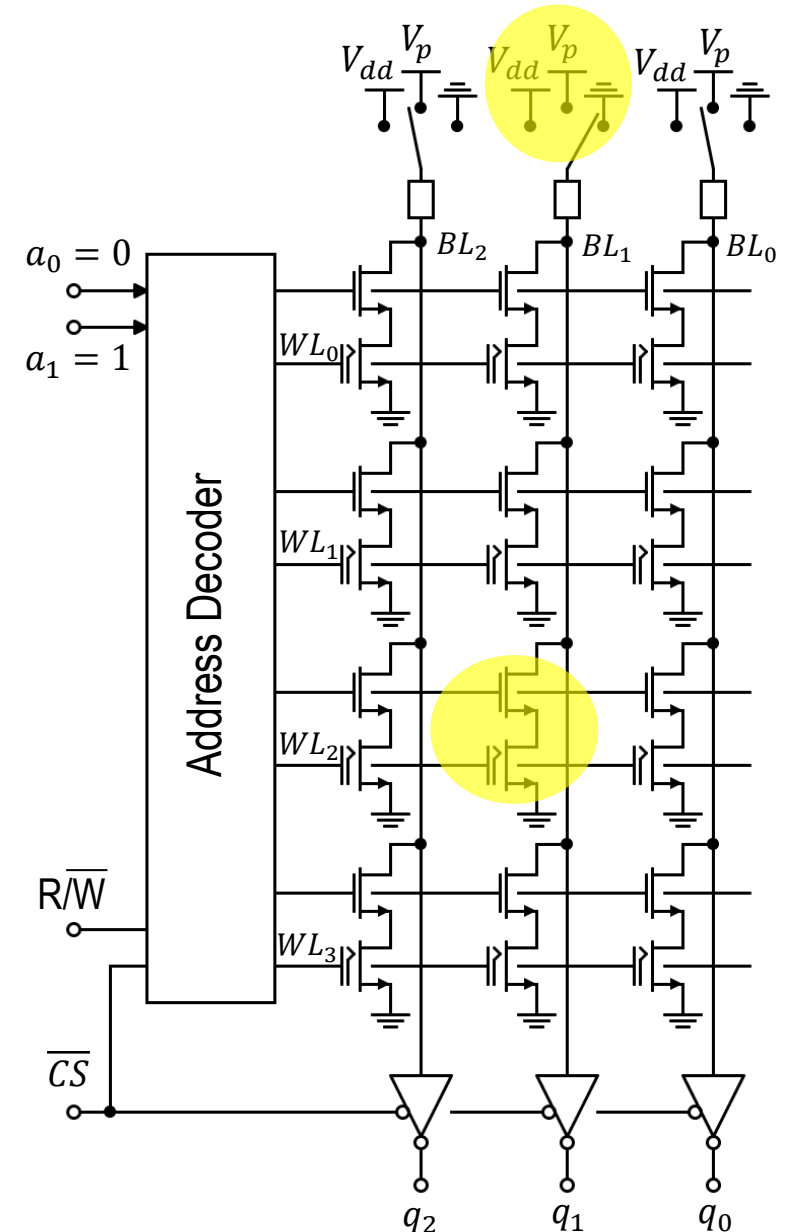
Programmable ROM

- Electrically Erasable Programmable ROM (E²PROM)
 - Based on FLOTOX transistors
 - All bit cells have a FLOTOX transistor
 - Thin oxide on the Drain side
 - All bit cells have an access transistor
 - Drain connected to the Bit Line, Source connected to the FLOTOX
 - Gate driven by the Address Decoder
 - Access transistor activated when word is selected
 - All FLOTOX transistors are not programmed after fabrication
 - All transistors behave normally
 - The initial content of the memory is all 1s
 - Programming a transistor turns a logic 1 into a logic 0
 - The programmed transistor is always OFF at normal voltages
 - Erasing a transistor turns a logic 0 into a logic 1



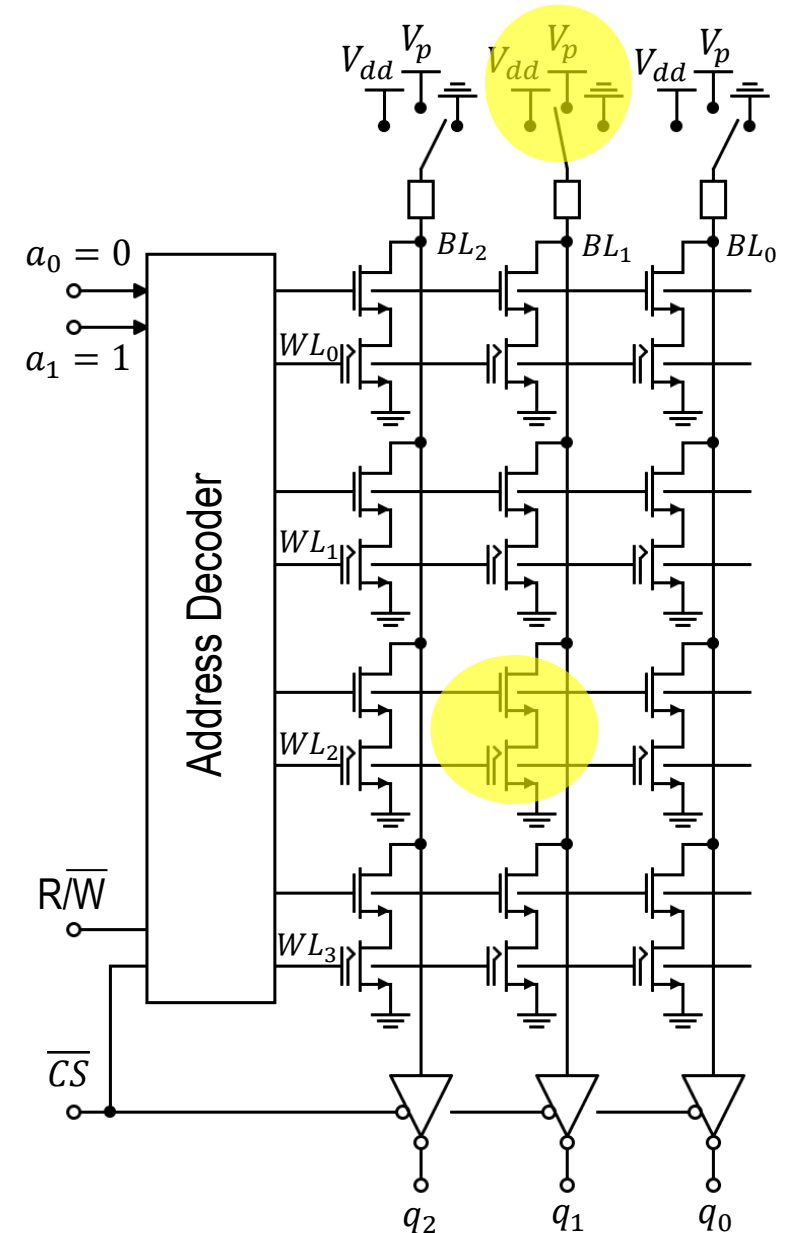
Programmable ROM

- Electrically Erasable Programmable ROM (E²PROM)
 - How to program a transistor
 - Select the word to be changed by providing its address
 - Example: $A = 10_2 = 2_{10}$
 - The selected Word Line is set to a high programming voltage
 - The access transistors for that Word Line are activated
 - The desired Bit Line is set to ground
 - All other Bit Lines are set to the programming voltage V_p
 - The corresponding transistor is programmed
 - Repeat the operation for all bits that need to become logic 0
 - In parallel if they belong to the same word
 - Avoid affecting other transistors
 - Transistors in other words have the Access transistor OFF
 - Other transistors in the same word have the Drain at V_p
 - Hot Electron Programming does not occur because it needs higher voltages than tunneling



Programmable ROM

- Electrically Erasable Programmable ROM (E²PROM)
 - How to erase a transistor
 - Select the word to be changed by providing its address
 - Example: $A = 10_2 = 2_{10}$
 - The selected Word Line is set to ground
 - The access transistors for that Word Line are activated
 - The desired Bit Line is set to a programming voltage V_p
 - All other Bit Lines are set to ground
 - The corresponding transistor is erased
 - The transistor can now be turned ON and corresponds to a logic 1
 - Repeat the operation for all bits that need to return to logic 1
 - Avoid affecting other transistors
 - Transistors in other words have the Access transistor OFF
 - Other transistors in the same word have the Drain grounded

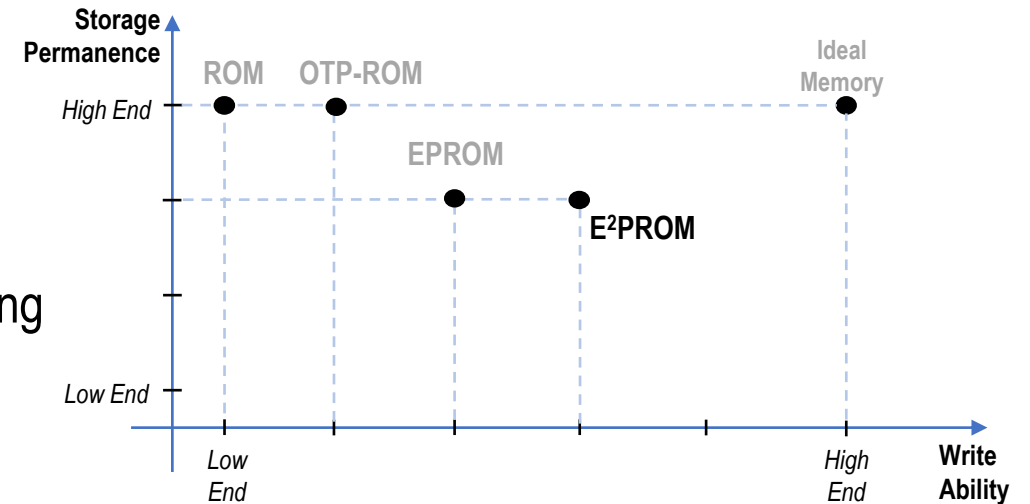


Programmable ROM

- Electrically Erasable Programmable ROM (E²PROM)
 - Read operation
 - Identical to Mask Programmable ROM, OTP-ROM and EPROM
 - External interface
 - Provide programming voltages
 - Signal to select programming
 - Signal to select erasing
 - External interface often simplified by providing internal programming and erasing mechanisms
 - Widely used
 - Much more flexible than EPROMs
 - Can be erased by a micro-processor in circuit
 - Expensive
 - The bit cell is larger than in an EPROM

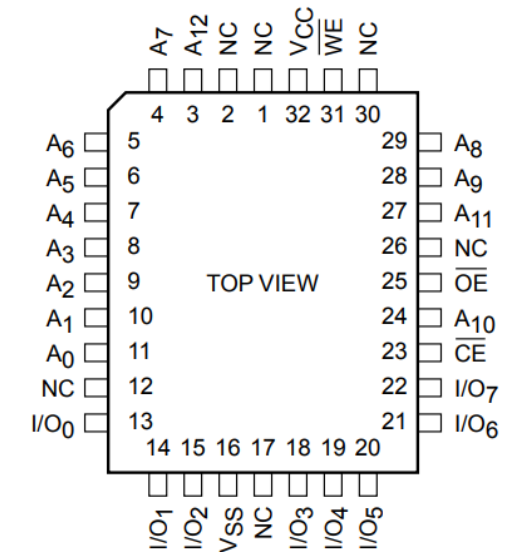
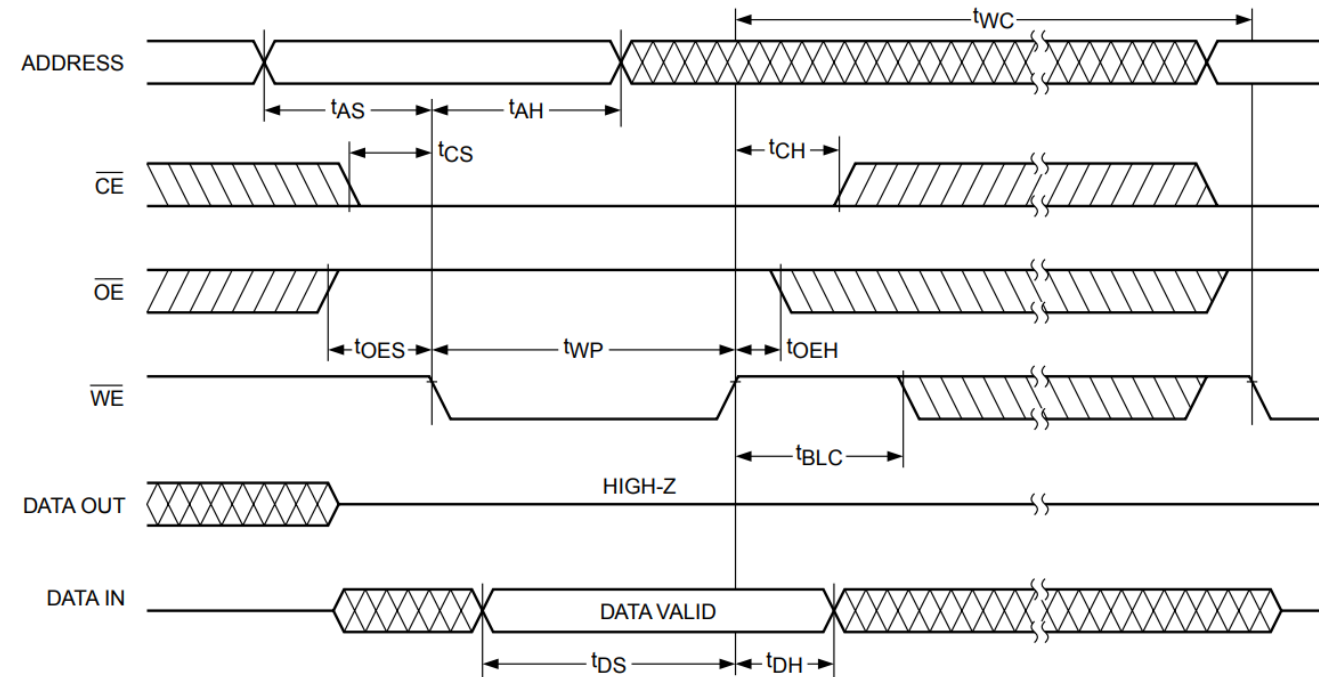
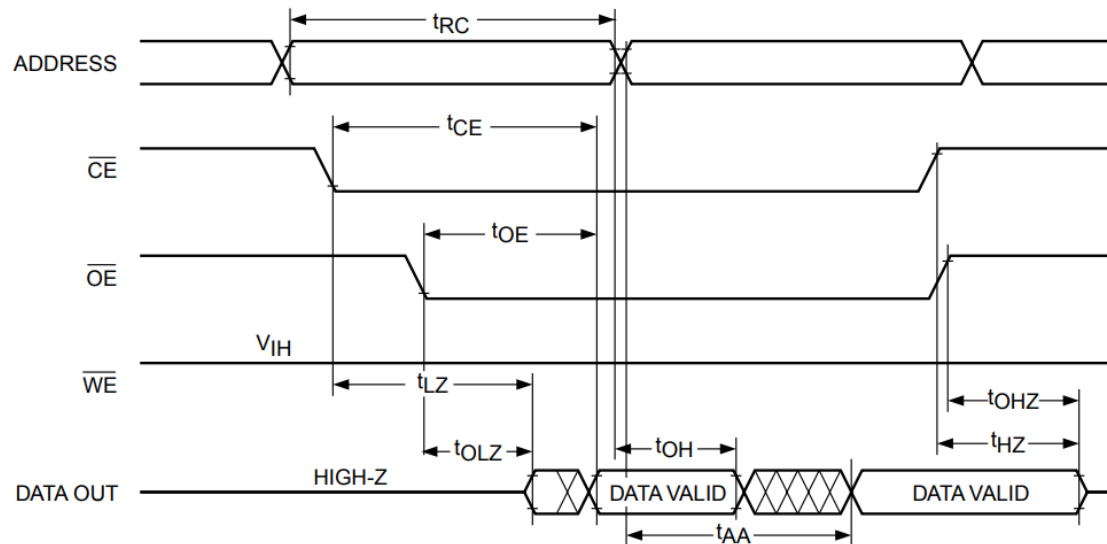
Programmable ROM

- Electrically Erasable Programmable ROM (E²PROM)
 - Very good write ability
 - Can be written thousands of times
 - Programmed in circuit
 - Erased in circuit
 - Single bit access in both writing (programming) and erasing
 - Directly by a micro-processor or other digital hardware
 - High storage permanence
 - Tens of years, like EPROMs
 - Can be rewritten to restore the content before it is lost
 - Costs
 - Low NRE costs
 - Low per unit costs, but higher than an EPROM of the same size



Programmable ROM

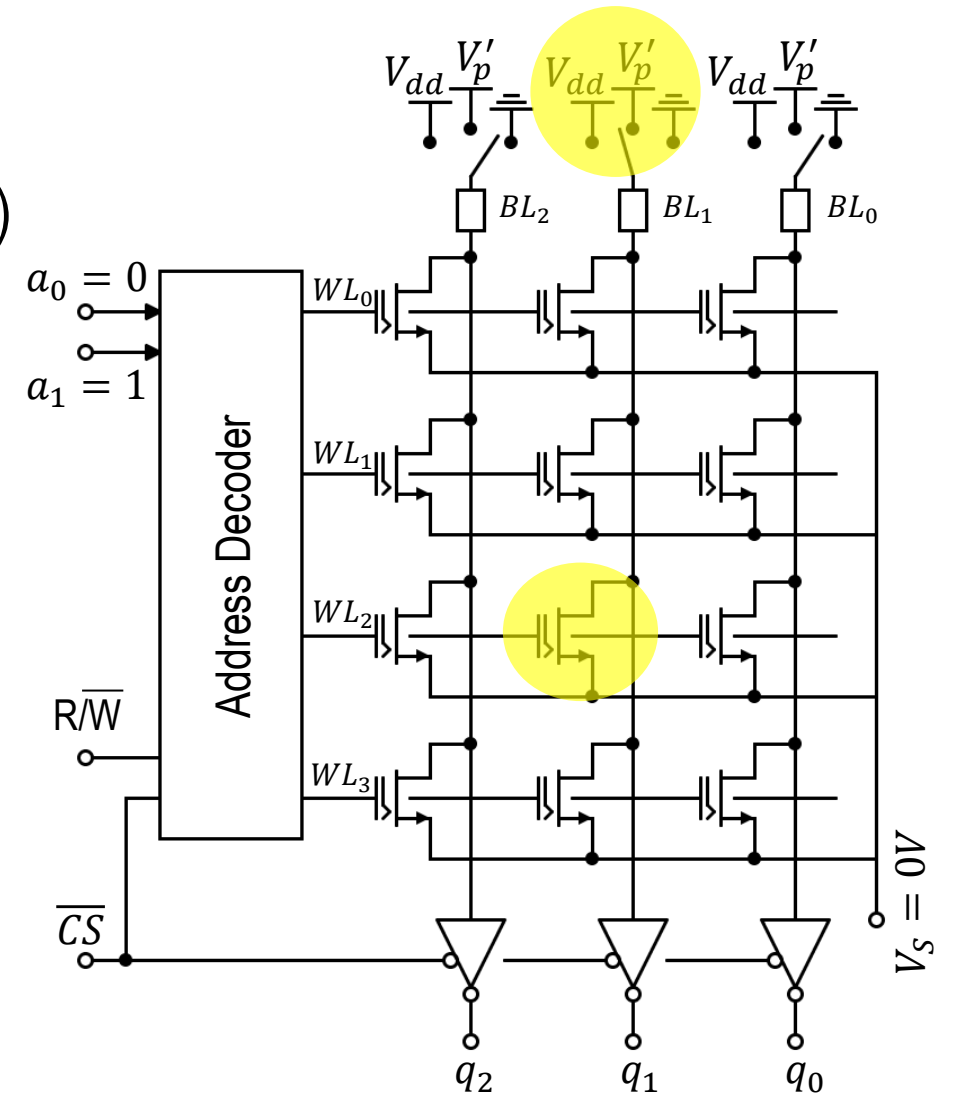
- Electrically Erasable Programmable ROM (E²PROM)
 - Datasheet of On Semiconductor CAT28C64B
 - 64K (8K x 8) CMOS E²PROM
 - Read cycle time: $t_{AA} = 150 \text{ ns}$
 - Write cycle time: $t_{WC} = 5 \text{ ms}$
 - To write, a word is first erased and then written



Flash Memory

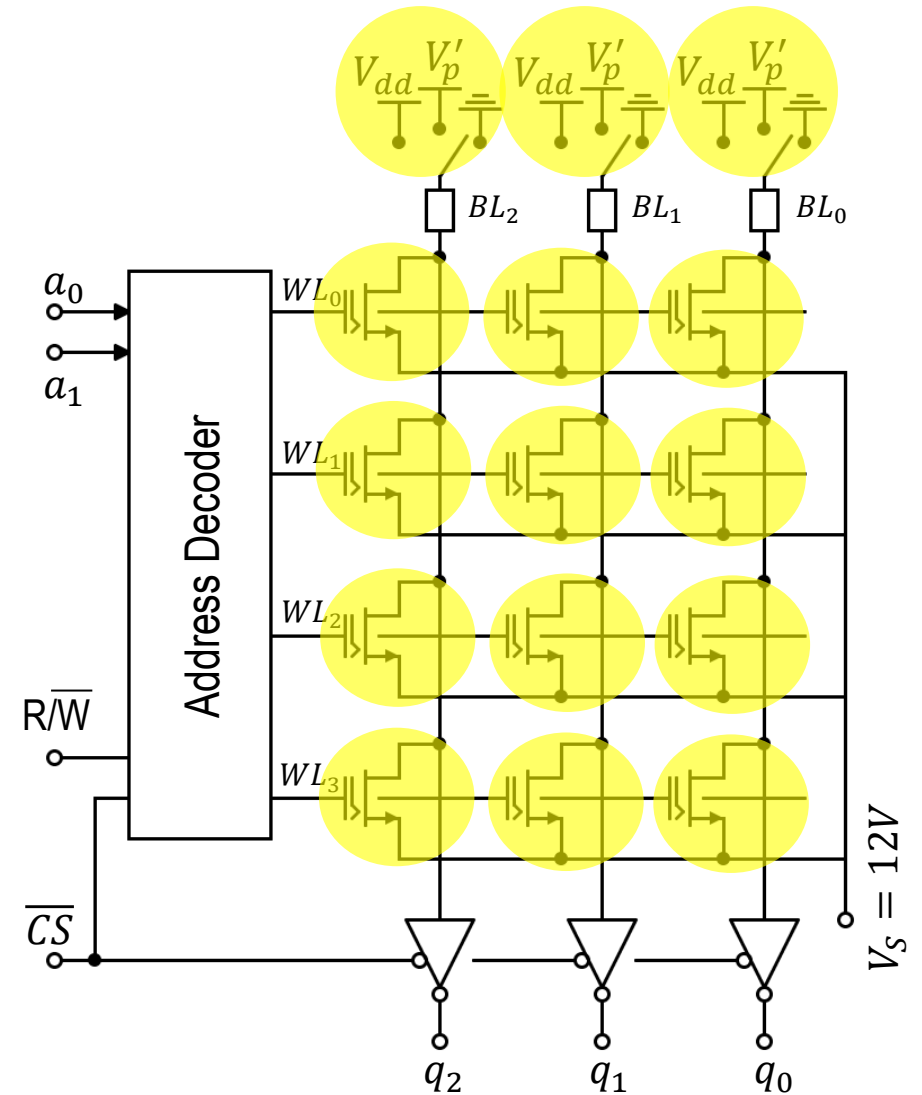
• NOR Flash Memory

- How to program a transistor (hot electrons programming)
 - Select the word to be changed by providing its address
 - Example: $A = 10_2 = 2_{10}$
 - The selected Word Line is set to a high programming voltage
 - The desired Bit Line is set to a high programming voltage
 - All other Bit Lines are set to ground
 - Connect all Sources to ground with $V_S = 0V$
 - The corresponding transistor is programmed
 - Repeat the operation for all bits that need to become logic 0
 - In parallel if they belong to the same word
- Avoid affecting other transistors
 - Transistors in other words (row) have the Gate grounded
 - Other transistors in the same word have the Drain grounded
 - They experience a short tunneling on the Source side



Flash Memory

- NOR Flash Memory
 - How to erase all transistors (tunneling)
 - All Word Lines are grounded
 - All Bit Lines are grounded (or left floating)
 - Connect all Sources to a high programming voltage
 - $V_S = 12V$
 - All transistors are erased
 - All transistor can now be turned ON and correspond to a logic 1



Flash Memory

- NOR Flash Memory

- How to erase all transistors (tunneling)

- All Word Lines are grounded
 - All Bit Lines are grounded (or left floating)
 - Connect all Sources to a high programming voltage

- $V_s = 12V$

- All transistors are erased

- All transistor can now be turned ON and correspond to a logic 1

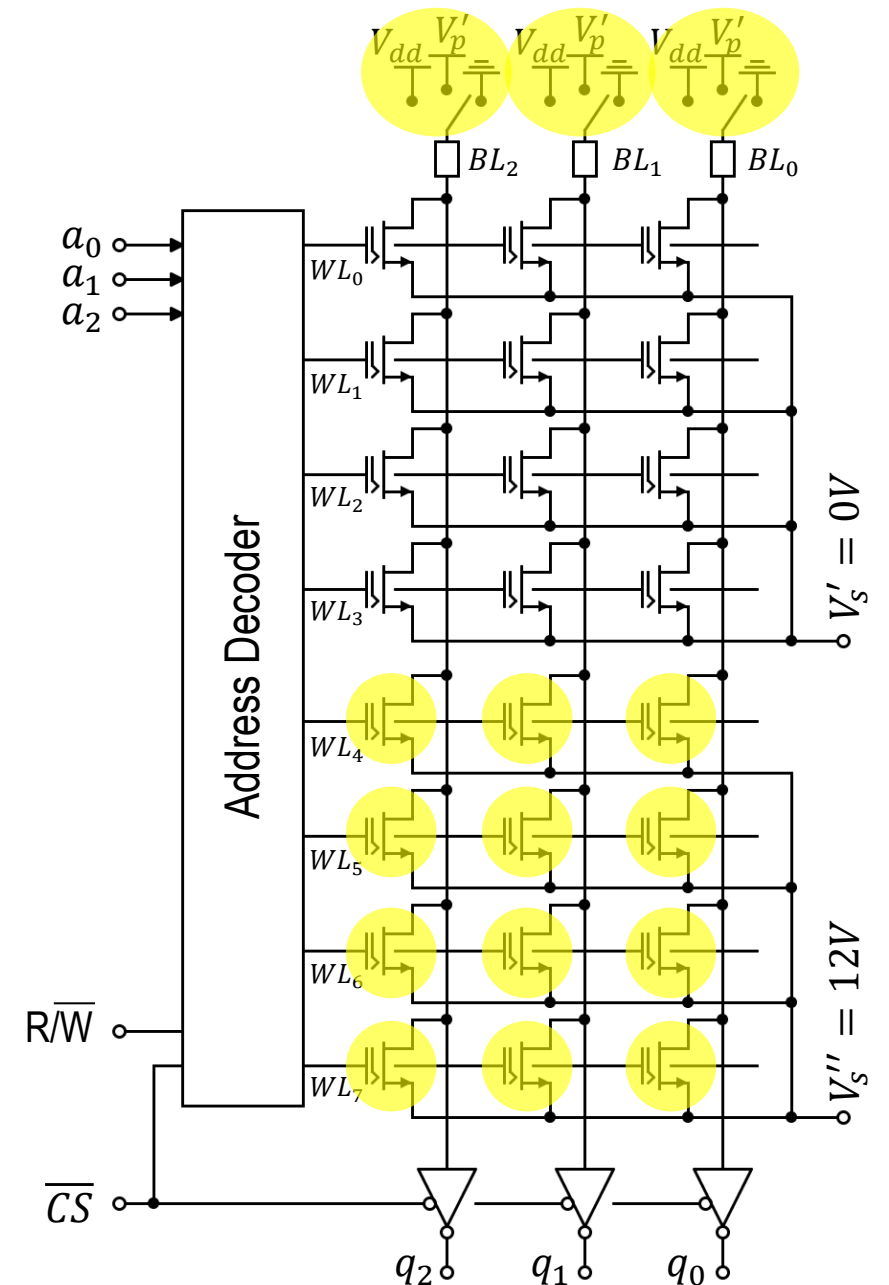
- Block level erase

- The memory is divided in blocks

- Block size is from a few Kbytes to hundreds of Kbytes
 - Sources of transistors in different blocks can be independently driven

- Erase is applied to a single block at a time

- Rather than the entire memory



Flash Memory

- NOR Flash Memory

- Read operation

- Identical to Mask Programmable ROM, OTP-ROM, EPROM and E²PROM

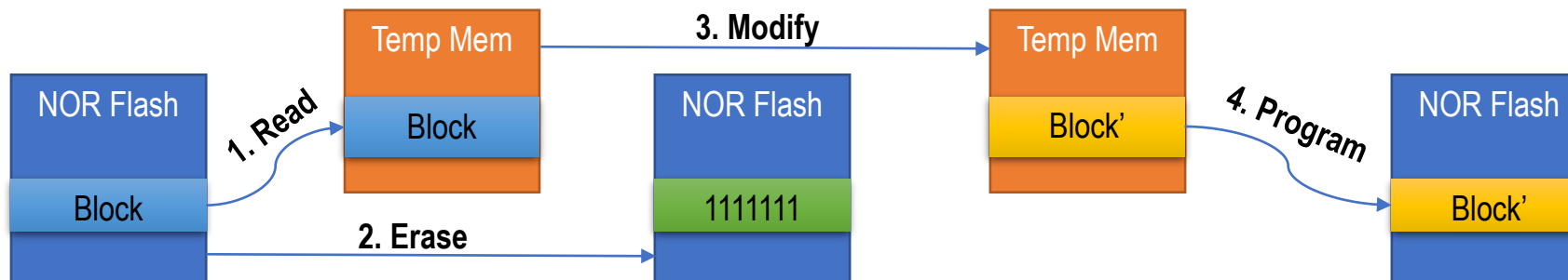
- Performance in writing

- Sacrifice single bit access while erasing
 - When writing, an entire block is erased and rewritten
 - Even when writing a single word

- Example: write 1K words

- E²PROM: $t_{WC} \cdot 1024 = 5 \text{ ms} \cdot 1024 = 5,12 \text{ s}$

- NOR Flash: $t_{erase} \cdot n_{blocks} + t_W \cdot 1024 = 5 \text{ ms} \cdot 1 + 240 \text{ ns} \cdot 1024 = 5.24 \text{ ms}$



Flash Memory

- NOR Flash Memory
 - Common Flash Memory Interface (CFI)
 - Reading as in a ROM
 - Writing is based on commands
 - Standard set of commands
 - All manufacturers adhere to this standard interface
 - A software driver is used
 - Example
 - Write 0xA1F4 @ 0x28EC560
 - AA @ 555
 - 55 @ 2AA
 - A0 @ 555
 - A1F4 @ 0x28EC560

Command Sequence (Note 1)		Cycles	Bus Cycles (Notes 2–5)											
			First		Second		Third		Fourth		Fifth		Sixth	
			Addr	Data	Addr	Data	Addr	Data	Addr	Data	Addr	Data	Addr	Data
Read (Note 5)		1	RA	RD										
Reset (Note 6)		1	XXX	F0										
Status Register Read		2	555	70	XXX	RD								
Status Register Clear		1	555	71										
Autoselect (Note 7)	Manufacturer ID	4	555	AA	2AA	55	555	90	X00	0001				
	Device ID (Note 8)	6	555	AA	2AA	55	555	90	X01	227E	X0E	(18)	X0F	(18)
	Device ID	4	555	AA	2AA	55	555	90	X01	(17)				
	Secure Silicon Region Factory Protect	4	555	AA	2AA	55	555	90	X03	(9)				
	Sector Protect Verify (Note 10)	4	555	AA	2AA	55	555	90	(SA) X02	00/01				
	Reset / ASO Exit (Note 6)	1	XXX	F0										
Program		4	555	AA	2AA	55	555	A0	PA	PD				
Write to Buffer (Note 11)		3	555	AA	2AA	55	SA	25	SA	WC	PA	PD	WBL	PD
Program Buffer to Flash		1	SA	29										
Write to Buffer Abort Reset (Note 12)		3	555	AA	2AA	55	555	F0						
Unlock Bypass	Enter	3	555	AA	2AA	55	555	20						
	Program (Note 13)	2	XXX	A0	PA	PD								
	Write to Buffer (Note 13)	4	SA	25	SA	WC	PA	PD	WBL	PD				
	Sector Erase	2	XXX	80	SA	30								
	Chip Erase	2	XXX	80	XXX	10								
	Reset (Note 14)	2	XXX	90	XXX	00								
Chip Erase		6	555	AA	2AA	55	555	80	555	AA	2AA	55	555	10
Sector Erase		6	555	AA	2AA	55	555	80	555	AA	2AA	55	SA	30

Flash Memory

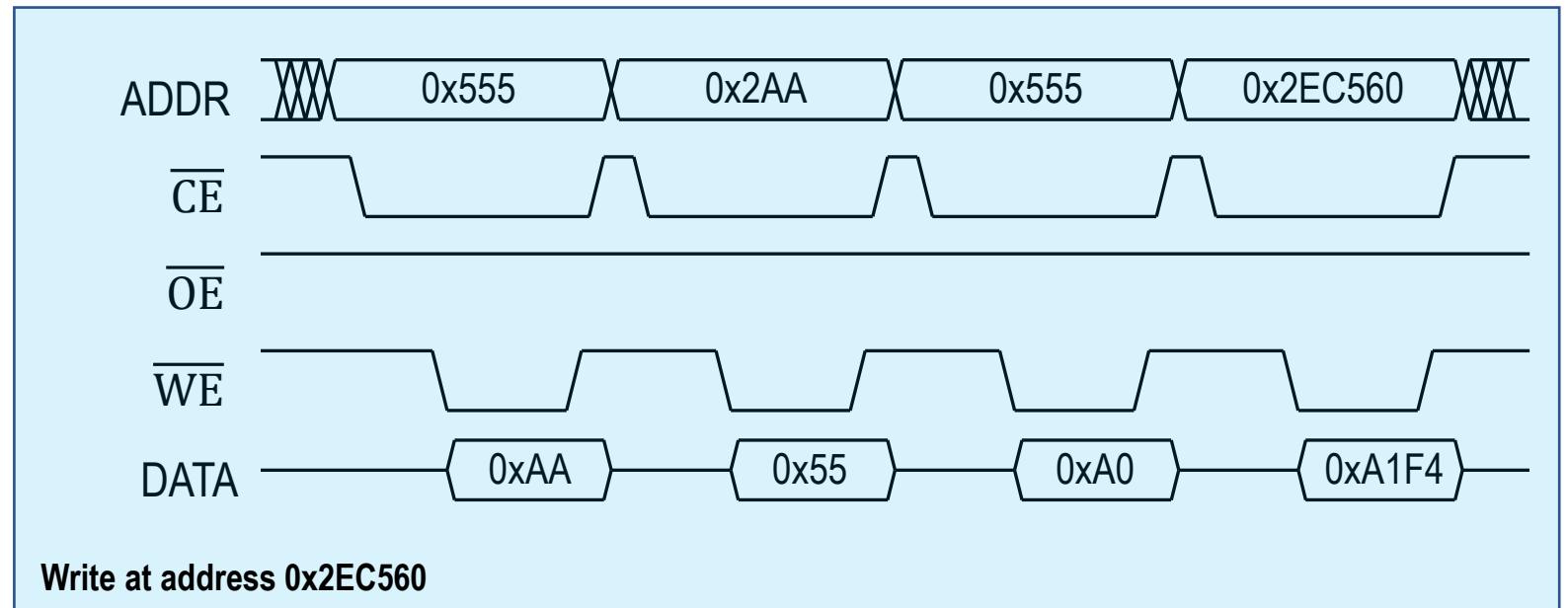
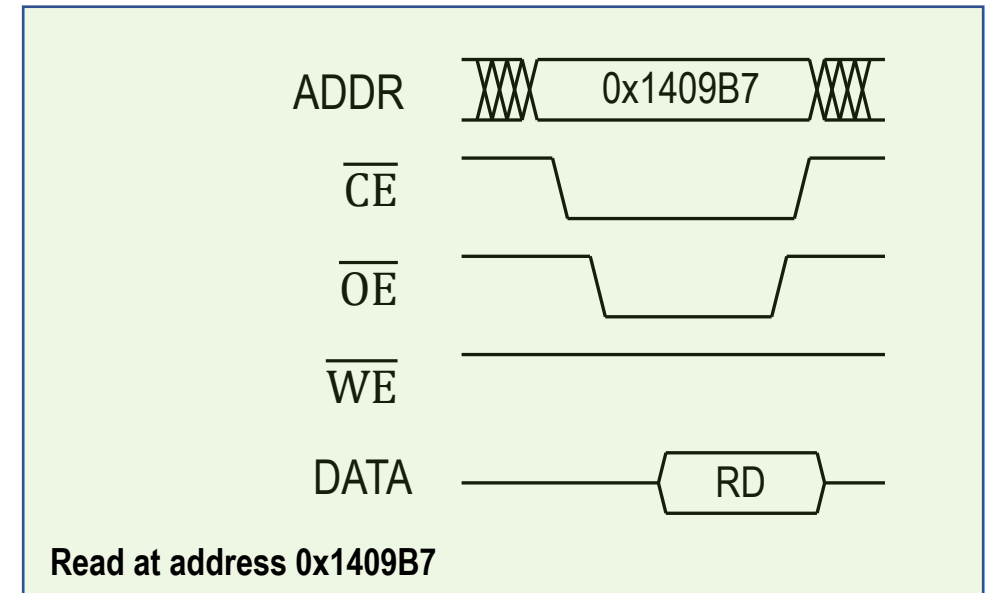
- NOR Flash Memory

- Example

- Read at address 0x1409B7
 - Normal operation
 - Write 0xA1F4 @ 0x2EC560
 - 4 write cycles
 - AA @ 555
 - 55 @ 2AA
 - A0 @ 555
 - A1F4 @ 0x2EC560

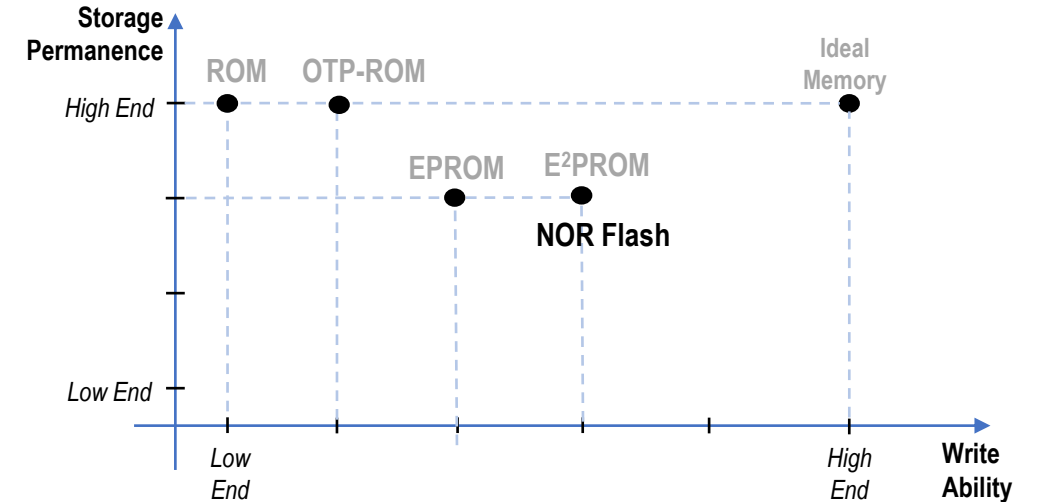
- Exercise

- Timing diagram for sector erase
 - Timing diagram for chip erase



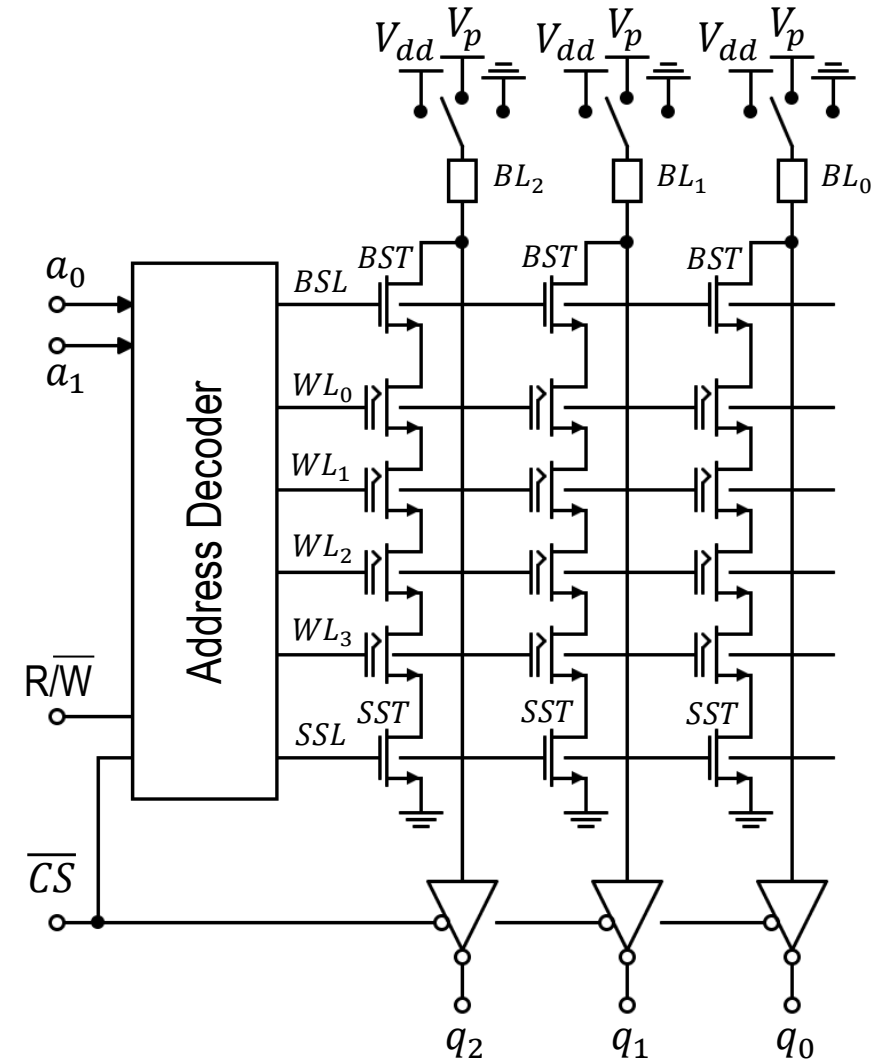
Flash Memory

- NOR Flash Memory
 - Very good write ability
 - Can be written thousands of time
 - Programmed in circuit
 - Erased in circuit
 - Word level access in reading
 - Can be used to store and execute code (BIOS)
 - High storage permanence
 - Tens of years, like EPROMs and E²PROMs
 - Costs
 - Low NRE costs
 - Low per unit costs, cheaper than E²PROM of the same size



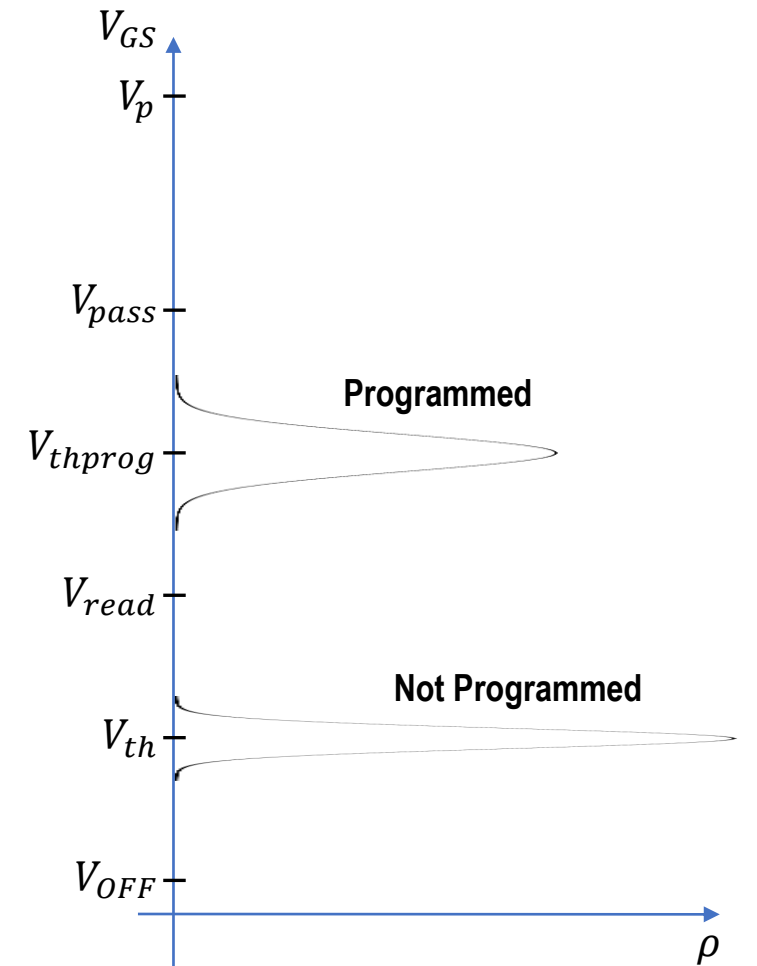
Flash Memory

- E²PROM and NOR Flash Problem
 - E²PROMs and NOR Flashes are expensive
- NAND Flash Memory
 - Based on FLOTOX transistors
 - All bit cells have a FLOTOX transistor
 - Thin oxide on the Drain side
 - Tunneling effect for both programming and erasing
 - NAND topology
 - Slower (higher parasitic capacitances and resistances)
 - Denser (lower costs)
 - Access transistors at the top and bottom of a series
 - BitLine Select Transistor (BST) and Source Select Transistor (SST)
 - All FLOTOX transistors are not programmed after fabrication
 - The initial content of the memory is all 1s



Flash Memory

- NAND Flash Memory
 - NAND topology
 - All gate voltages are complemented compared to NOR topology
 - Voltage values used in NAND Flash memories
 - V_{th} is the normal threshold voltage of a not programmed transistor
 - V_{thprog} is the threshold voltage of a programmed transistor
 - V_{pass} is a gate voltage such that all transistors are conducting, regardless of their programming state
 - $V_{pass} > V_{thprog}$
 - V_{read} is a gate voltage such that only non programmed transistors are conducting
 - $V_{th} < V_{read} < V_{thprog}$
 - V_{OFF} is a gate voltage such that all transistors are not conducting
 - $V_{OFF} < V_{th}$



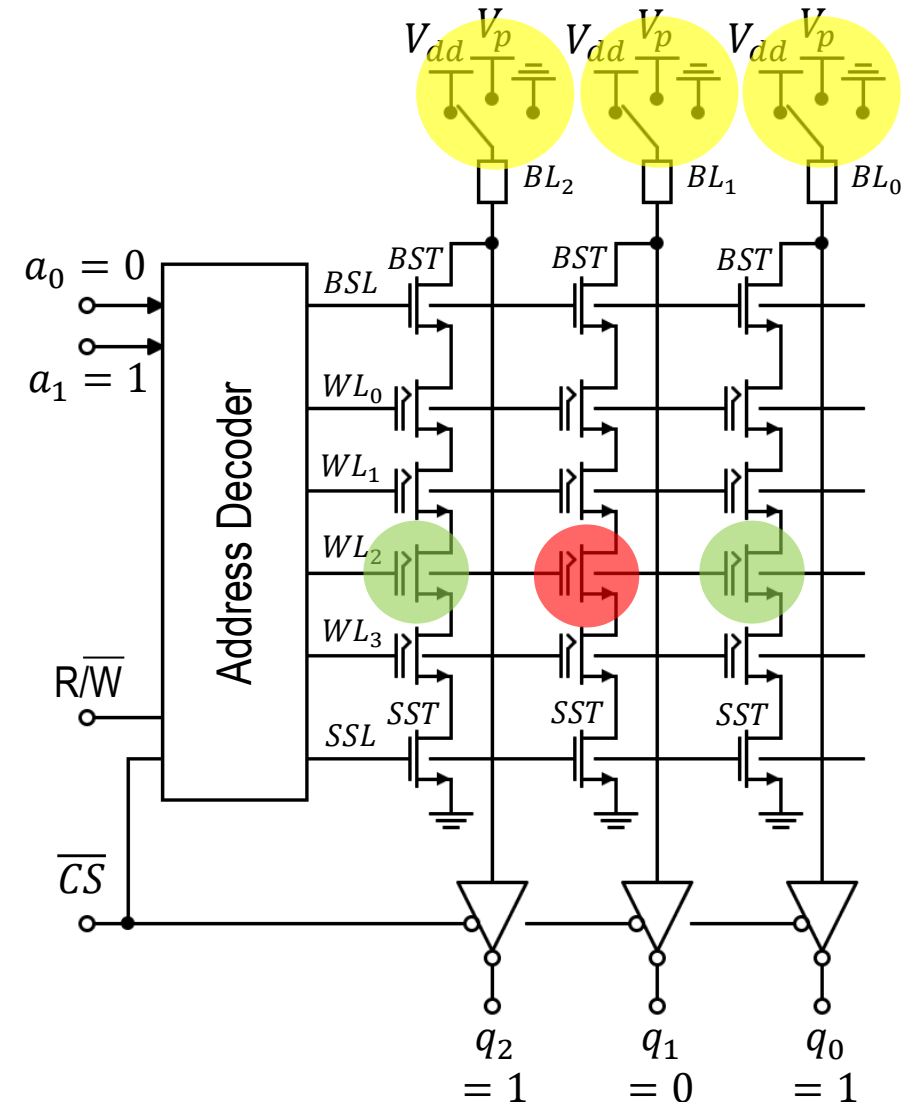
Distribution of threshold voltage for programmed and not programmed transistors

Flash Memory

- NAND Flash Memory

- How to read a transistor

- All Bit Lines are connected to V_{dd}
 - Select the word to be read by providing its address
 - Example: $A = 10_2 = 2_{10}$
 - The selected Word Line is set to V_{read}
 - All other Word Lines are set to V_{pass}
 - All transistors, except in the word to be read, are conducting
 - BST and SST access transistors are activated
 - If the transistor in the word is not programmed, it is conducting
 - The Bit Line is grounded, the output is logic 1
 - If the transistor in the word is programmed, it is not conducting
 - The series is interrupted, the Bit Line remains high, the output is logic 0

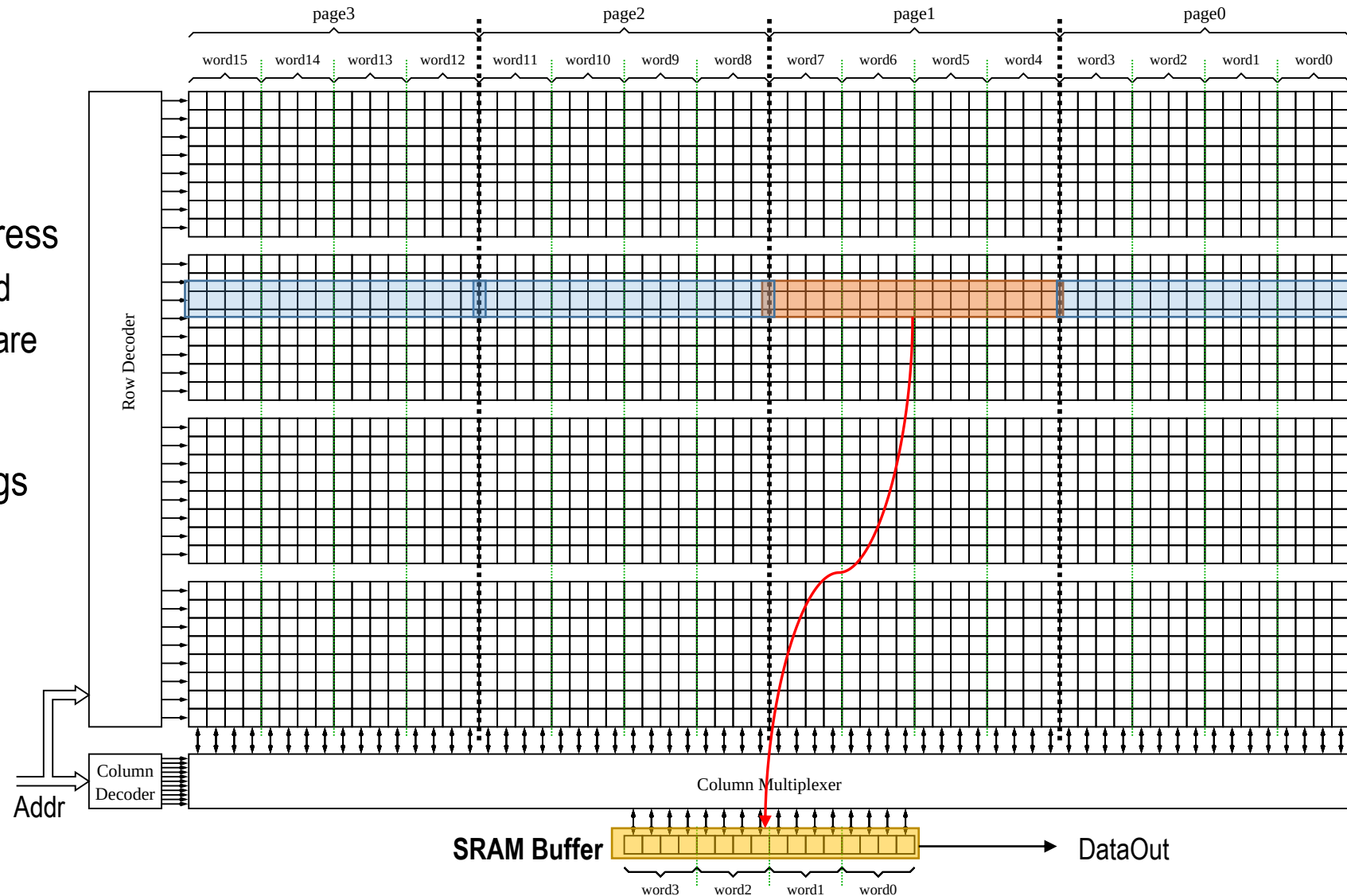


Flash Memory

- NAND Flash Memory
 - Reading is slow
 - The NAND topology has much higher delays than the NOR topology
 - The memory is divided in blocks
 - Each block is tens or hundreds of Kbytes
 - Each block is divided in pages
 - Each page is tens or hundreds of bytes
 - A page is a row, or a fraction of a row, of the bit cell array
 - When some data is read, an entire page is moved to a fast memory (usually an SRAM)
 - Most NAND Flash Memories are read page by page
 - Words are read sequentially within a page
 - No random access to individual words
 - Otherwise performance would be very low
 - A NAND Flash Memory is not a drop-in replacement of a ROM
 - Cannot be used to store and execute code directly

Flash Memory

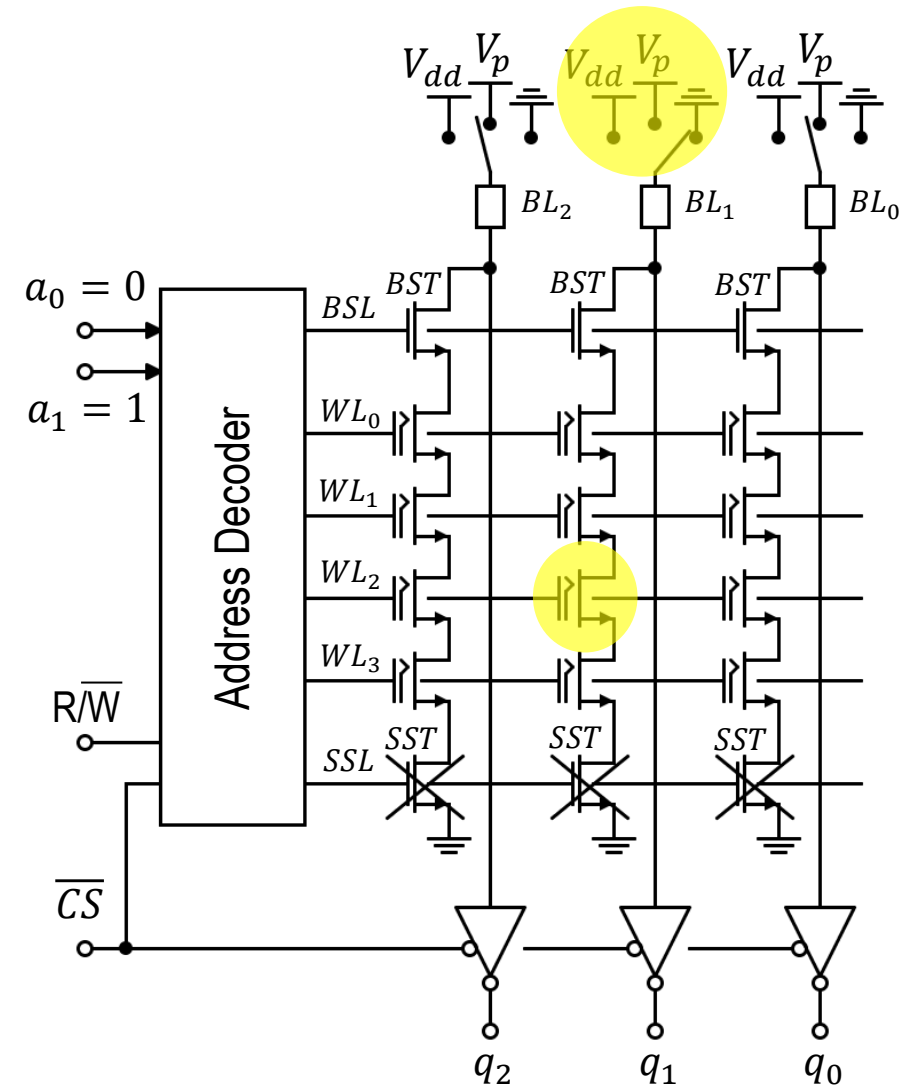
- NAND Flash Memory
 - Reading is slow
 - Select a row from the address
 - Only some bits are used
 - All bits cells of the row are sensed
 - Copy the page to which the word to be read belongs to the SRAM buffer
 - Transfer the words of the page one by one sequentially to the output



Flash Memory

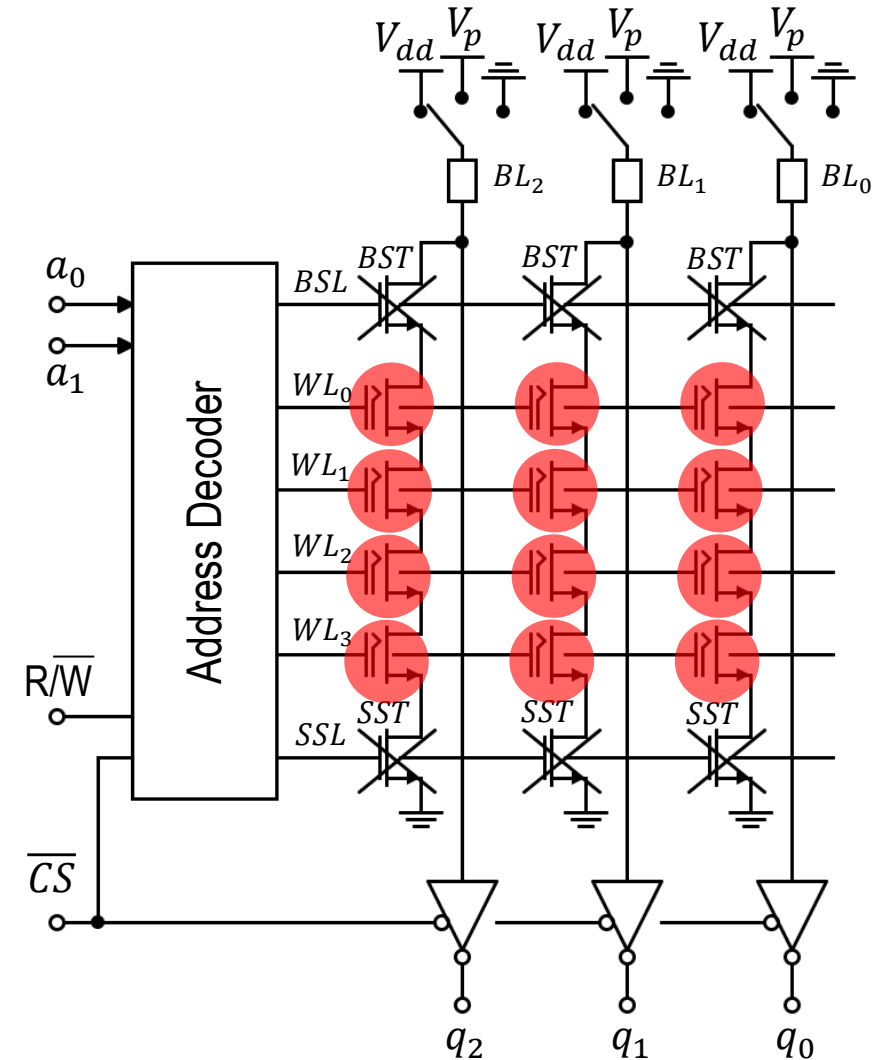
• NAND Flash Memory

- How to program a transistor
 - Select the word to be changed by providing its address
 - Example: $A = 10_2 = 2_{10}$
 - The selected Word Line is set to a high programming voltage V_p
 - All other Word Lines are set to V_{pass}
 - The desired Bit Line is set to ground
 - All other Bit Lines are set to the programming voltage V_p
 - All BitLine Select Transistors are activated
 - All Source Select Transistors are deactivated
 - Sources are floating
 - The corresponding transistor is programmed
 - Repeat the operation for all bits that need to become logic 0
- Avoid affecting other transistors
 - Transistors in other words (row) have a small V_{GD}
 - Other transistors in the same word have $V_{GD} = 0V$



Flash Memory

- NAND Flash Memory
 - How to erase transistors
 - Erasing happens at the block level
 - BST and SST access transistors in a block are deactivated
 - Drains and Sources are floating
 - The Bit Line voltages do not affect the operation
 - All Word Lines are set to ground
 - The substrate of the block is set to a programming voltage V_p
 - All transistors in the block are erased through tunneling
 - Directly to the substrate, not to Drain or Source
 - Erase block or entire memory
 - Blocks are in isolated substrate wells
 - All blocks can be erased concurrently, affecting the entire memory



Flash Memory

- NAND Flash Memory
 - Serial interface
 - Similar to SPI (see interconnections)
 - Up to 4 parallel bits (non-SPI mode)
 - Increase performance
 - A page is first transferred to an internal cache
 - The cache is an SRAM
 - Reads and writes always occur from/to the cache
 - The protocol defines a number of commands

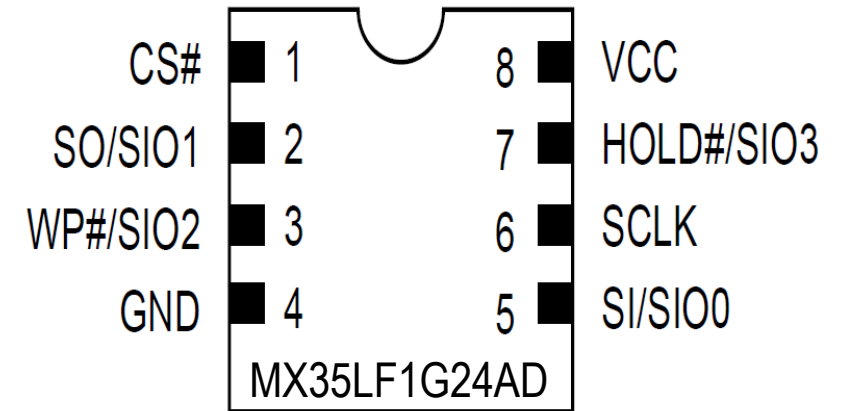
Mode	1st Byte	2nd Byte	3rd Byte	4th Byte	5th Byte	6th Byte
Get Feature	0Fh	ADD	Data			
Set Feature	1Fh	ADD	Data			
Page Read	13h	RADD2	RADD1	RADD0		
Page Read cache random	30h	RADD2	RADD1	RADD0		
Page Read cache sequential	31h					
Page Read cache last	3Fh					
Read From Cache x1	03h	CADD1	CADD0	DUMMY	DATA~	
Read From Cache x1 (alternative)	0Bh	CADD1	CADD0	DUMMY	DATA~	
Read From Cache x2 ^{Note}	3Bh	CADD1	CADD0	DUMMY	DATA~ ⁽²⁾	
Read From Cache x4 ^{Note}	6Bh	CADD1	CADD0	DUMMY	DATA~ ⁽⁴⁾	
Read From Cache Dual IO 1-2-2 ^{Note}	BBh	CADD1 ⁽²⁾	CADD0 ⁽²⁾	DUMMY ⁽²⁾	DATA~ ⁽²⁾	
Read From Cache Quad IO 1-4-4 ^{Note}	EBh	CADD1 ⁽⁴⁾	CADD0 ⁽⁴⁾	DUMMY ⁽⁴⁾	DUMMY ⁽⁴⁾	DATA~ ⁽⁴⁾
Read ID	9Fh	DUMMY	MID	DID1	DID2	
Block Erase	D8h	RADD2	RADD1	RADD0		
Program execute	10h	RADD2	RADD1	RADD0		
Program Load x1	02h	CADD1	CADD0	DATA~		
Program Load random data x1	84h	CADD1	CADD0	DATA~		
Program Load x4 ^{Note}	32h	CADD1	CADD0	DATA~ ⁽⁴⁾		
Program Load random data x4 ^{Note}	34h	CADD1	CADD0	DATA~ ⁽⁴⁾		
Write enable	06h					
Write disable	04h					
Reset	FFh					

Flash Memory

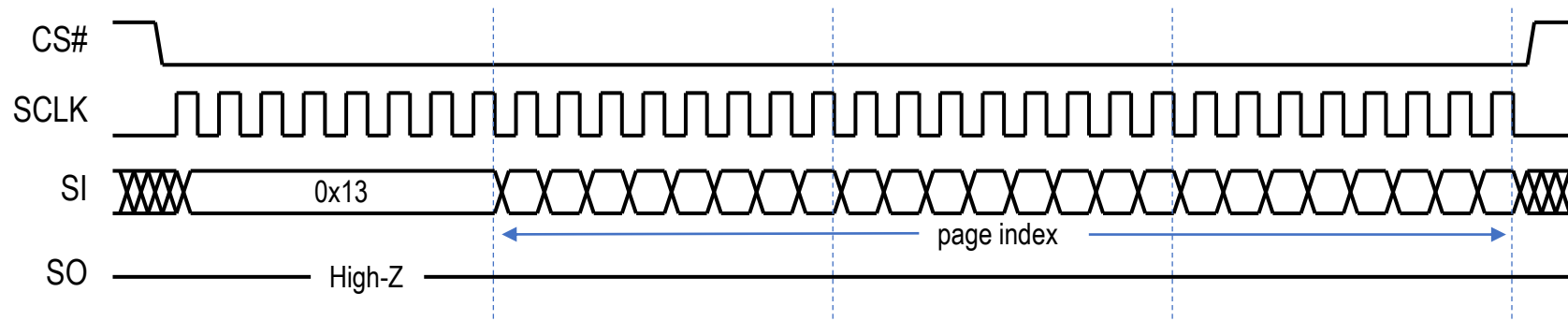
- NAND Flash Memory

- Serial interface

- Similar to SPI (see interconnections)
 - Up to 4 parallel bits (non-SPI mode)
 - A page is first transferred to a cache
 - Reads and writes always occur from/to the cache
 - The protocol defines a number of commands



- 1024 Blocks
 - 64 pages per block
 - 2048 bytes per page
- 128 MB = 1 Gb



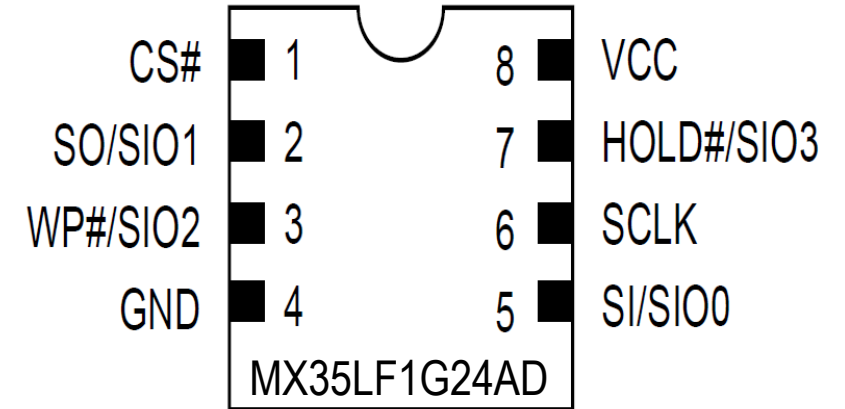
Page Read: transfer a page to cache (command 0x13 + 24bit index of the page)

Flash Memory

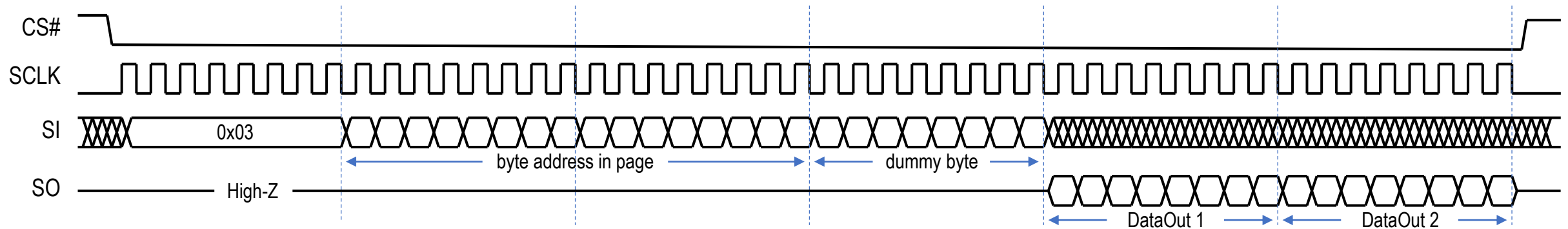
- NAND Flash Memory

- Serial interface

- Similar to SPI (see interconnections)
 - Up to 4 parallel bits (non-SPI mode)
 - A page is first transferred to a cache
 - Reads and writes always occur from/to the cache
 - The protocol defines a number of commands



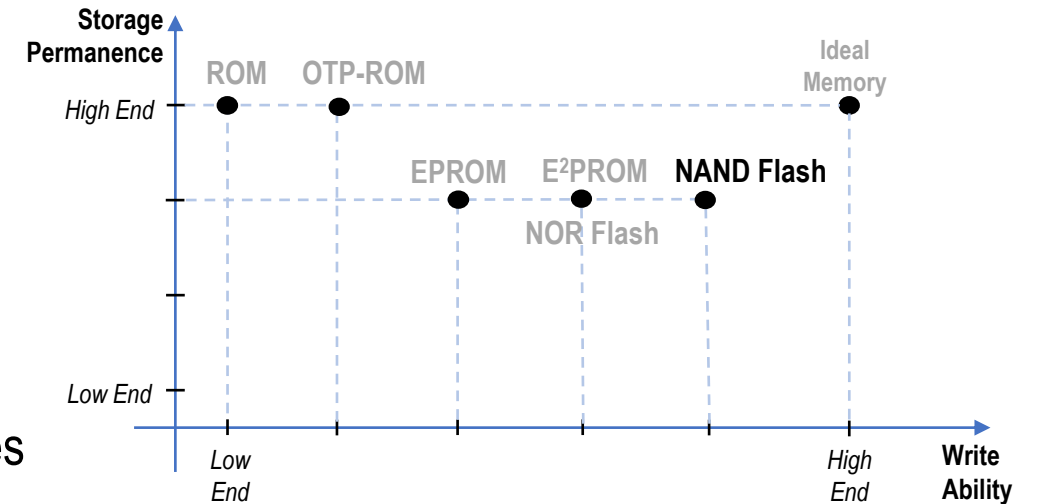
- 1024 Blocks
 - 64 pages per block
 - 2048 bytes per page
- 128 MB = 1 Gb



Read from cache: transfer bytes from the cache to the microprocessor (command 0x03 + 16bit byte address in the page + 1 dummy byte + n data bytes)

Flash Memory

- NAND Flash Memory
 - Very good write ability
 - Can be written tens of thousands of time
 - Tunneling cause less wear than hot electrons
 - Better than E²PROMs and NOR Flashes
 - Programmed and erased in circuit
 - High storage permanence
 - Tens of years, like EPROMs, E²PROMs and NOR Flashes
 - Costs
 - Low NRE costs
 - Low per unit costs, much cheaper than NOR Flashes of the same size
 - Used in SSD Drives and portable memories (USB and SD cards)



Flash Memory

- NAND Flash Memory
 - Disadvantages
 - Cannot be used to directly execute code
 - Code shall be loaded in another memory first
 - Slow access
 - Originally designed to substitute hard disks
 - Fast internal buffers to speed up operations
 - Sequential access not considered a problem
 - Access is often serial
 - It may become the performance bottleneck
 - Writing is always at the block level
 - Need to read – modify – rewrite, like in the NOR Flash Memory case
 - It's easier to write the updated block to an empty block, marking the initial one for erasing
 - Need address translation
 - Allows wearing the memory uniformly

Memory Classification

- Graph of Write Ability and Storage Permanence

- All ROM memories have a large penalty in writing

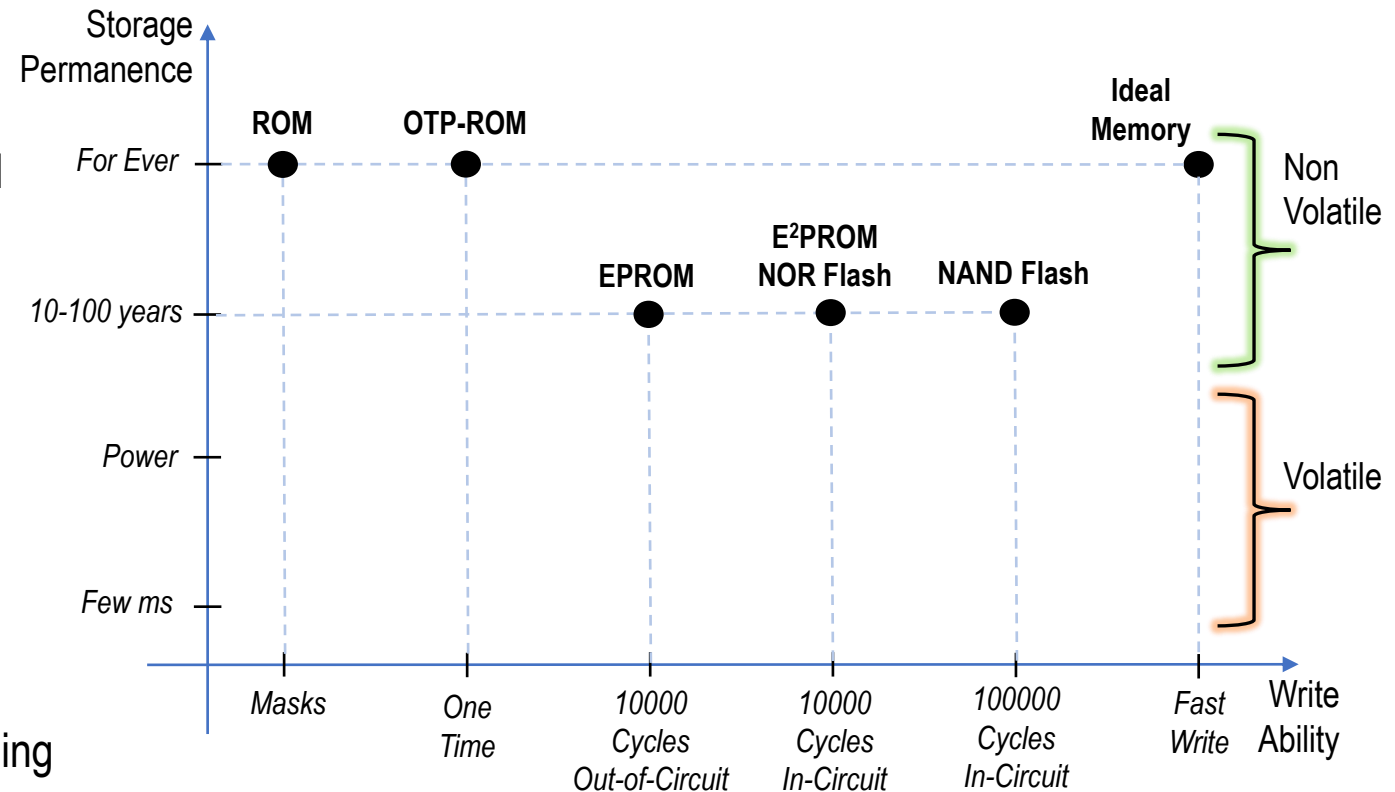
- The graph is not to scale
 - Used mostly to read data
 - Flashes are faster
 - But erasing and writing not at word level

- All ROM memories are non volatile

- In many cases it is not necessary

- To increase writing performance

- Sacrifice storage permanence
 - Write as fast as read
 - Write at word and bit levels
 - Can use standard access methods
 - No need of programming and erasing

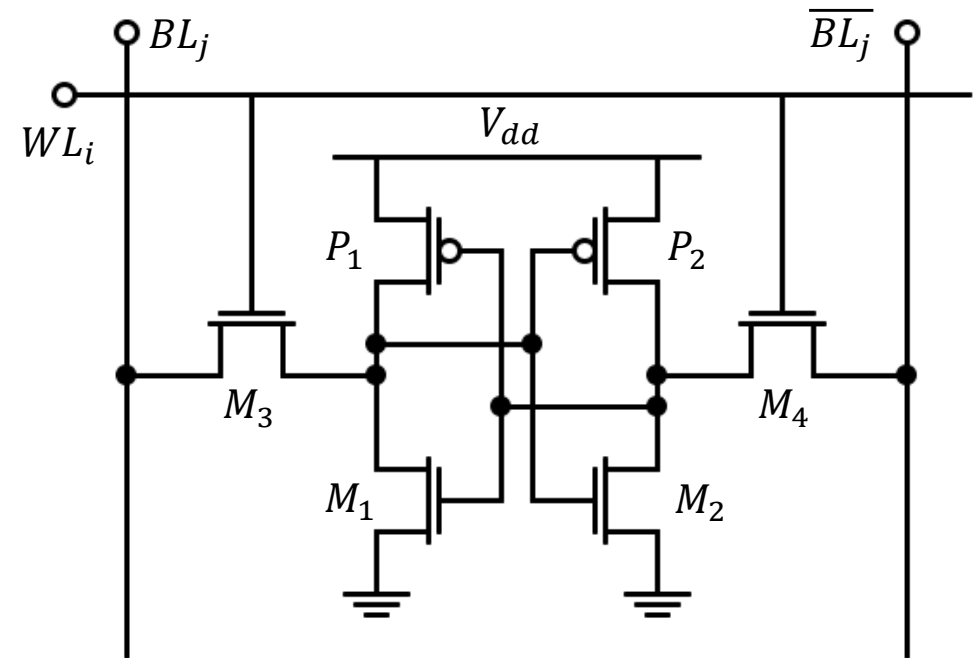
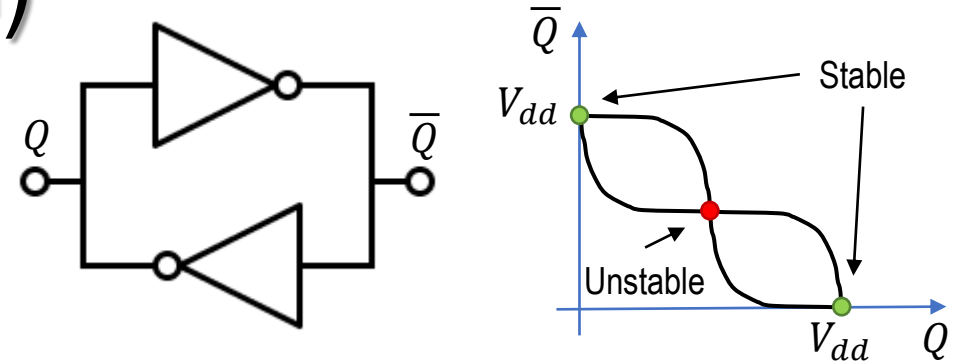


Random Access Memories (RAM)

- RAM denotes memories that are fast to write
 - ROMs are also random access
 - But RAMs are much faster than E²PROMs and Flashes
 - Several order of magnitude faster in writing
 - Example: 100 ns (10^{-7} s) versus 1 ms (10^{-3} s)
 - Single bit and single word access in reading and writing
 - No distinction between programming and erasing
 - Write is identical in the two cases ($0 \rightarrow 1$ and $1 \rightarrow 0$)
 - Two kind of RAMs
 - Static RAM (SRAM, CAM)
 - Dynamic RAM (DRAM, SDRAM, DDR)

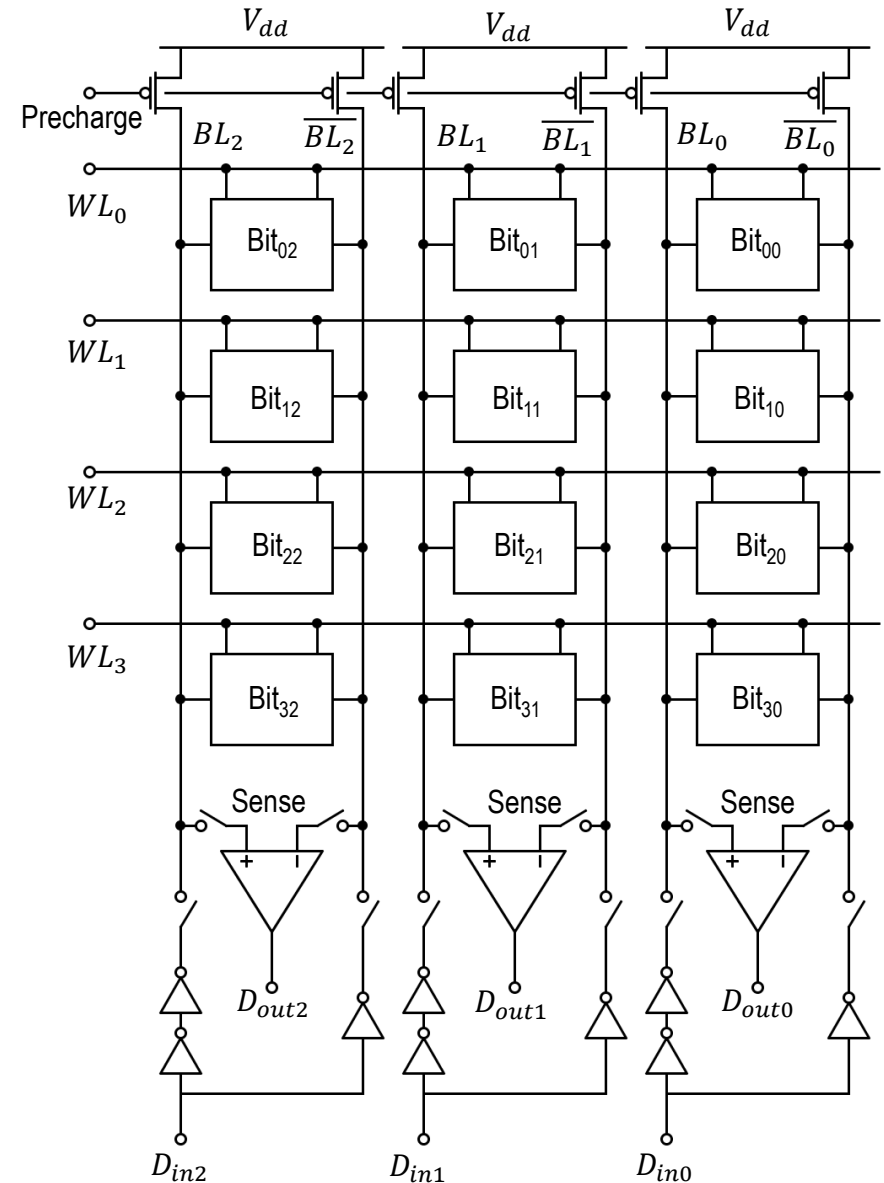
Random Access Memories (RAM)

- Static RAM (SRAM)
 - Bistable circuit (two inverters in a loop)
 - Three states, two stable states and one unstable
 - When switched on it goes into one of the two stable states
 - Need a way to force a certain stable state
 - SRAM bit cell
 - Bistable circuit (M_1, P_1, M_2, P_2)
 - Access transistors (M_3, M_4)
 - Pass transistors
 - Two Bit Lines for each bit cell
 - Affirmed and complemented
 - A single Word Line
 - Driving the gate of the access transistors



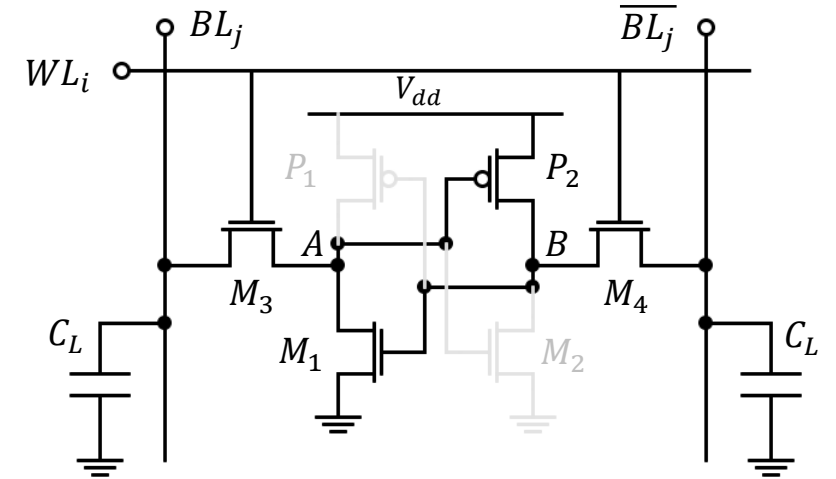
Random Access Memories (RAM)

- Static RAM (SRAM)
 - SRAM structure
 - Matrix of bit cells
 - Vertical Bit Lines and complemented Bit Lines
 - Horizontal Word Lines
 - Precharging circuit
 - To connect Bit Lines to V_{dd}
 - Sense amplifier
 - To detect small voltage differences
 - Differential amplifier
 - I/O circuits
 - Inputs and outputs, switches driven by column decoder and mux
 - When switched ON the content is random
 - When switched OFF the content is lost



Random Access Memories (RAM)

- Static RAM (SRAM)
 - Read operation
 - Don't want to change the data stored in the bit cell
 - Assume the bit cell stores the logic value 0
 - M_1 and P_2 are ON, M_2 and P_1 are OFF
 - $V_A = 0V$, $V_B = V_{dd}$
 - Word Lines are initially inactive
 - Precharge all Bit Lines BL and $\overline{BL}$ of a word to V_{dd}
 - $V_{BL_j} = V_{\overline{BL_j}} = V_{dd}$
 - Voltage is held by the parasitic capacitance C_L
 - Activate the Word Line
 - M_3 and M_4 start conducting

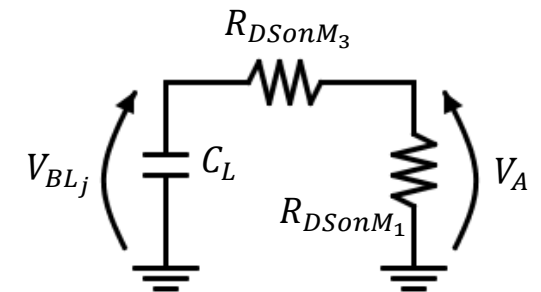
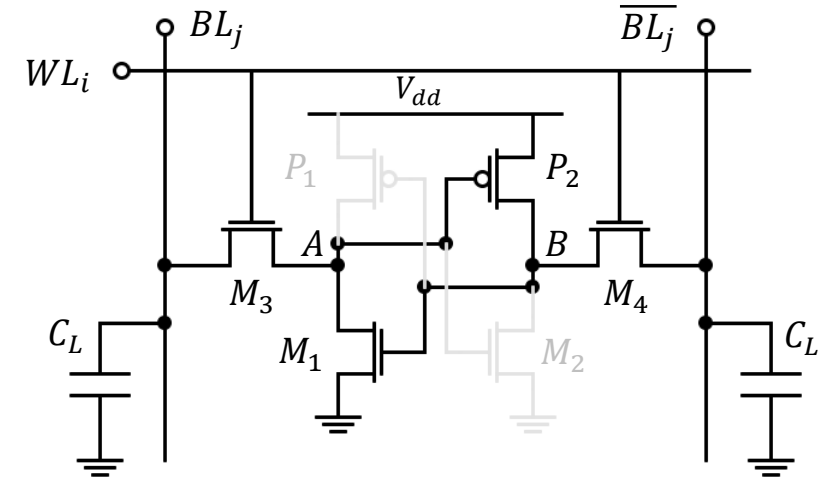


Random Access Memories (RAM)

- Static RAM (SRAM)

- Read operation (continued)

- Reading the value 0
- Since $V_B \simeq V_{\overline{BL}_j}$, nothing happens on the right side
 - No current flows in M_4
- Since $V_A \neq V_{BL_j}$, there is current flowing in M_3
 - V_{BL_j} starts decreasing
 - V_A starts increasing
 - V_A may reach the threshold voltage of M_2
 - It may cause the flip of the bistable circuit
- To avoid changing the bit cell content, M_1 should be larger than M_3
- The sense amplifier detects that BL_j is discharging, while $\overline{BL}_j$ is not
 - ΔV is negative
 - It produces the value 0 as output



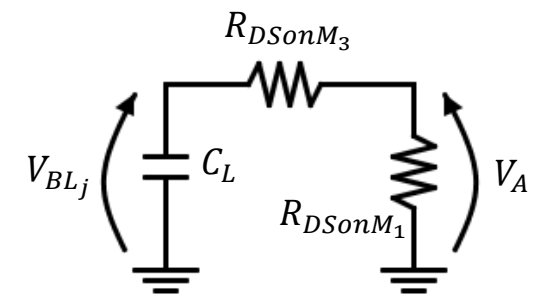
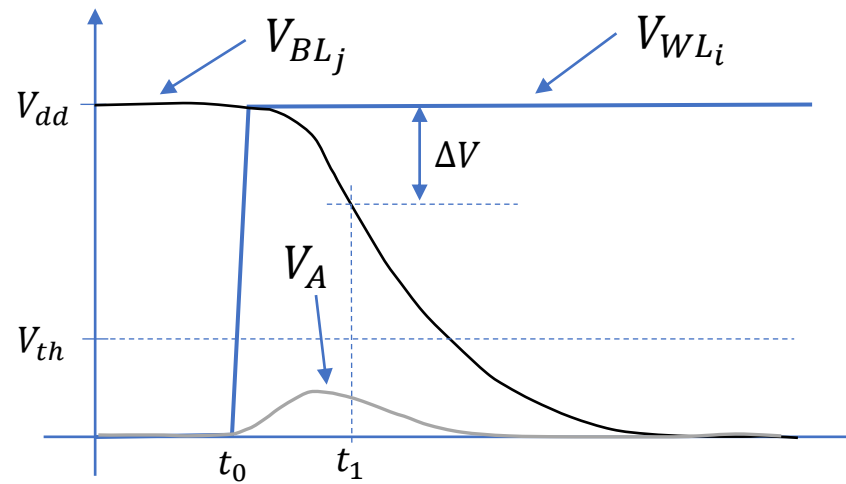
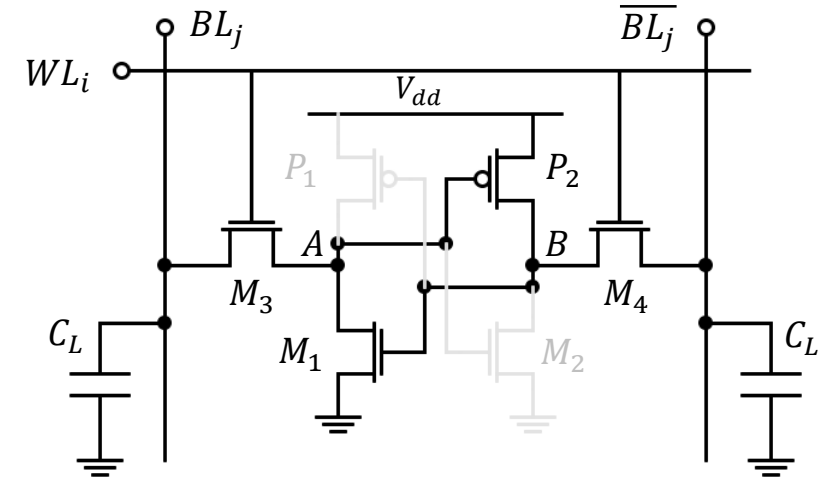
$$R_{DSon} = \frac{1}{\mu C_{ox} \frac{W}{L} (v_{GS} - v_{th})}$$

Random Access Memories (RAM)

- Static RAM (SRAM)

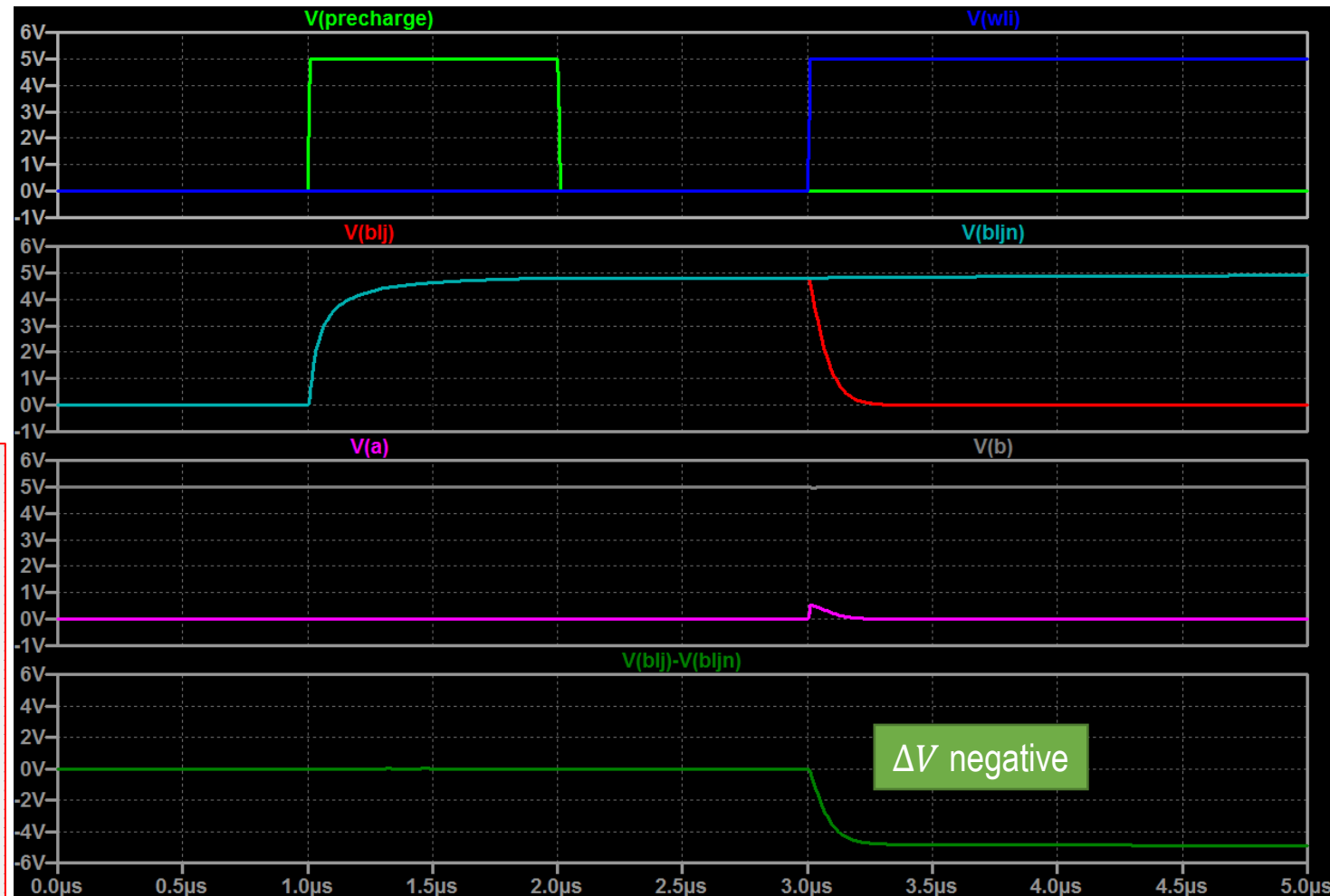
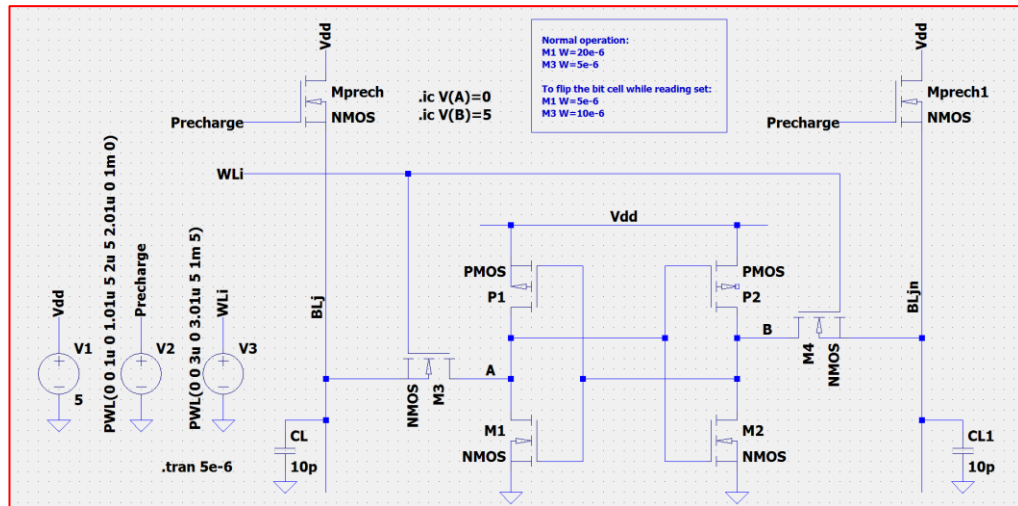
- Read operation (continued)

- Reading the value 0
- Waveforms of voltages when activating the Word Line at time t_0
 - V_B and $V_{\overline{BL_j}}$ remain at V_{dd} (not shown in the graph)
 - V_A should not exceed transistor M_2 threshold
 - Make R_{DSonM_1} suitably smaller than R_{DSonM_3}
 - M_1 larger than M_3
 - When ΔV is sufficiently large the sense amplifier detects the bit cell content (time t_1)
 - $\Delta V = V_{BL_j} - V_{\overline{BL_j}} =$
 $= V_{BL_j} - V_{dd} < 0$



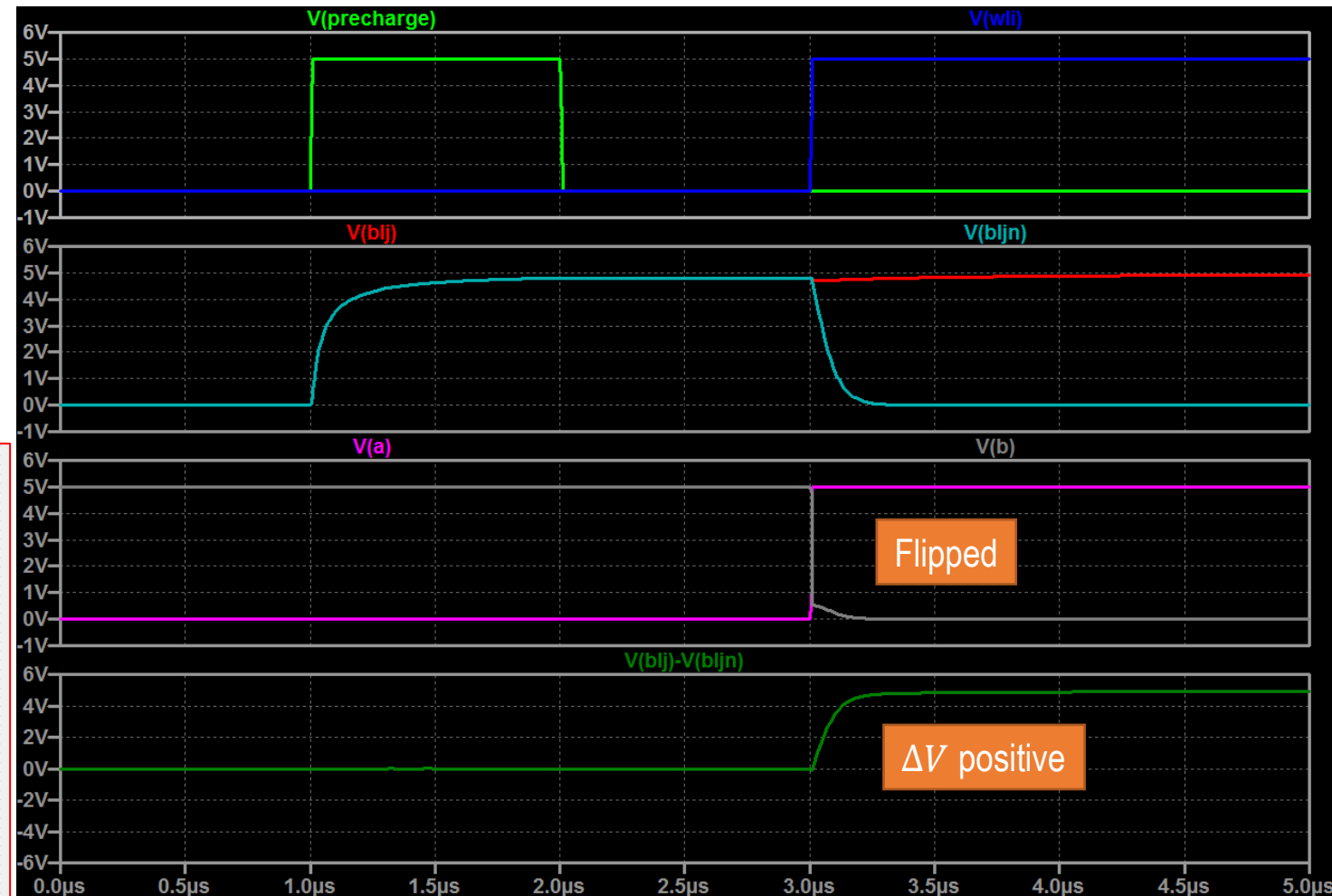
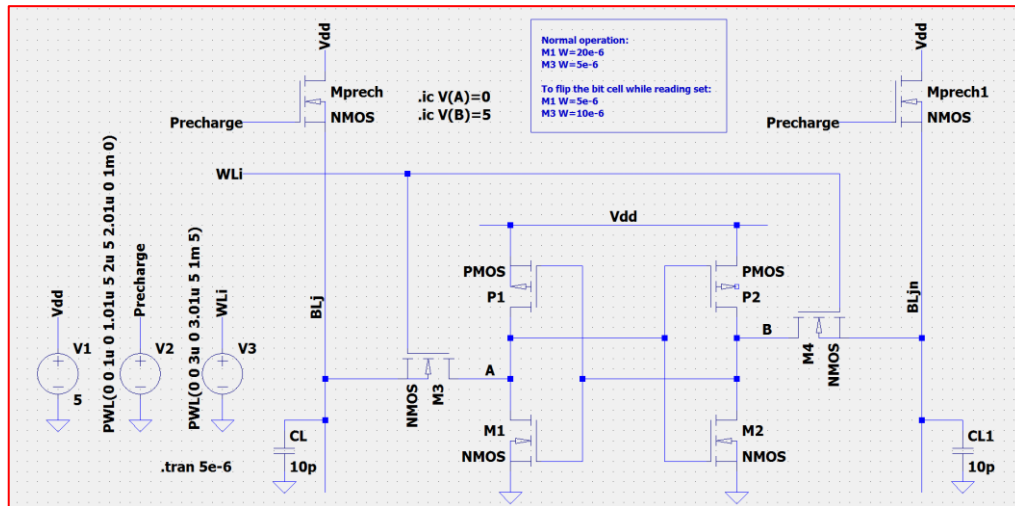
Random Access Memories (RAM)

- Static RAM (SRAM)
 - Read operation (continued)
 - Reading the value 0
 - LTSpice simulation
 - M1: $W = 20 \mu m$, $L = 2 \mu m$
 - M3: $W = 5 \mu m$, $L = 2 \mu m$
 - Correct



Random Access Memories (RAM)

- Static RAM (SRAM)
 - Read operation (continued)
 - Reading the value 0
 - LTSpice simulation
 - M1: $W = 5 \mu m$, $L = 2 \mu m$
 - M3: $W = 10 \mu m$, $L = 2 \mu m$
 - Failed and flipped the content

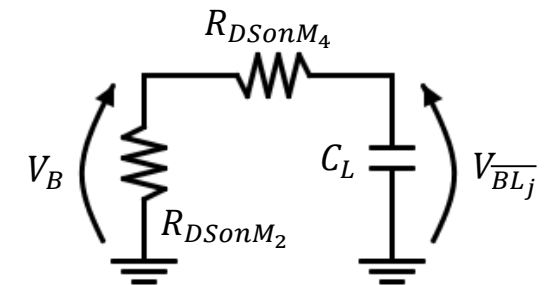
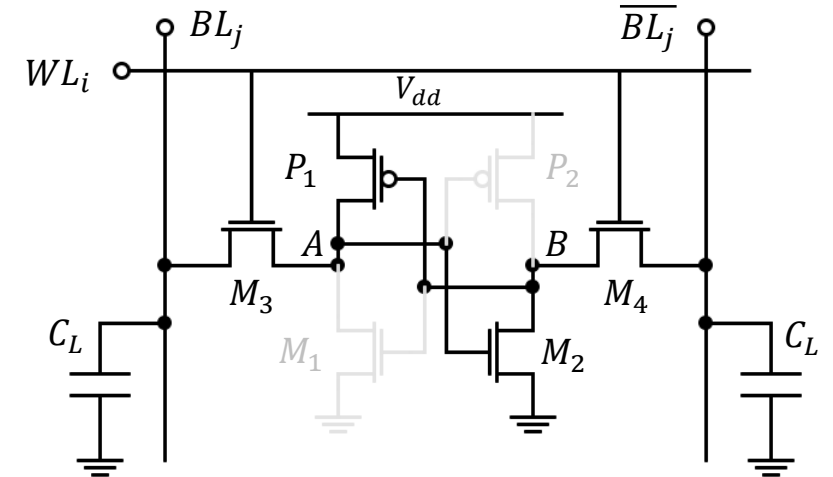


Random Access Memories (RAM)

- Static RAM (SRAM)

- Read operation (continued)

- Reading a value 1 is complementary
 - Since $V_A \approx V_{BL_j}$, nothing happens on the left side
 - No current flows in M_3
 - Since $V_B \neq V_{\overline{BL_j}}$, there is current flowing in M_4
 - $V_{\overline{BL_j}}$ starts decreasing
 - V_B starts increasing
 - V_B may reach the threshold voltage of M_1
 - It may cause the flip of the bistable circuit
 - To avoid changing the bit cell content, M_2 should be larger than M_4
 - The sense amplifier detects that $\overline{BL_j}$ is discharging, while BL_j is not
 - ΔV is positive: $\Delta V = V_{BL_j} - V_{\overline{BL_j}} = V_{dd} - V_{\overline{BL_j}} > 0$
 - It produces the value 1 as output



Random Access Memories (RAM)

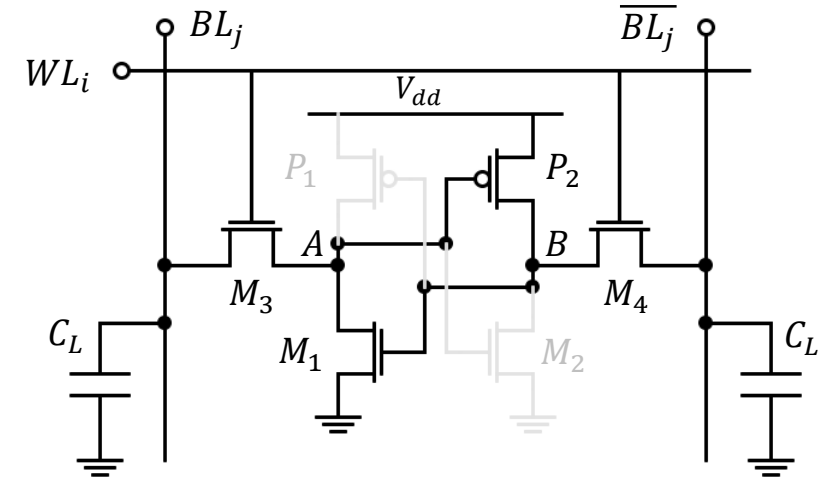
- Static RAM (SRAM)
 - Read operation (continued)
 - The low side nMOS transistors in the bistable circuit should be larger than the corresponding nMOS access transistors
 - $M_1 \equiv M_2 > M_3 \equiv M_4$
 - The ON resistance is lower for the low side nMOS transistors
 - The sense amplifier detects very small changes of the input voltages
 - ΔV in the order of a few hundreds mV
 - It often uses a positive feedback network
 - Performance
 - Reading is very fast
 - A few nanoseconds to tens of nanoseconds

Random Access Memories (RAM)

- Static RAM (SRAM)

- Write operation (continued)

- To force a flip, either:
 - Increase V_A over the threshold of M_2
 - Decrease V_B below the threshold of M_1
 - Voltage V_A cannot be increased
 - Because of transistor sizing for the read operation
 - Decrease V_B so that transistor M_1 goes OFF
 - Make transistor M_4 larger than transistor P_2
 - The entire bistable circuit flips
 - Writing logic 0 when the cell contains logic 1 requires transistor M_3 larger than transistor P_1
 - If the value to be written equals the bit cell content
 - $V_A \simeq V_{BL_j}, V_B \simeq V_{\overline{BL_j}}$
 - Nothing happens on both sides of the bit cell



Random Access Memories (RAM)

- Static RAM (SRAM)
 - Write operation (continued)
 - The nMOS access transistors should be larger than the high side pMOS transistors in the bistable circuit
 - $M_3 \equiv M_4 > P_1 \equiv P_2$
 - Sometimes pMOS implemented using resistances
 - Or pMOS that are always conducting, designed to have a high R_{DSon}
 - The exact ratio of all transistors depends on technological parameters
 - Transistors are not minimum size
 - Performance
 - Writing is very fast
 - A few nanoseconds to tens of nanoseconds
 - Comparable to read
 - No limits in the number of write operations
 - No distinction between writing 0 or 1

Memories

- # Memories



Memories

Random Access Memories (RAM)

- Static RAM (SRAM)

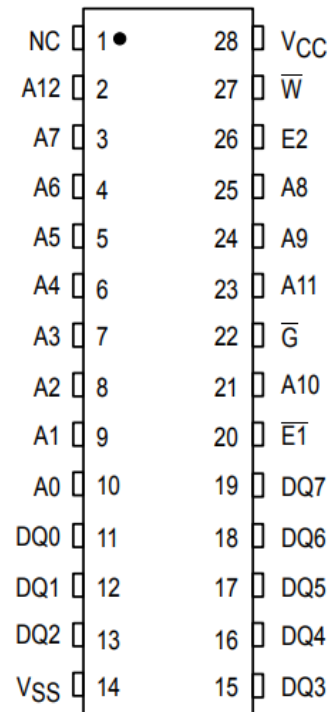
- Timing Diagrams

- MCM6264-12 (8K x 8)

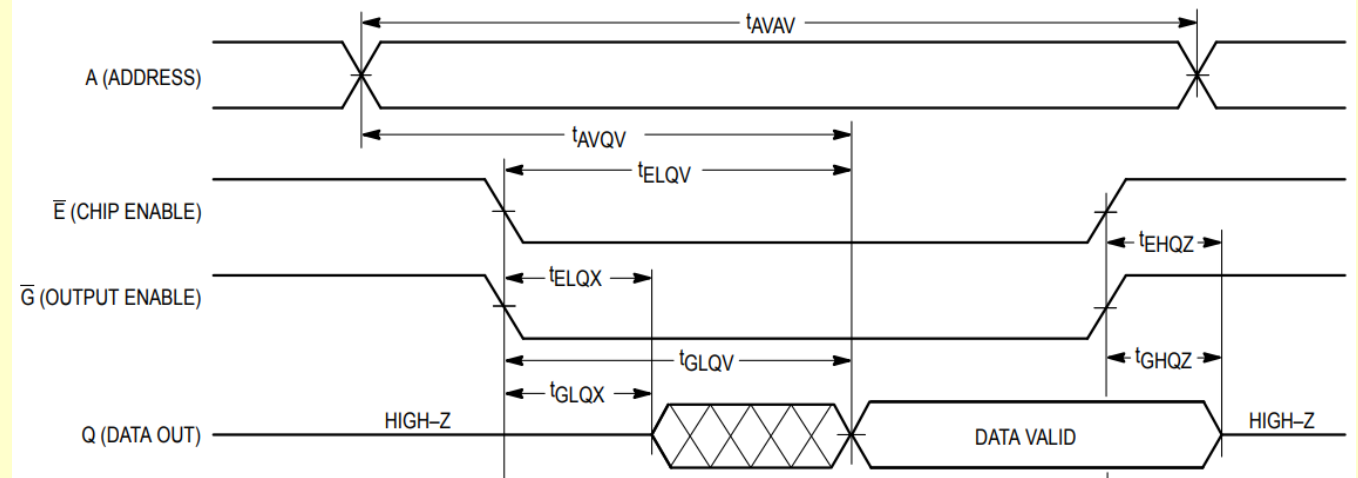
- $t_{AVAV} = 12 \text{ ns}$

- Asynchronous Memory

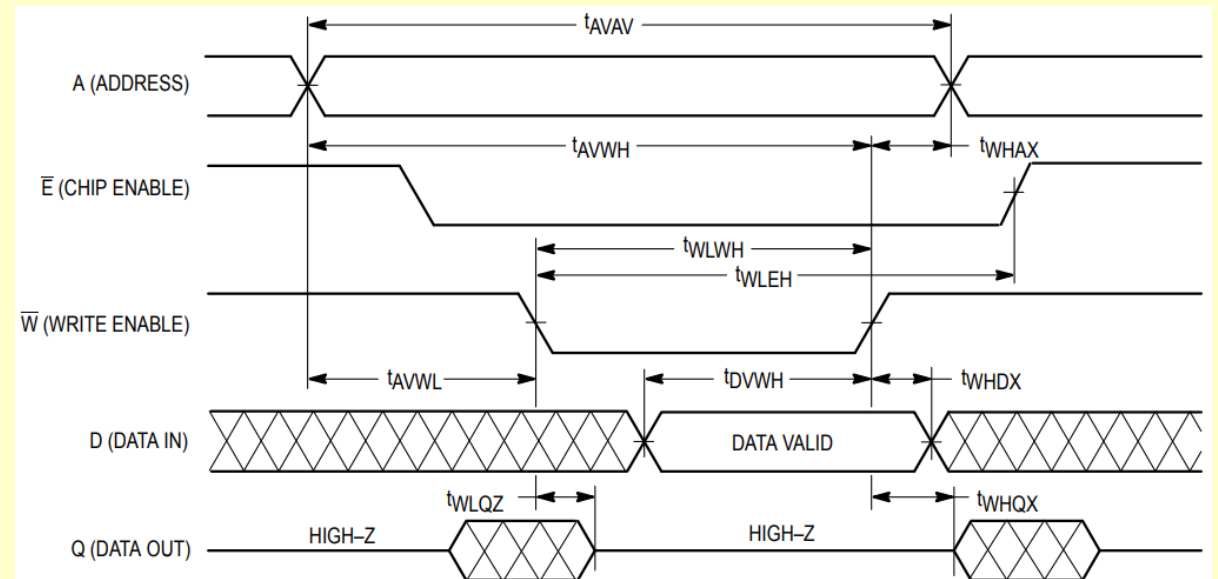
- Convert timing in clock cycles when used with synchronous circuits



Read Cycle



Write Cycle



Random Access Memories (RAM)

- Static RAM (SRAM)
 - Timing Diagrams
 - MCM6264-12 (8K x 8)
 - $t_{AVAV} = 12 \text{ ns}$
 - There is a family with different performance
 - $t_{AVAV} = 15 \text{ ns}$
 - $t_{AVAV} = 20 \text{ ns}$
 - $t_{AVAV} = 25 \text{ ns}$
 - $t_{AVAV} = 35 \text{ ns}$
 - They have different costs

READ CYCLE (See Notes 1 and 2)

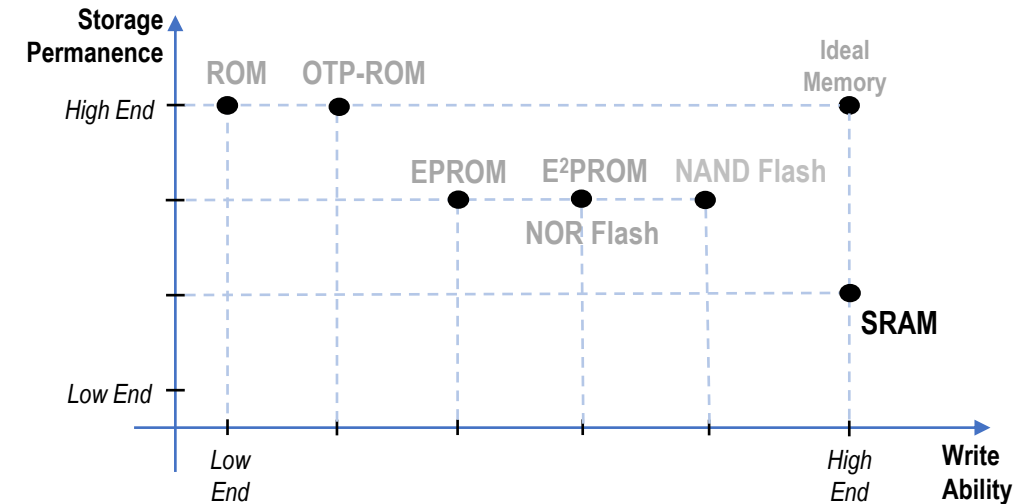
Parameter	Symbol	- 12		- 15		- 20		- 25		- 35		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Read Cycle Time	t_{AVAV}	12	—	15	—	20	—	25	—	35	—	ns	3
Address Access Time	t_{AVQV}	—	12	—	15	—	20	—	25	—	35	ns	
Enable Access Time	t_{ELQV}	—	12	—	15	—	20	—	25	—	35	ns	4
Output Enable Access Time	t_{GLQV}	—	6	—	8	—	10	—	11	—	12	ns	
Output Hold from Address Change	t_{AXQX}	4	—	4	—	4	—	4	—	4	—	ns	
Enable Low to Output Active	t_{ELQX}	4	—	4	—	4	—	4	—	4	—	ns	5, 6, 7
Enable High to Output High-Z	t_{EHQZ}	0	6	0	8	0	9	0	10	0	11	ns	5, 6, 7
Output Enable Low to Output Active	t_{GLQX}	0	—	0	—	0	—	0	—	0	—	ns	5, 6, 7
Output Enable High to Output High-Z	t_{GHQZ}	0	6	0	7	0	8	0	9	0	10	ns	5, 6, 7
Power Up Time	t_{ELICCH}	0	—	0	—	0	—	0	—	0	—	ns	
Power Down Time	t_{EHICCL}	—	12	—	15	—	20	—	25	—	35	ns	

WRITE CYCLE 1 ($\overline{W}$ Controlled, See Notes 1, 2, and 3)

Parameter	Symbol	- 12		- 15		- 20		- 25		- 35		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
Write Cycle Time	t_{AVAV}	12	—	15	—	20	—	25	—	35	—	ns	4
Address Setup Time	t_{AVWL}	0	—	0	—	0	—	0	—	0	—	ns	
Address Valid to End of Write	t_{AVWH}	10	—	12	—	15	—	17	—	20	—	ns	
Write Pulse Width	t_{WLWH} t_{WLEH}	10	—	12	—	15	—	17	—	20	—	ns	
Write Pulse Width, $\overline{G}$ High	t_{WLWH} t_{WLEH}	8	—	10	—	12	—	15	—	17	—	ns	5
Data Valid to End of Write	t_{DVWH}	6	—	7	—	8	—	10	—	12	—	ns	
Data Hold Time	t_{WHDX}	0	—	0	—	0	—	0	—	0	—	ns	
Write Low to Output High-Z	t_{WLQZ}	0	6	0	7	0	8	0	10	0	12	ns	6, 7, 8
Write High to Output Active	t_{WHQX}	4	—	4	—	4	—	4	—	4	—	ns	6, 7, 8
Write Recovery Time	t_{WHAX}	0	—	0	—	0	—	0	—	0	—	ns	

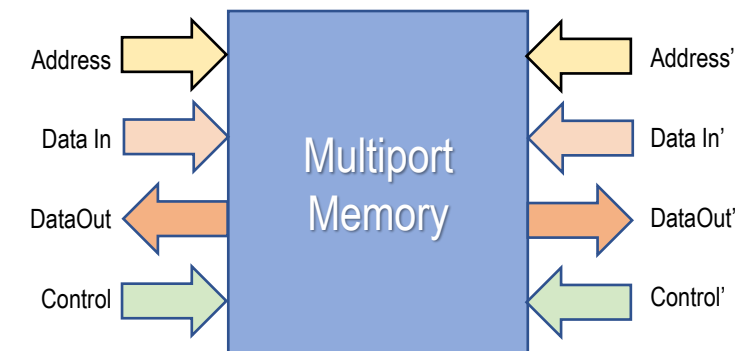
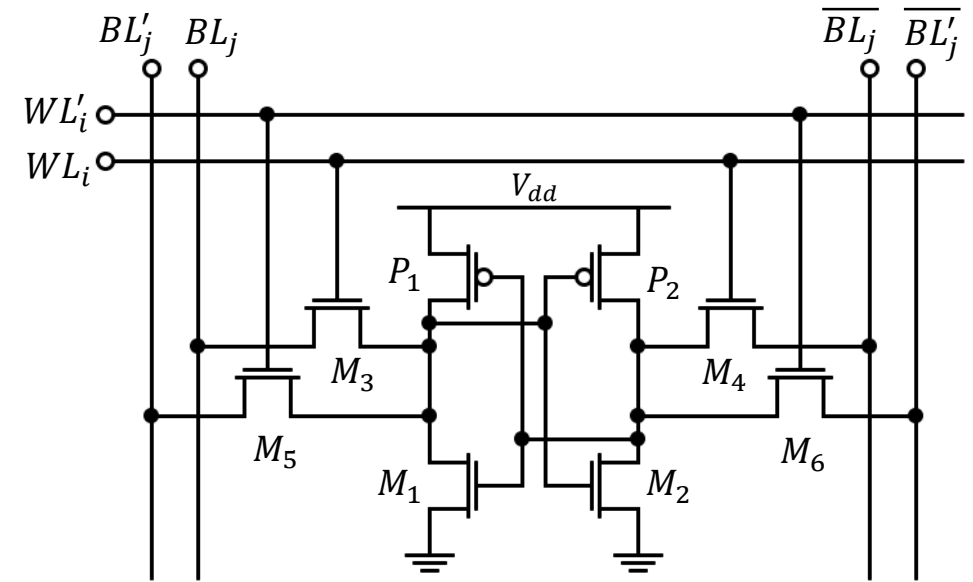
Random Access Memories (RAM)

- Static RAM (SRAM)
 - Excellent write ability
 - Very fast write
 - Word and bit level writing
 - Can write an infinite number of times
 - Low storage permanence
 - As long as power is applied
 - Costs
 - Low NRE costs
 - High per unit costs
 - Bit cell has 6 transistors (compared to 1 or 2 for programmable ROMs)
 - Transistors are not minimum size



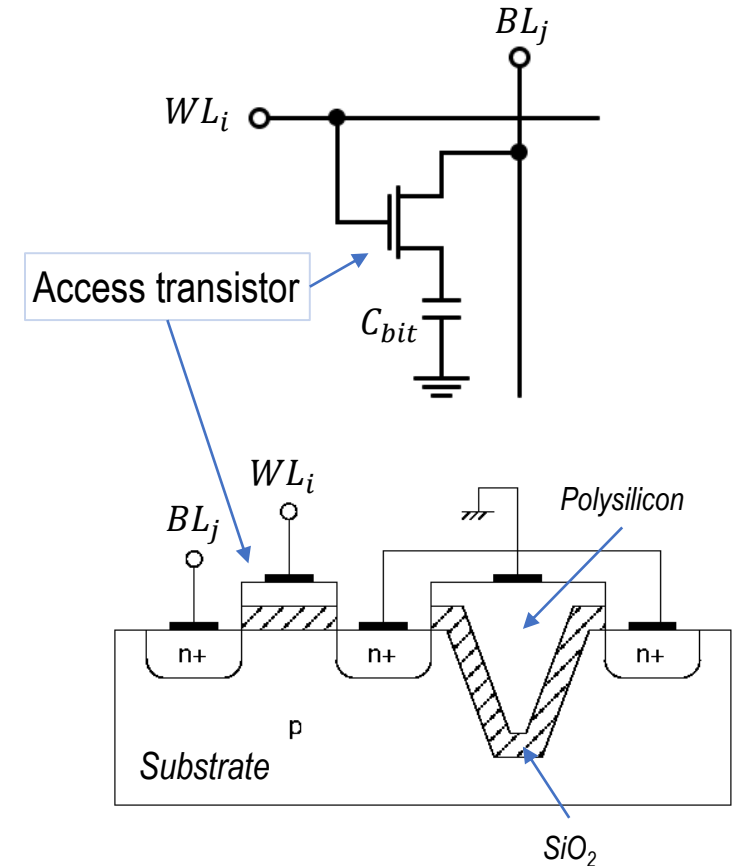
Random Access Memories (RAM)

- SRAM variations
 - Permanent memory NVRAM (non volatile)
 - A battery keeps power for several years
 - More expensive, larger physical size, heavy
 - Multiport SRAM
 - Independent ports with separate busses
 - Independent access transistor and row and column circuits
 - Can read concurrently multiple location (even the same one)
 - Can read a location while writing another location
 - If reading a location while writing the same location, result of reading is not defined
 - Can write multiple different locations concurrently
 - Writing the same location from different ports yields an undefined stored value



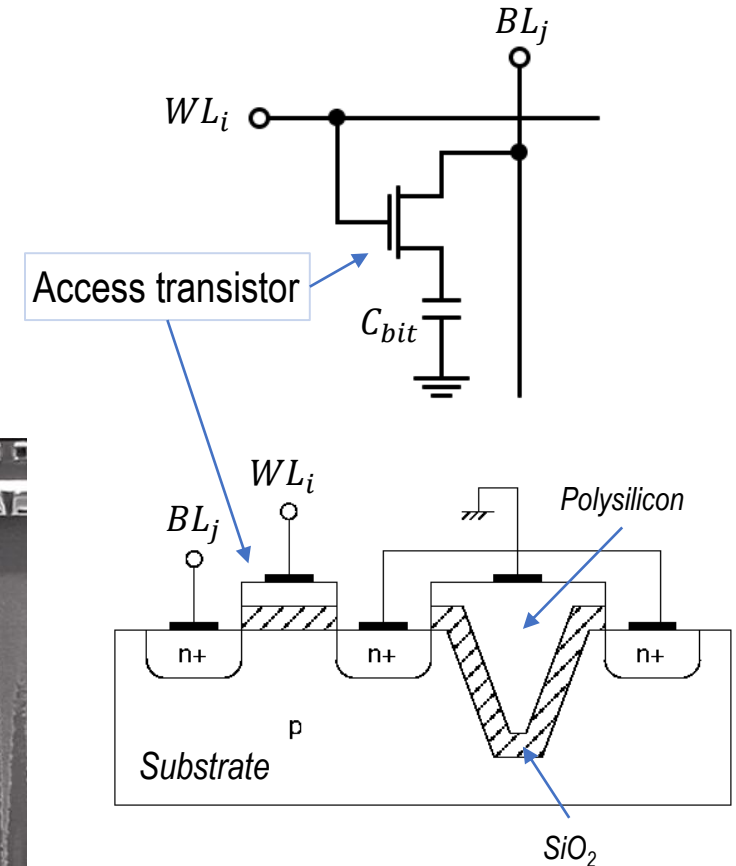
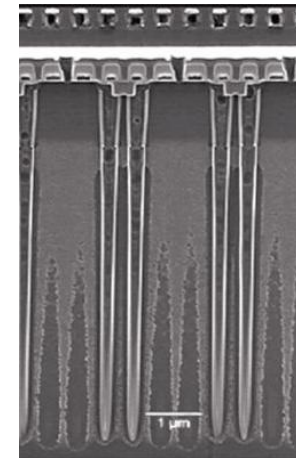
Random Access Memories (RAM)

- Dynamic RAM (DRAM)
 - Goal: design a much smaller bit cell
 - Make a fast and cheap memory
 - Store information as charge on a capacitor
 - A capacitor is an analog memory
 - Charges present $\rightarrow$ Logic 1, Charges not present $\rightarrow$ Logic 0
- DRAM bit cell
 - Storage capacitor C_{bit}
 - Parasitic capacitance of a special transistor
 - Access transistors
 - nMOS pass transistors
 - Single Bit Line
 - Single Word Line
 - Driving the gate of the access transistors



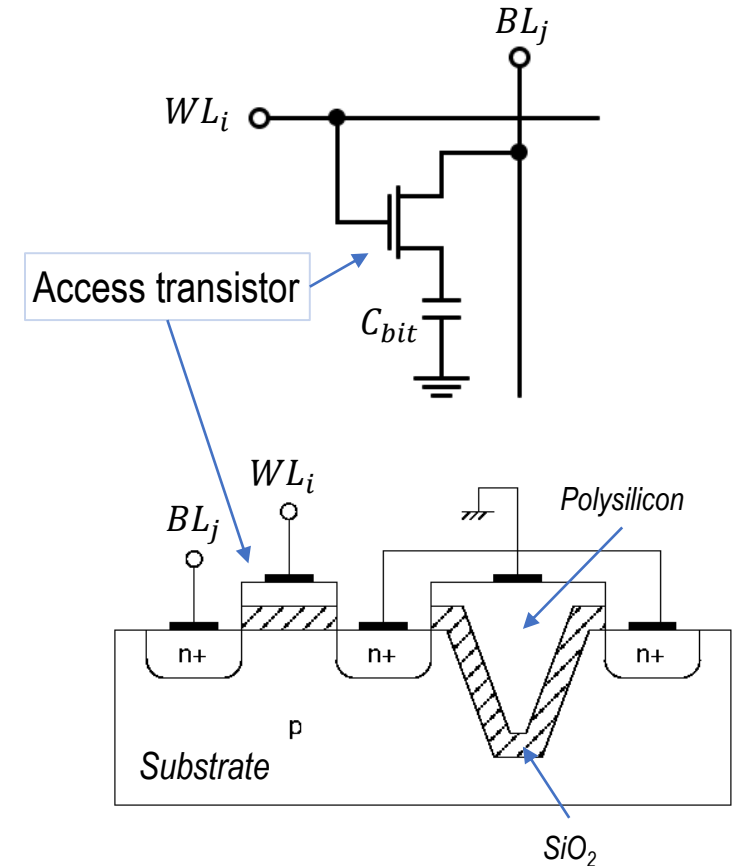
Random Access Memories (RAM)

- Dynamic RAM (DRAM)
 - Layout cross-section
 - The nMOS access transistor is a standard one
 - Trench capacitor
 - An nMOS transistor with Source and Drain connected together
 - Connected to the Source of the access transistor
 - The polysilicon Gate goes inside the substrate
 - Increases the area A of the gate
 - The gate is connected to ground
 - The equivalent capacitance is $C_{bit} = C_{ox} \cdot A$
 - To improve the analog memory
 - C_{bit} should be the highest possible
 - Technology to increase the parasitic capacitance
 - Cannot be embedded on the same integrated circuit as standard logic



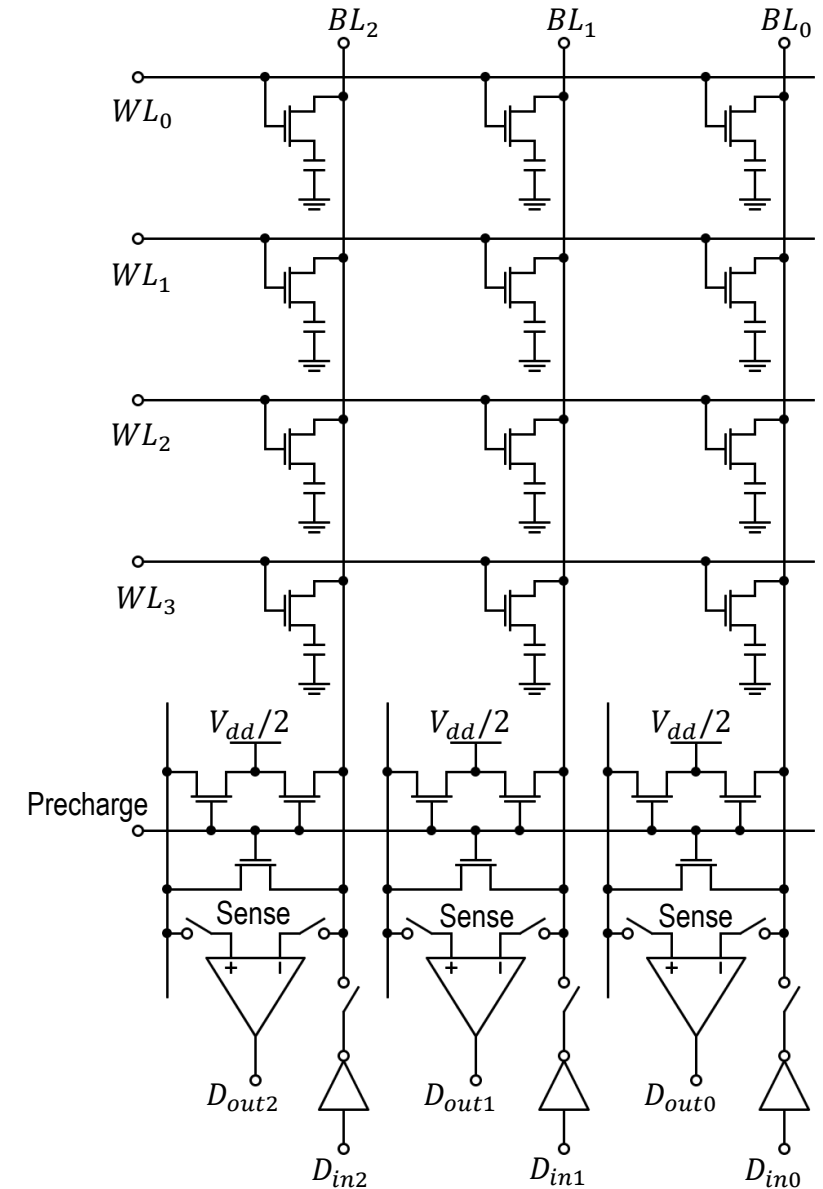
Random Access Memories (RAM)

- Dynamic RAM (DRAM)
 - Layout cross-section
 - The bit cell is small
 - Much smaller than SRAM
 - Cheaper memory
 - Stacked capacitor for smaller footprint in planar technology
 - Problems
 - Leakage currents destroy the information after some time (milliseconds)
 - Need to periodically refresh the data stored
 - Reading the charge on the capacitor requires to remove some charges
 - Need to restore data after reading



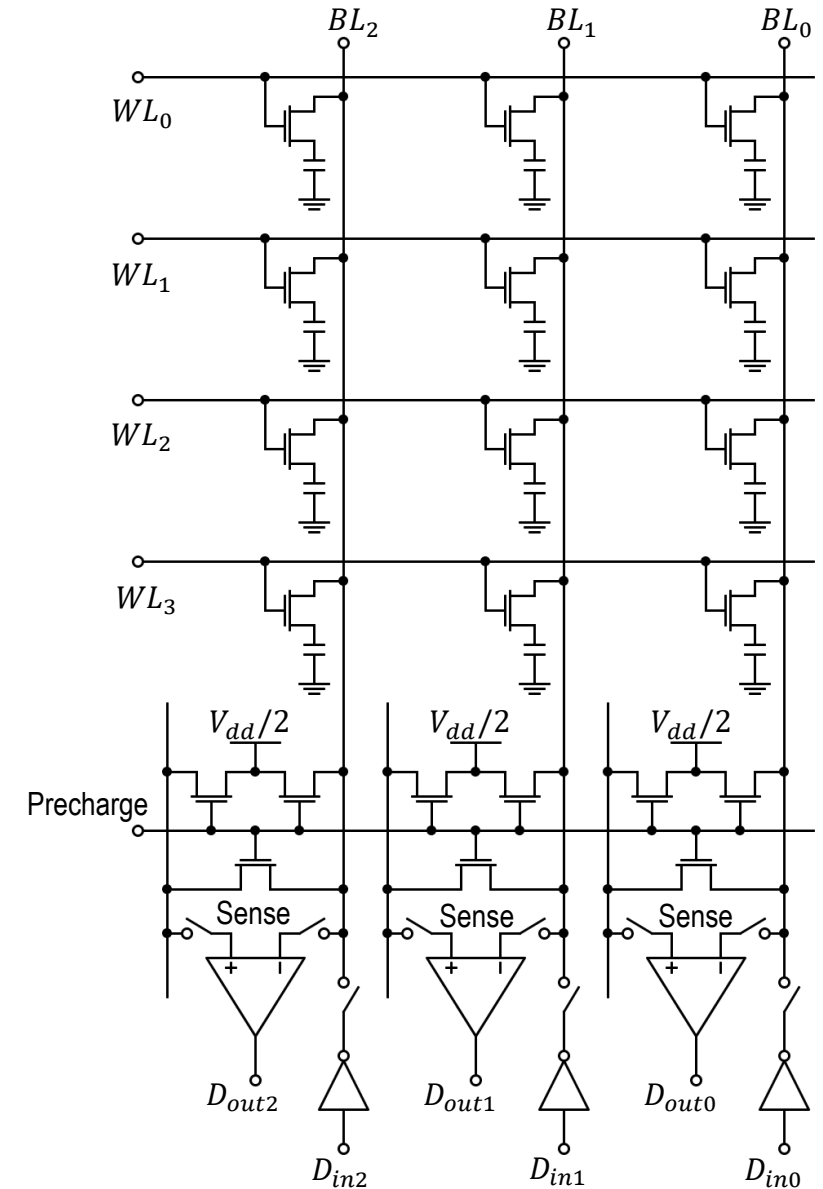
Random Access Memories (RAM)

- Dynamic RAM (DRAM)
 - DRAM structure
 - Matrix of bit cells
 - Vertical affirmed Bit Lines
 - Parasitic capacitance C_L
 - Horizontal Word Lines
 - Precharging circuit
 - To connect Bit Lines to $V_{dd}/2$
 - Sense amplifier
 - Differential amplifier to detect small voltage differences
 - Need an additional comparison line
 - I/O circuits
 - Inputs and outputs, switches driven by column decoder and mux



Random Access Memories (RAM)

- Dynamic RAM (DRAM)
 - Read operation
 - All Word Line are inactive
 - Precharge Bit Lines to $V_{dd}/2$ (or in general to a certain V_{ref})
 - Activate the selected Word Line
 - Charge sharing between C_{bit} and C_L
 - Compute voltage change on the Bit Line
 - Activate the sense amplifiers and select the desired word
 - Sense amplifiers detect the small change, positive or negative
 - The final voltage on the storage capacitor is close to V_{ref}
 - Information is lost
 - Write after reading
 - Performed internally by the memory
 - Not necessary to issue a write command
 - Necessary also in other words of the same row



Random Access Memories (RAM)

- Dynamic RAM (DRAM)
 - Read operation (continued)
 - Bit cell contains logic 1
 - Charge in the storage capacitor: $Q_{bit} = (V_{dd} - V_{th}) \cdot C_{bit} \simeq V_{dd} \cdot C_{bit}$
 - Charge in the Bit Line parasitic capacitance: $Q_L = V_{ref} \cdot C_L$
 - Total charge after activating the Word Line
 - $Q_{tot} = Q_L + Q_{bit} = V_x(C_L + C_{bit}) = V_{ref} \cdot C_L + V_{dd} \cdot C_{bit}$
 - Final voltage (on the Bit Line and on the storage capacitor)
 - $V_x = V_{ref} \frac{C_L}{C_L + C_{bit}} + V_{dd} \frac{C_{bit}}{C_L + C_{bit}}$
 - Voltage difference at the sense amplifier
 - $\Delta V = V_x - V_{ref} = (V_{dd} - V_{ref}) \frac{C_{bit}}{C_L + C_{bit}}$
 - Voltage difference at the sense amplifier with $V_{ref} = V_{dd}/2$
 - $\Delta V = \frac{V_{dd}}{2} \frac{C_{bit}}{C_L + C_{bit}}$

Random Access Memories (RAM)

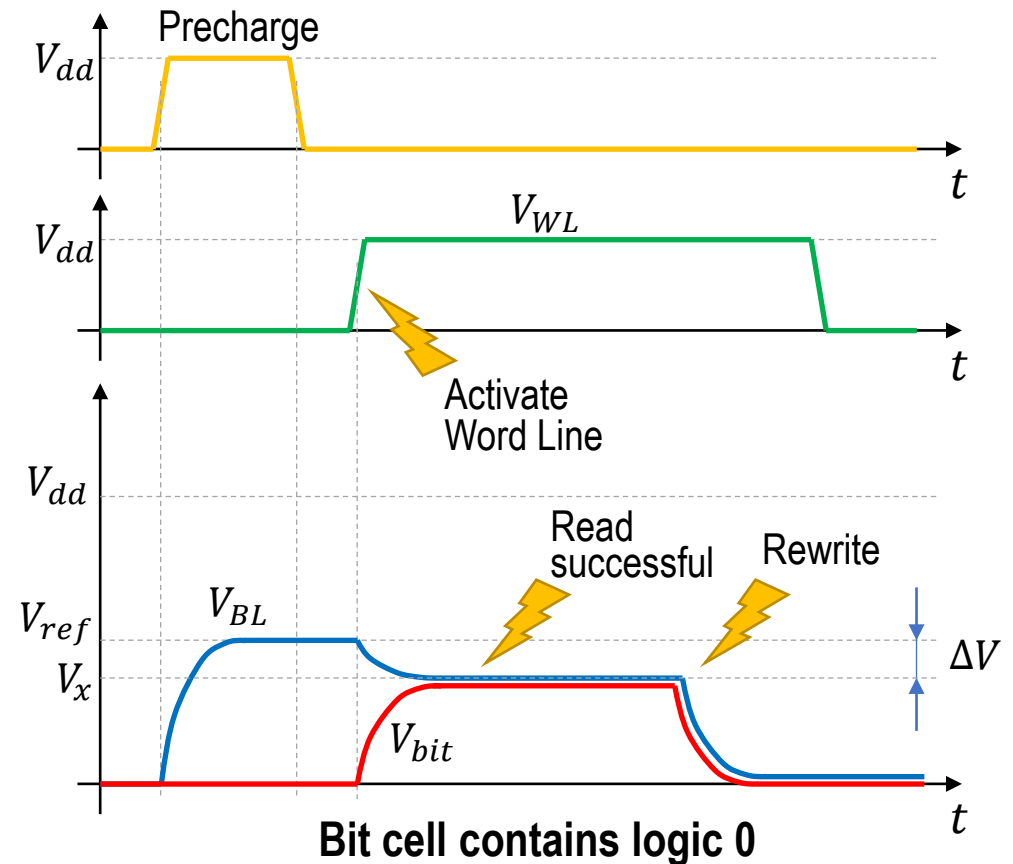
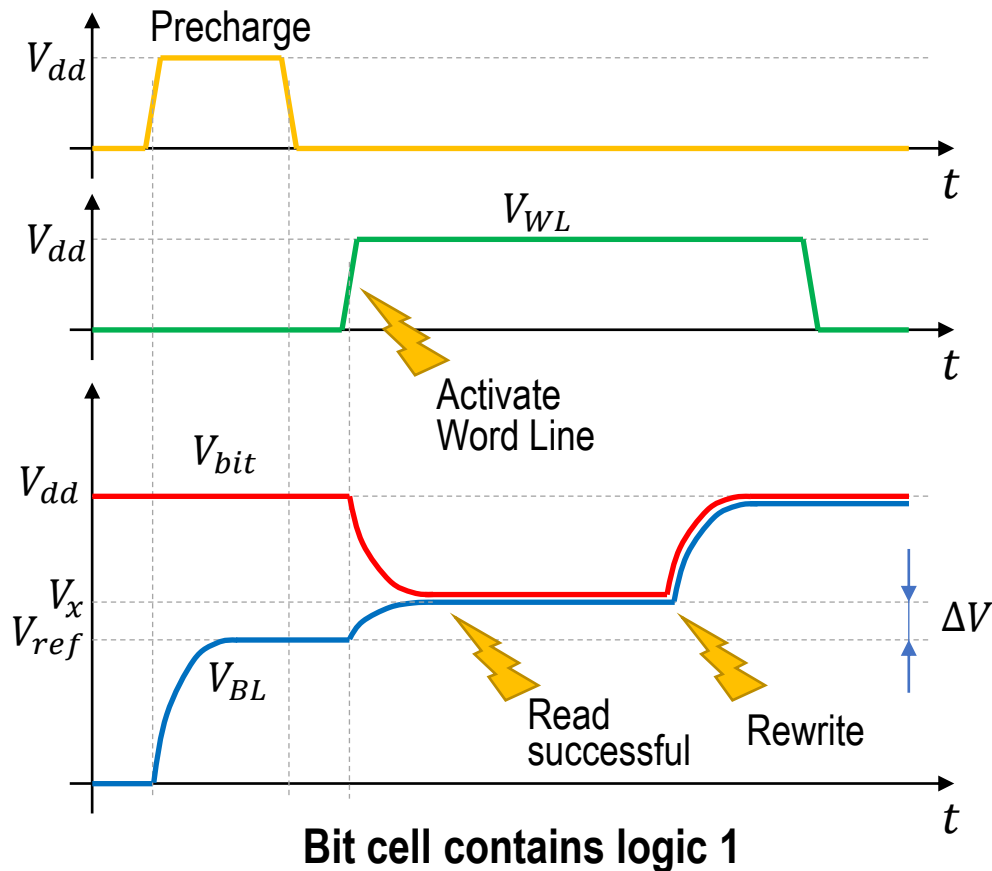
- Dynamic RAM (DRAM)
 - Read operation (continued)
 - Bit cell contains logic 0
 - Charge in the storage capacitor: $Q_{bit} = 0$
 - Charge in the Bit Line parasitic capacitance: $Q_L = V_{ref} \cdot C_L$
 - Total charge after activating the Word Line
 - $Q_{tot} = Q_L = V_x(C_L + C_{bit}) = V_{ref} \cdot C_L$
 - Final voltage (on the Bit Line and on the storage capacitor)
 - $V_x = V_{ref} \frac{C_L}{C_L + C_{bit}}$
 - Voltage difference at the sense amplifier
 - $\Delta V = V_x - V_{ref} = -V_{ref} \frac{C_{bit}}{C_L + C_{bit}}$
 - Voltage difference at the sense amplifier with $V_{ref} = V_{dd}/2$
 - $\Delta V = -\frac{V_{dd}}{2} \frac{C_{bit}}{C_L + C_{bit}}$

Random Access Memories (RAM)

- Dynamic RAM (DRAM)
 - Read operation (continued)
 - At the end of the read operation the voltage on the storage capacitor is V_x
 - It is very close to V_{ref}
 - The original logic value is lost
 - All bits in the same row are lost
 - Even if they are not part of the word that is being read
 - But the information is known at the output of the sense amplifiers!
 - This is true for all columns, even those that were not selected
 - The logic value is written back
 - Drive the Bit Line with an appropriate voltage
 - The original logic value is restored on C_{bit}
 - It applies to all bits in the row
 - The value is taken from the sense amplifier of the corresponding column

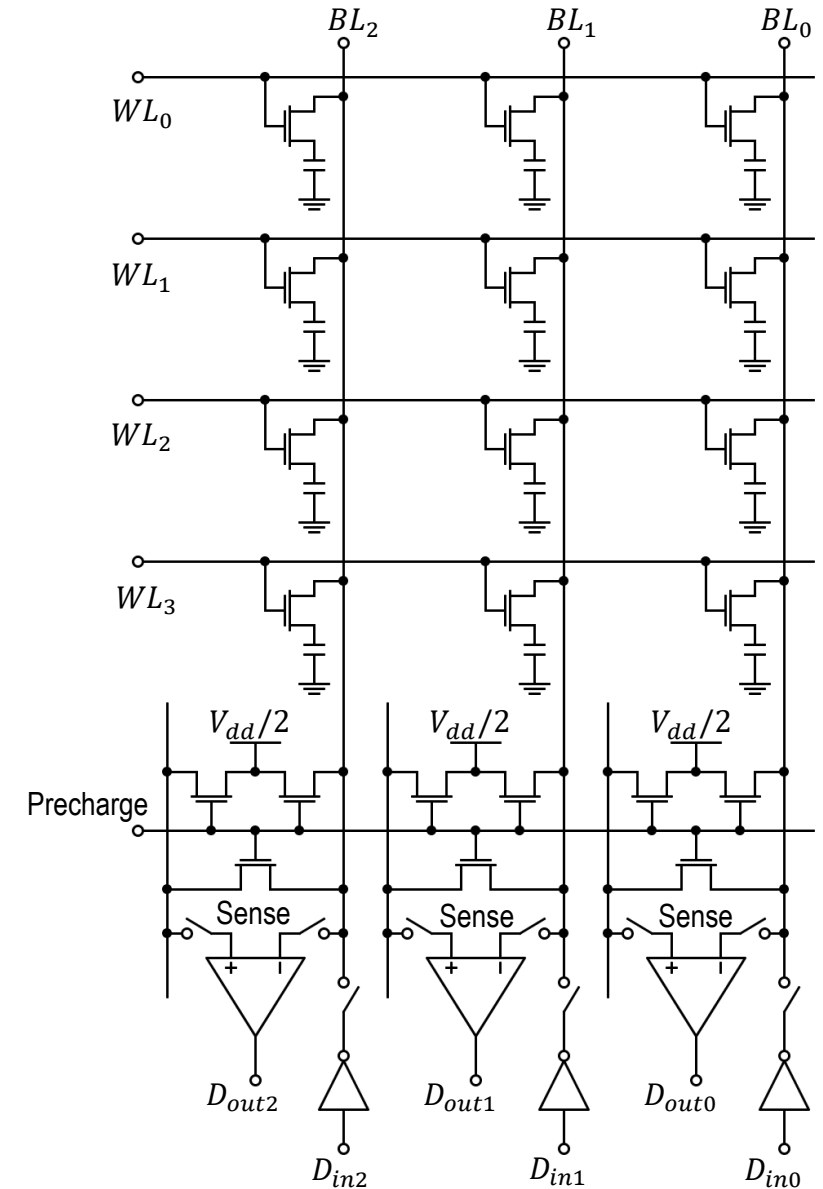
Random Access Memories (RAM)

- Dynamic RAM (DRAM)
 - Read operation (continued)



Random Access Memories (RAM)

- Dynamic RAM (DRAM)
 - Write operation
 - All Word Lines are inactive
 - Apply the desired values to the Bit Lines
 - Precharge the Bit Lines of the other words to V_{ref}
 - Activate the selected Word Line
 - Storage capacitors are charged or discharged to the wanted values
 - For logic 1 the capacitor is charged to $V_{dd} - V_{th}$
 - Then $V_{GS} < V_{th}$ and the access transistor turns OFF
 - Seems fast, but...
 - There are other words in the same row
 - Their bit cells are activated and loose the information!
 - Like in reading
 - Need to rewrite the other bit cells
 - Same performance as reading

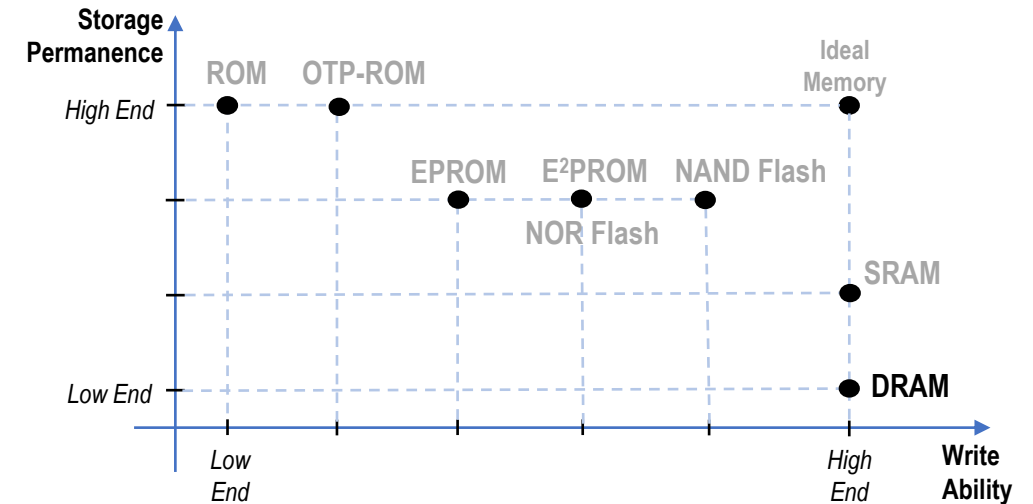


Memories

-
- The diagram illustrates a 1T1R1C1 array structure. It features eight word lines, labeled WL_0 through WL_7 , and two bit lines, labeled BL_0 at both the top and bottom. Each word line WL_i is connected to a series of access transistors (represented by small rectangles) that lead to a common vertical bit line. Each access transistor is in series with a storage capacitor (represented by a parallel plate capacitor symbol). The bit line BL_0 is connected to a sense amplifier circuit. This circuit includes a precharge stage with a PMOS transistor connected to $V_{dd}/2$ and an NMOS transistor connected to a 'Precharge' input. The sense amplifier also includes a differential pair of NMOS transistors connected to the bit line, with their gates controlled by a 'Sense' input. The outputs of the sense amplifier are labeled D_{out0} and D_{in0} .

Random Access Memories (RAM)

- Dynamic RAM (DRAM)
 - Excellent write ability
 - Very fast write
 - Word and bit level writing
 - Can write an infinite number of times
 - Very low storage permanence
 - Data is lost after a few milliseconds
 - Costs
 - Low NRE costs
 - Low per unit costs
 - Bit cell has 1 transistor (compared to 6 in SRAMs)
 - Transistors are minimum size
 - High speed, but slower than SRAM by an order of magnitude
 - A bistable circuit is much stronger and faster



Random Access Memories (RAM)

- Dynamic RAM (DRAM)

- Overall structure

- Multiplex two halves of the address in time

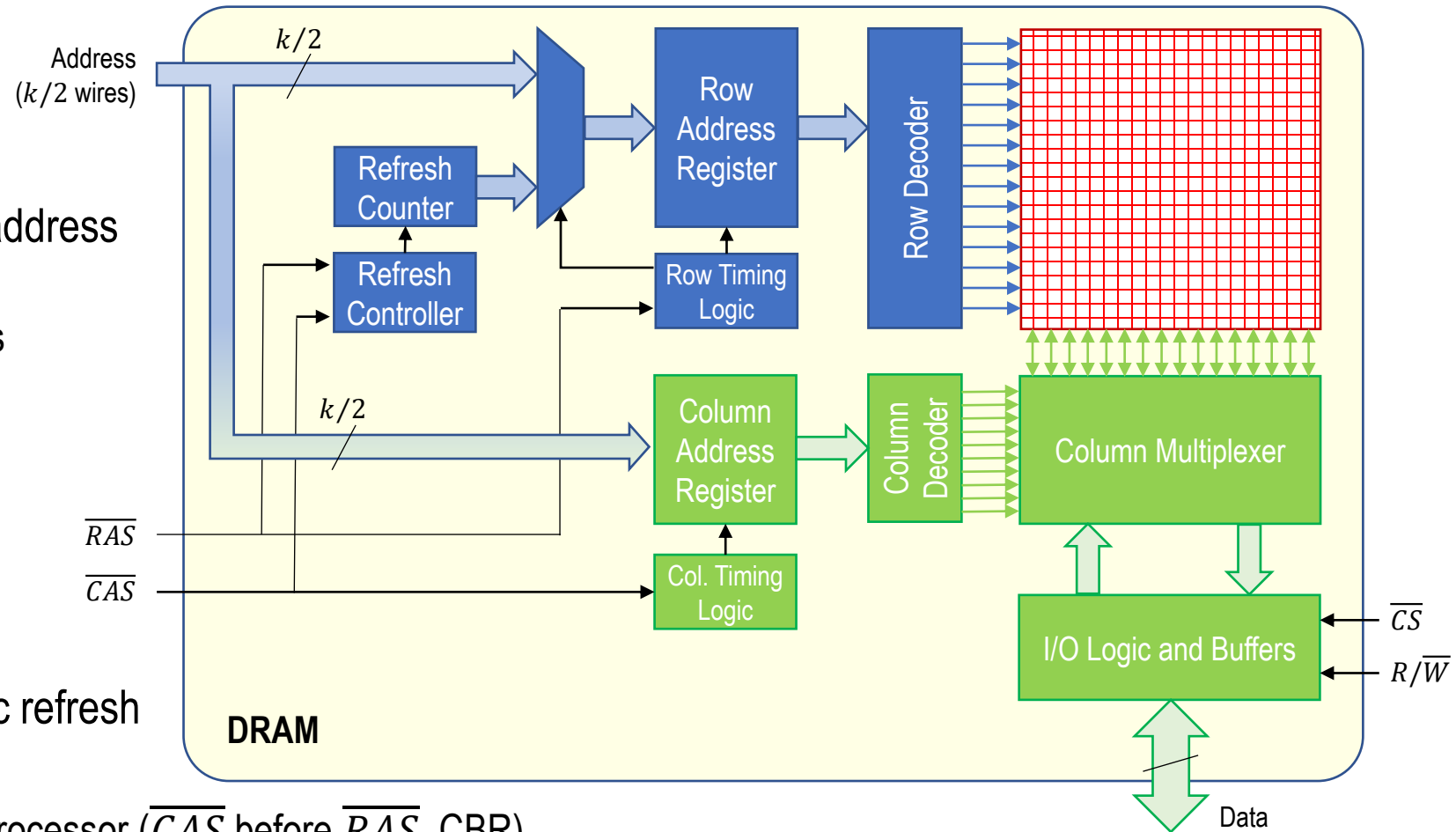
- Row and column registers

- Additional control signals

- $\overline{RAS}$ and $\overline{CAS}$

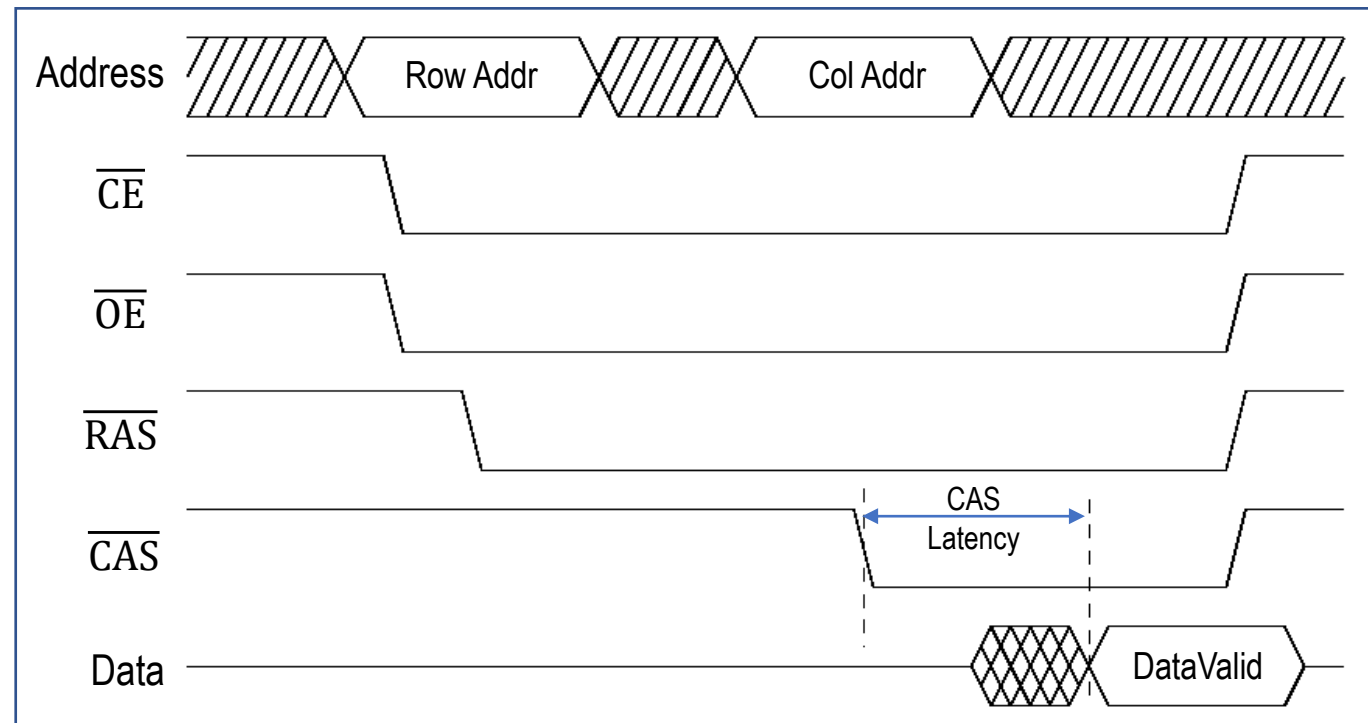
- Refresh

- When a row is opened, all data is refreshed
 - Special circuits for automatic refresh
 - Refresh automatic
 - Refresh issued by microprocessor ($\overline{CAS}$ before $\overline{RAS}$, CBR)
 - Overhead due to refresh



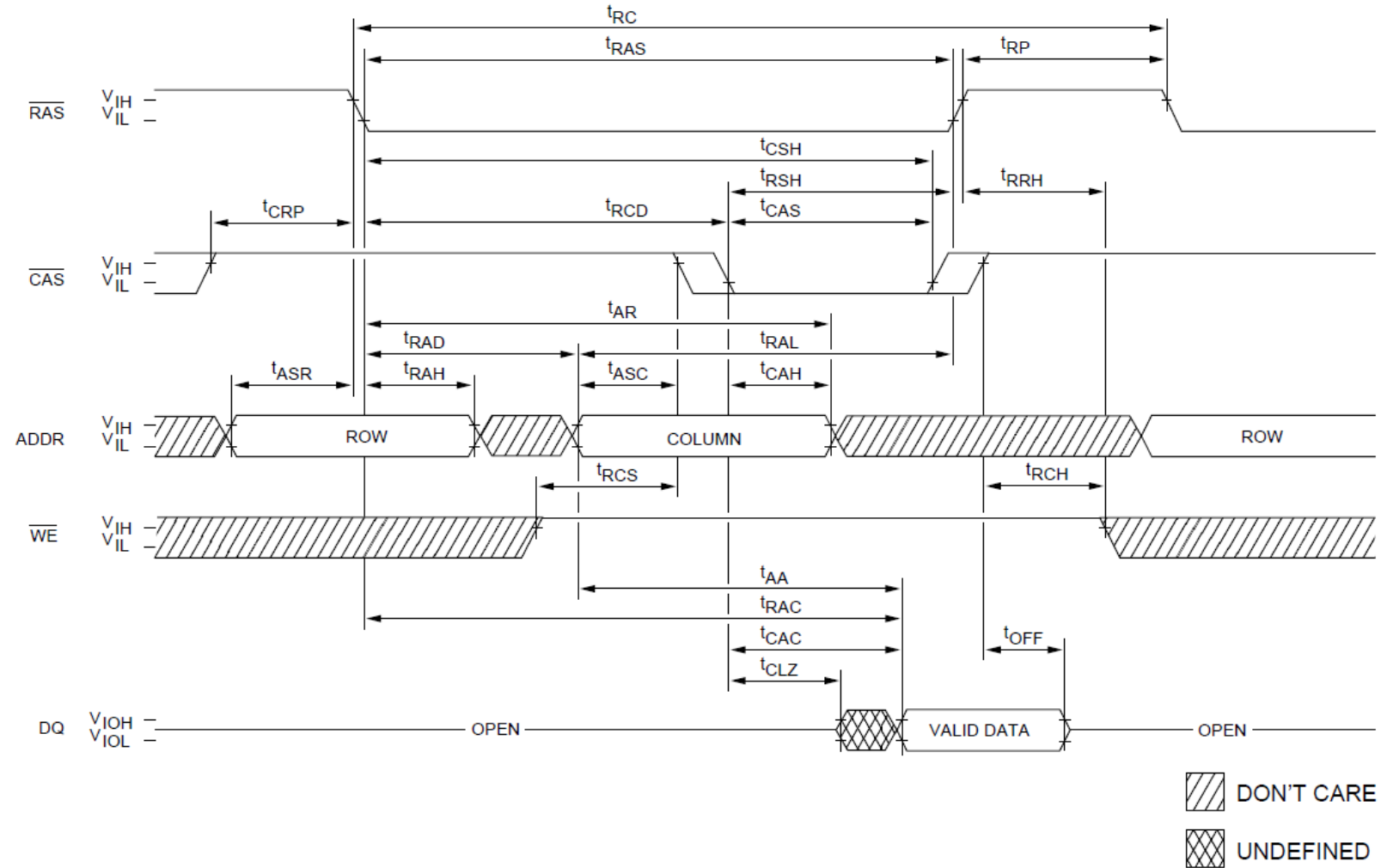
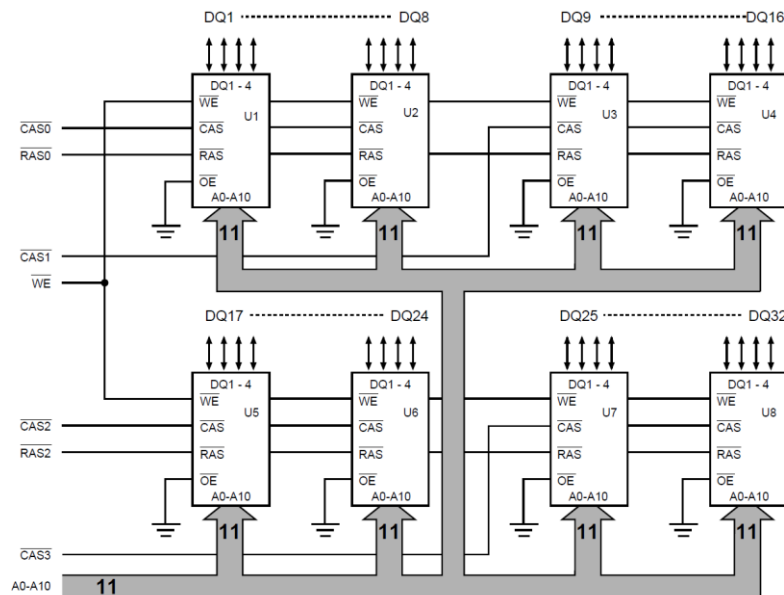
Random Access Memories (RAM)

- Dynamic RAM (DRAM)
 - Timing diagram for an asynchronous DRAM
 - Read operation ($\overline{CE}$ active - low, $\overline{WE}$ inactive - high)
 - The Row Address is presented at the address pins
 - Highest significant bits of the address
 - $\overline{RAS}$ is activated (low)
 - The Row Address is latched internally
 - The Column Address is presented at the address pins
 - Lowest significant bits of the address
 - $\overline{CAS}$ is activated (low)
 - The Column Address is latched internally
 - After a certain minimum amount of time
 - The output data is valid and stable



Random Access Memories (RAM)

- Dynamic RAM (DRAM)
 - Timing diagram
 - Asynchronous DRAM
 - Micron MT8D432-6
 - 4M x 32, 16 Mbytes
 - $t_{RC} = 110 \text{ ns}$ (minimum)



Random Access Memories (RAM)

- DRAM Variations

- DRAM is slow

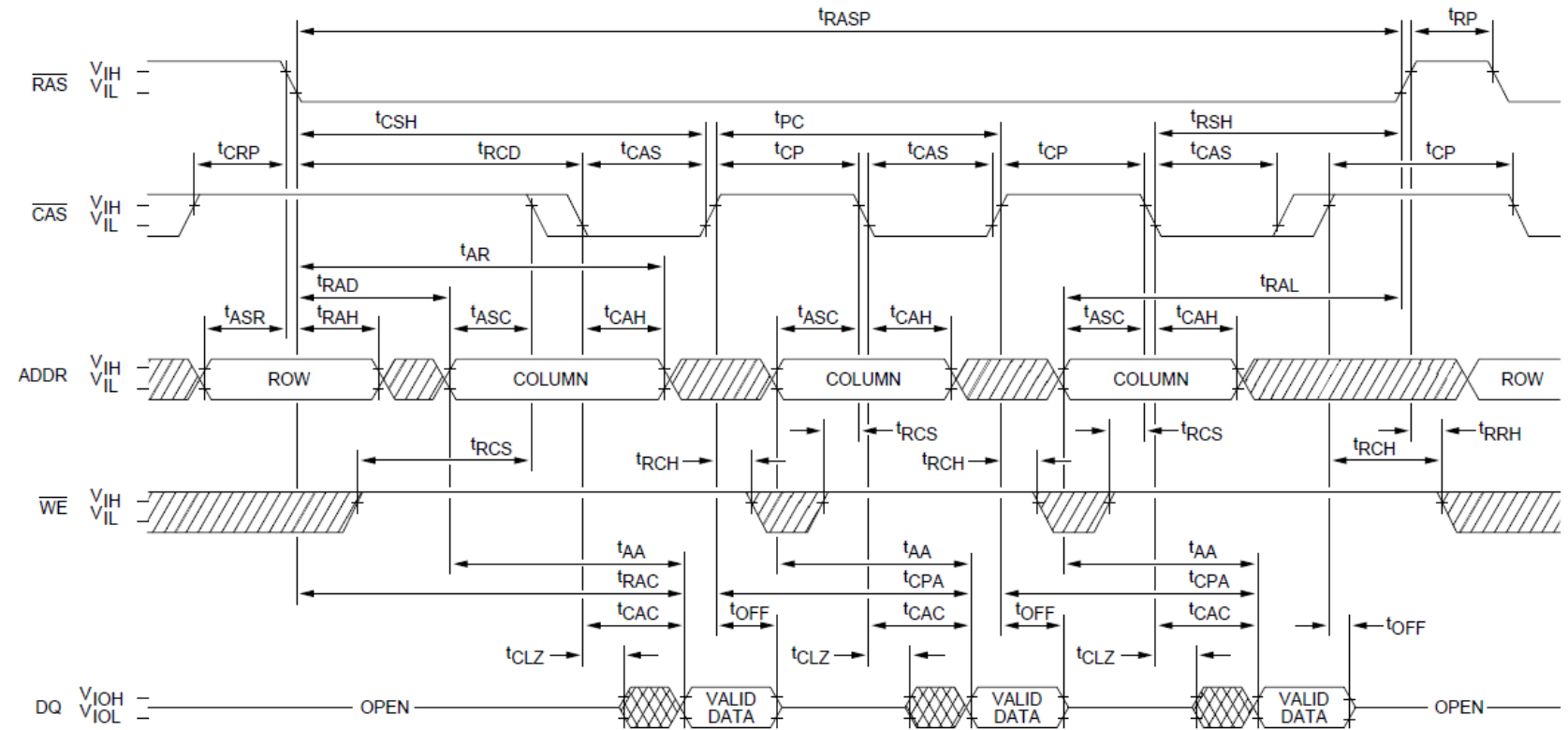
- Precharge bit lines, select (*open*) a word, wait for sense amplifiers to detect the voltage change
 - To *open* a row (page), the Row Address is sufficient

- Fast Page Mode

- All words in the same row are sensed, even those that are not read
 - They even need to be rewritten
 - To read other words in the same row (page), the Column Address is sufficient
 - Need not be contiguous, but within the same row (page)
 - Perform a normal read, but keep $\overline{\text{RAS}}$ active
 - $\overline{\text{CAS}}$ is deactivated (high)
 - A new Column Address is presented at the address pins
 - $\overline{\text{CAS}}$ is re-activated (low)
 - The new Column Address is latched internally
 - After a certain minimum amount of time the new output data is valid and stable

Random Access Memories (RAM)

- DRAM Variations
 - Fast Page Mode timing diagram
 - $t_{RC} = 110 \text{ ns}$ (minimum)
 - $t_{PC} = 35 \text{ ns}$ (minimum)



Random Access Memories (RAM)

- DRAM Variations
 - Extended Data Out (EDO) Mode
 - Similar to Fast Page Mode
 - An additional register added at the output of the memory
 - The Column Address can change while keeping the data of the previous operation on the DataOut Bus
 - Not possible in Fast Page Mode
 - The additional register implements a sort of small pipeline
 - Increased performance

Memories

- $t_{PC} = 25 \text{ ns}$ (minimum)



Random Access Memories (RAM)

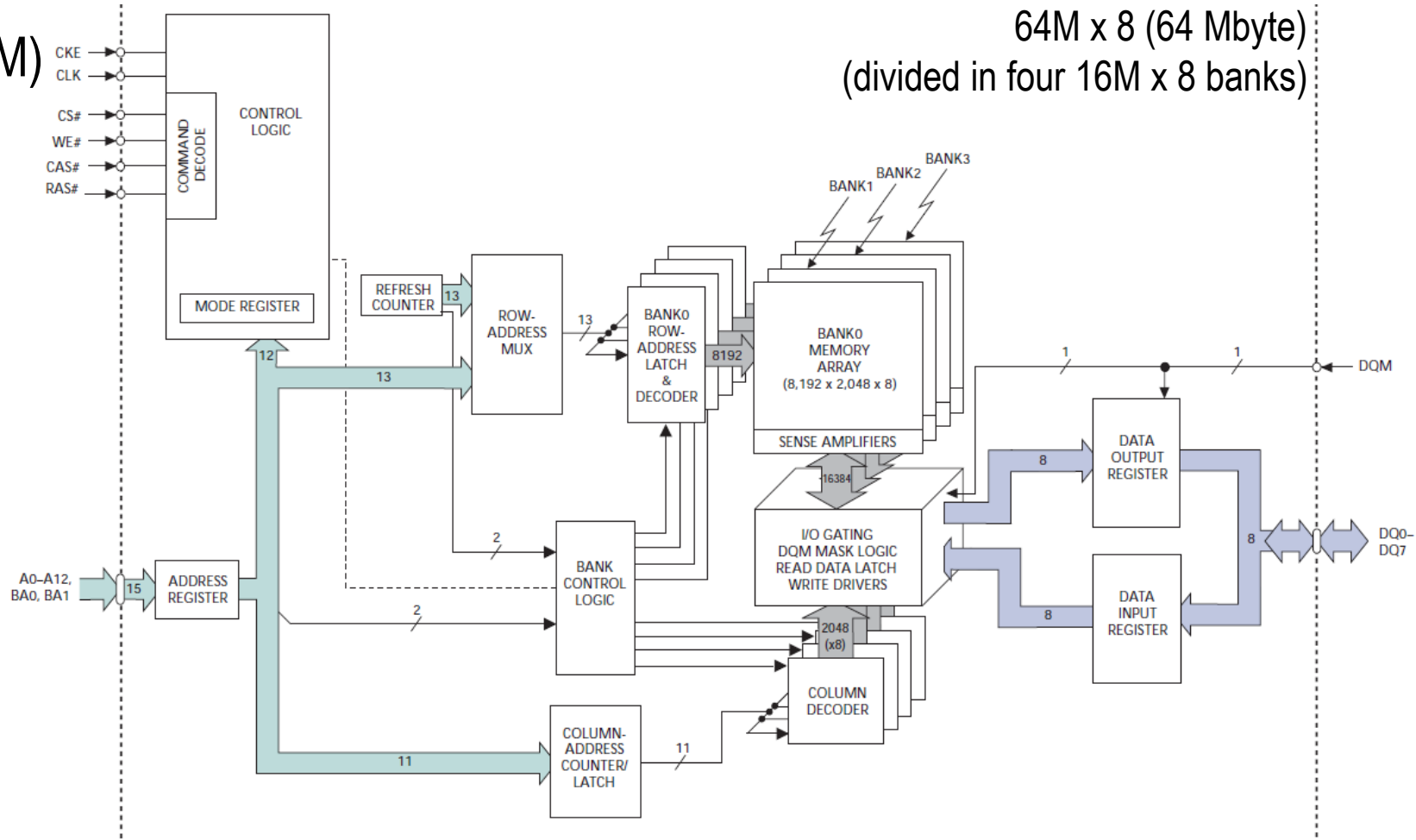
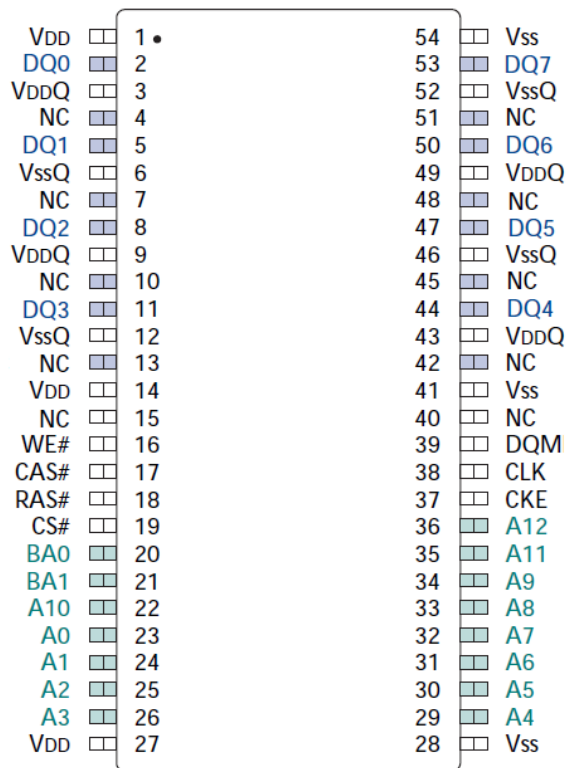
- DRAM Variations
 - Synchronous DRAM (SDRAM)
 - All operations synchronized on the active edge of a clock
 - Simplifies internal structure compared to asynchronous memories
 - Simple circuits improve performance
 - Contiguous addresses, column sent only once
 - Useful using caches
 - Double Data Rate
 - For contiguous data, output updated on both clock edges
 - Rising and falling
 - Can use multiple clock edges
 - Multiple clocks out of phase

Random Access Memories (RAM)

- DRAM Variations

- Synchronous DRAM (SDRAM)

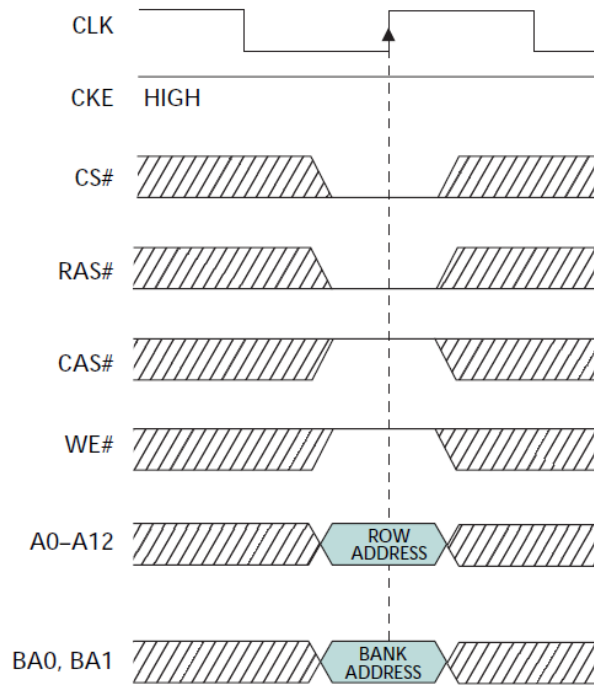
- Micron MT48LC64M8A2



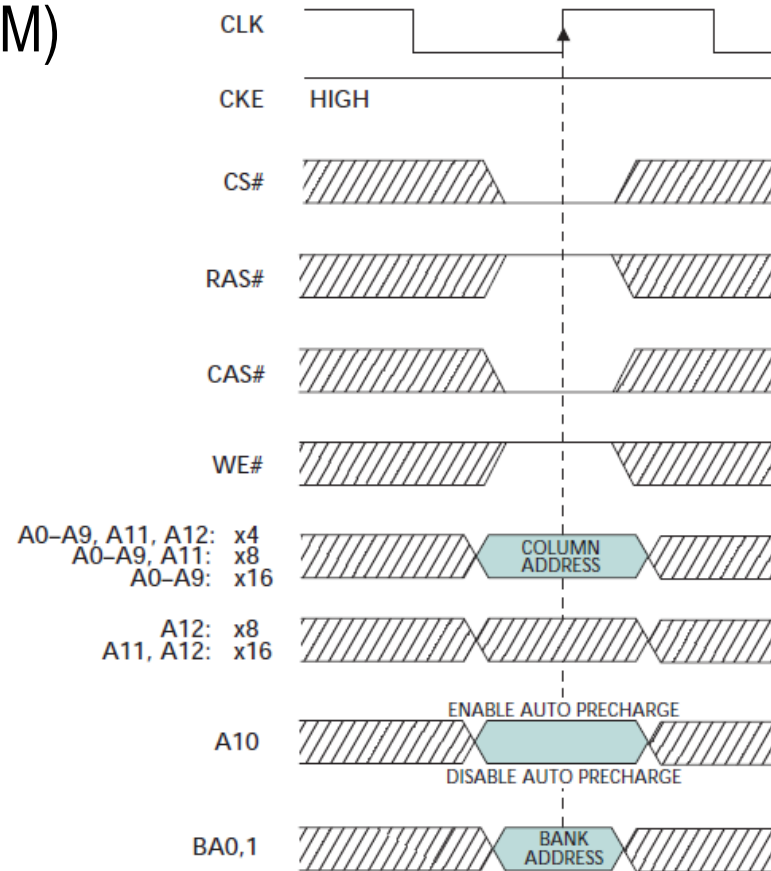
64M x 8 (64 Mbyte)
(divided in four 16M x 8 banks)

Random Access Memories (RAM)

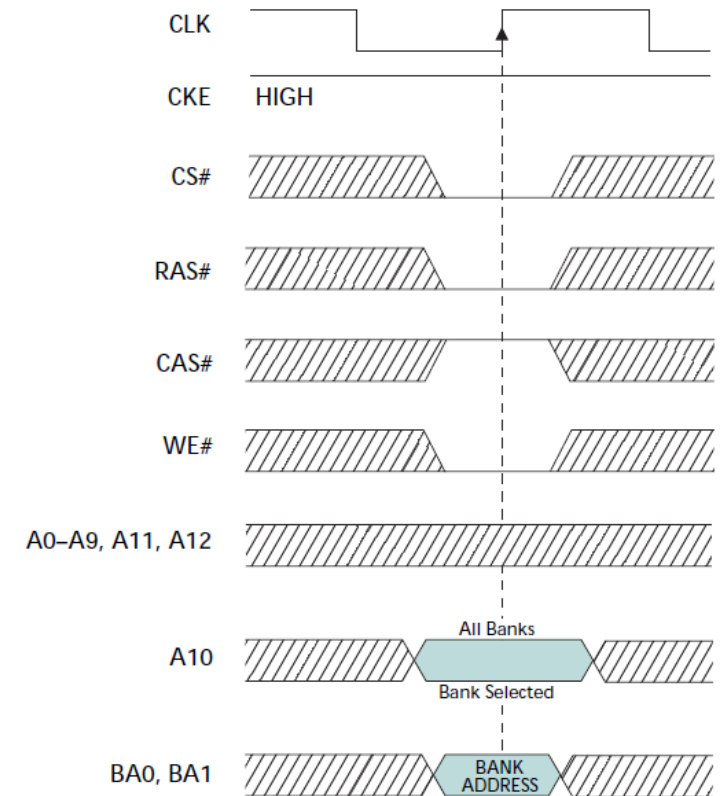
- DRAM Variations
 - Synchronous DRAM (SDRAM)
 - Micron MT48LC64M8A2



Active



Read



Precharge

Random Access Memories (RAM)

- DRAM Variations
 - Synchronous DRAM (SDRAM)
 - Micron MT48LC64M8A2
 - Signals $\overline{CS}$, $\overline{RAS}$, $\overline{CAS}$ and $\overline{WE}$ used to issue commands to the memory
 - Maximum of 16 commands
 - But only 8 used
 - When $\overline{CS} = 1$ the memory is disabled (NOP)

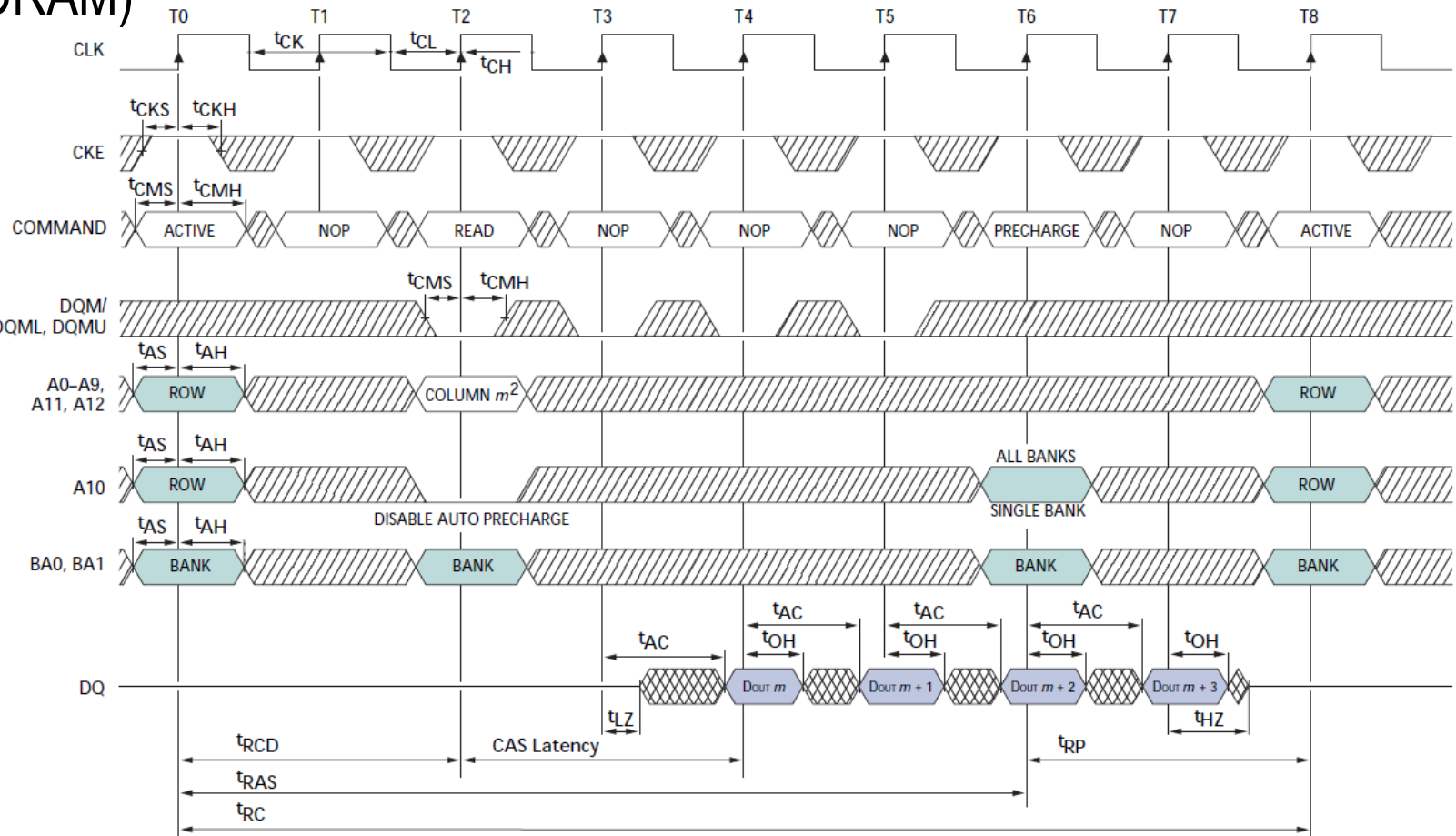
Command	$\overline{CS}$	$\overline{RAS}$	$\overline{CAS}$	$\overline{WE}$	Address
NOP	H	X	X	X	X
NOP	L	H	H	H	X
ACTIVE (select bank and activate row)	L	L	H	H	Bank/row
READ (select bank and column, start burst)	L	H	L	H	Bank/col
WRITE (select bank and column, start burst)	L	H	L	L	Bank/col
BURST TERMINATE	L	H	H	L	X
PRECHARGE (Deactivate row in banks)	L	L	H	L	Code
AUTO REFRESH	L	L	L	H	X
LOAD MODE REGISTER	L	L	L	L	Op-code

Random Access Memories (RAM)

- DRAM Variations

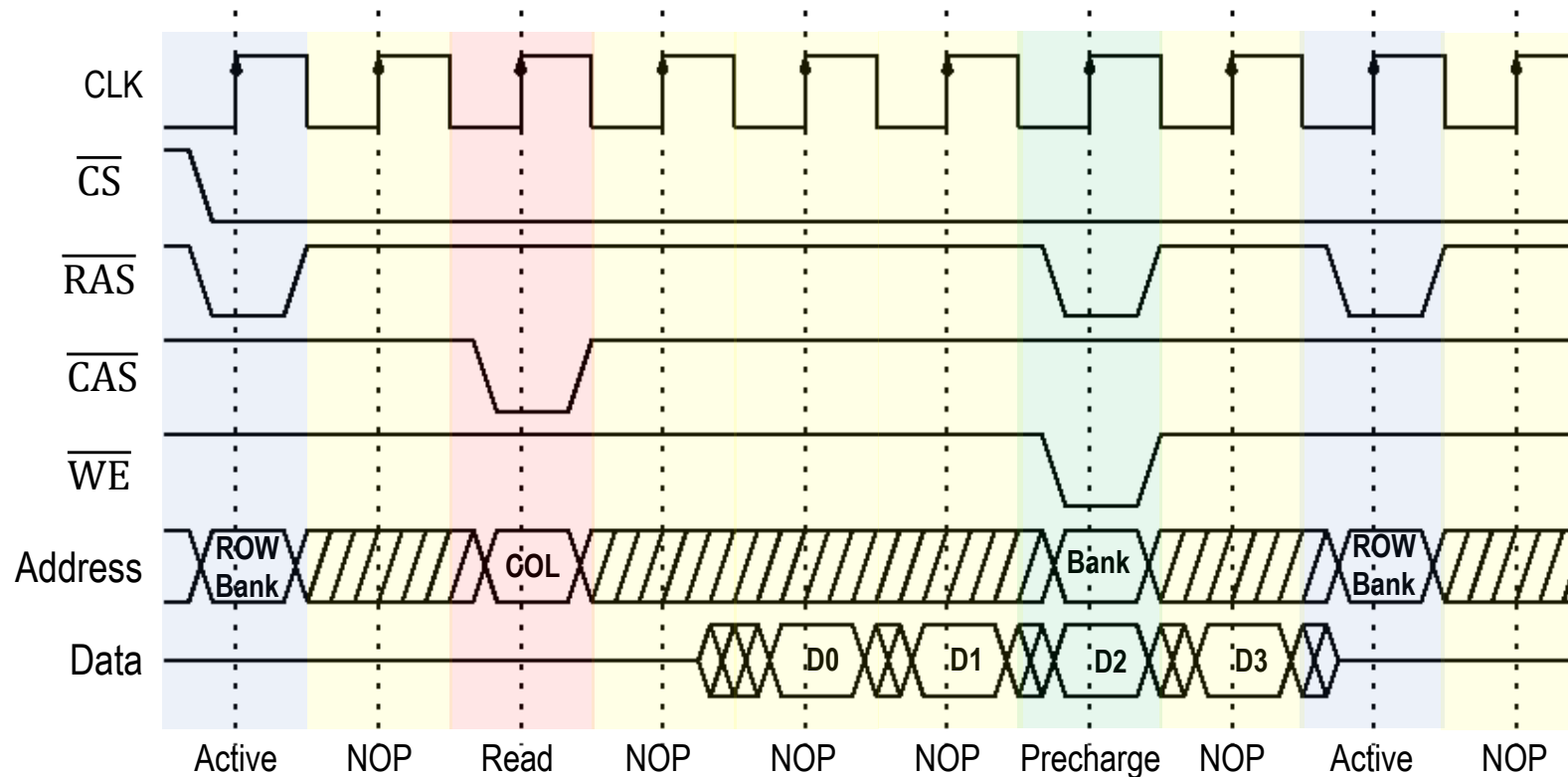
- Synchronous DRAM (SDRAM)

- Micron MT48LC64M8A2
 - $f_{ck} = 133 \text{ MHz}$
 - $t_{ck} = 7,5 \text{ ns}$
 - CAS Latency = 2
 - $t_{AC} = 5,4 \text{ ns}$ (maximum)
 - Single read
 - 7 clock cycles
 - $t_{TOT} = 52,5 \text{ ns}$



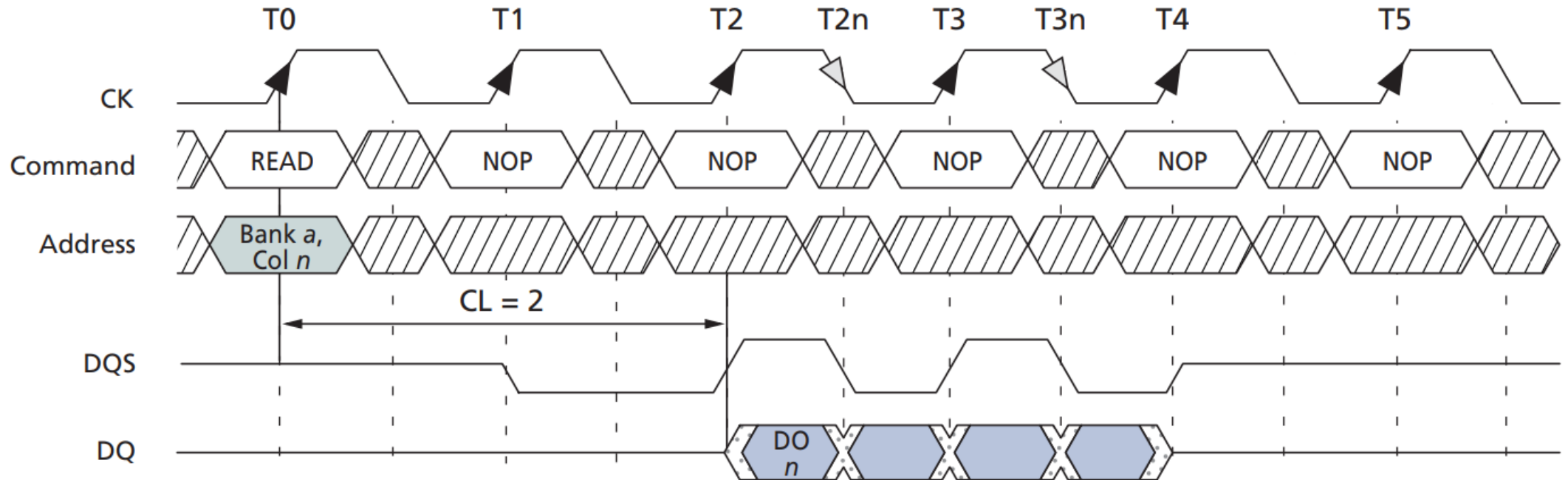
Random Access Memories (RAM)

- DRAM Variations
 - Synchronous DRAM (SDRAM)
 - Micron MT48LC64M8A2



Random Access Memories (RAM)

- DRAM Variations
 - Double Data Rate DRAM (DDR)
 - Micron MT46V32M8



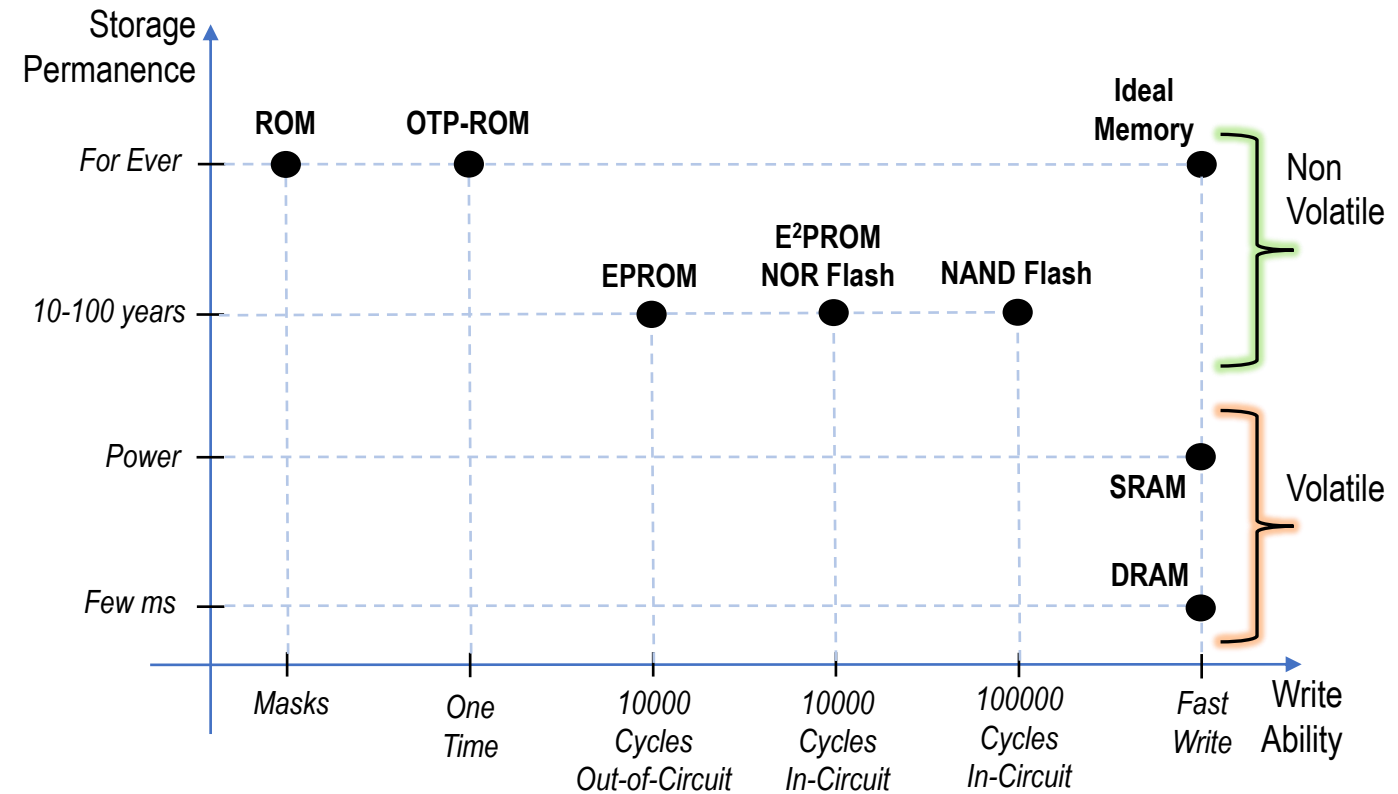
Random Access Memories (RAM)

- DRAM Variations
 - Normal applications
 - DDR, DDR2, DDR3, DDR4, ...
 - Low Power
 - For mobile applications
 - LPDDR1, LPDDR2, ...
 - High bandwidth
 - For graphic applications
 - GDDR2, GDDR3, GDDR4, GDDR5, ...
 - Special architectures
 - RAMBus, eDRAM

Memory Classification

- Graph of Write Ability and Storage Permanence

- RAM memories are fast to write
 - Used to read and write data
 - Word and bit level accesses
 - SRAM are faster
 - But they are more expensive
- RAM memories are volatile
 - Use different kind of memories in the same embedded system



Memory controllers

- Translate *standard* memory requests to DRAM command sequences
 - From address to row, column, bank, ...
- A flexible memory controller can support various DRAM types
 - Underlying architecture is very similar
- Schedule requests for higher performance
 - First Come First Served (FCFS)
 - First ready, first come first served (FR-FCFS)
 - Read/Write (column operations) on the same row first, oldest first
 - Active/Precharge (row operations) afterwards, oldest first
- Ensure correct operation of DRAM
 - Timing Constraints
 - Refresh

Content Addressable Memories

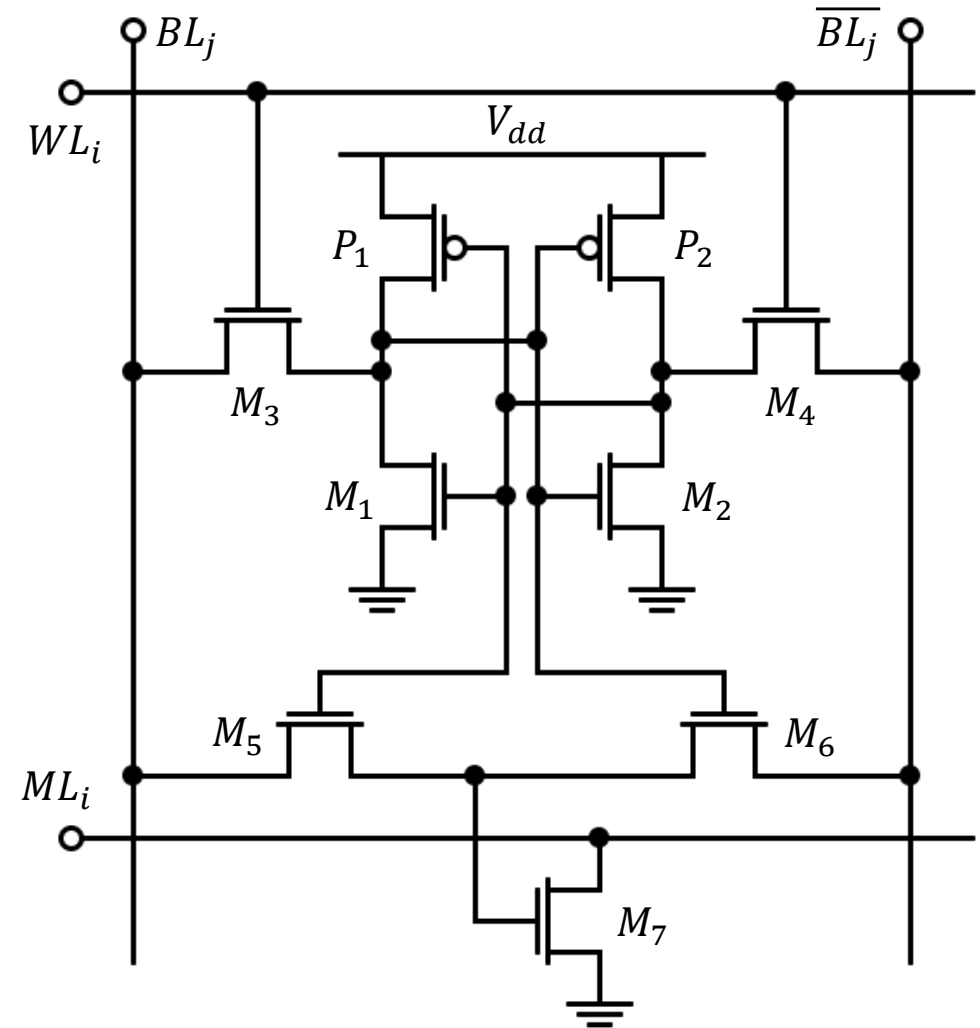
- Content Addressable Memories (CAM)
 - Also known as *Associative Memories*
 - Match operation
 - Instead of providing the address and read the data, look for the address of some given data
 - Straightforward way
 - Read all memory locations sequentially until one with the wanted value is found
 - Or the end of the memory is reached
 - Extremely slow
 - Implement matching in hardware inside the memory
 - Much faster
 - More expensive

7	00FE
6	E474
5	369C
4	9DB8
3	0001
2	FFC8
1	1260
0	5A71

- Where is FFC8?
Location 2
- Where is 369C?
Location 5
- Where is 0002?
Nowhere!

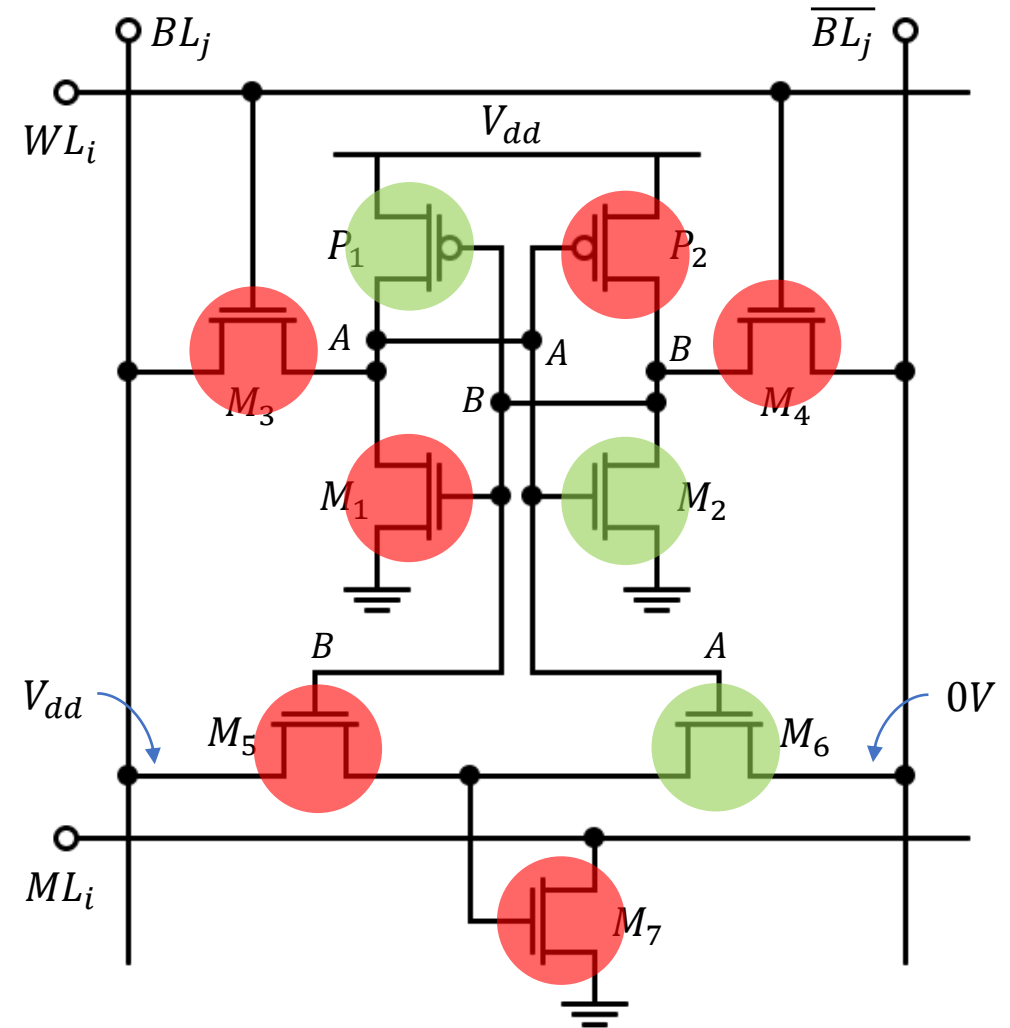
Content Addressable Memories

- Associative SRAM
 - Bit cell
 - Similar to an SRAM
 - But with three additional transistors
 - An additional Match Line
 - Read and write operations
 - Identical to SRAM
 - Match Line floating
 - Match operation
 - Word Line WL_i remains inactive for the entire operation
 - Don't want to change the bit cell content
 - Precharge Match Line ML_i to V_{dd}
 - Apply the value to BL_j and its complement to $\overline{BL_j}$
 - ML_i is discharged if there is no match



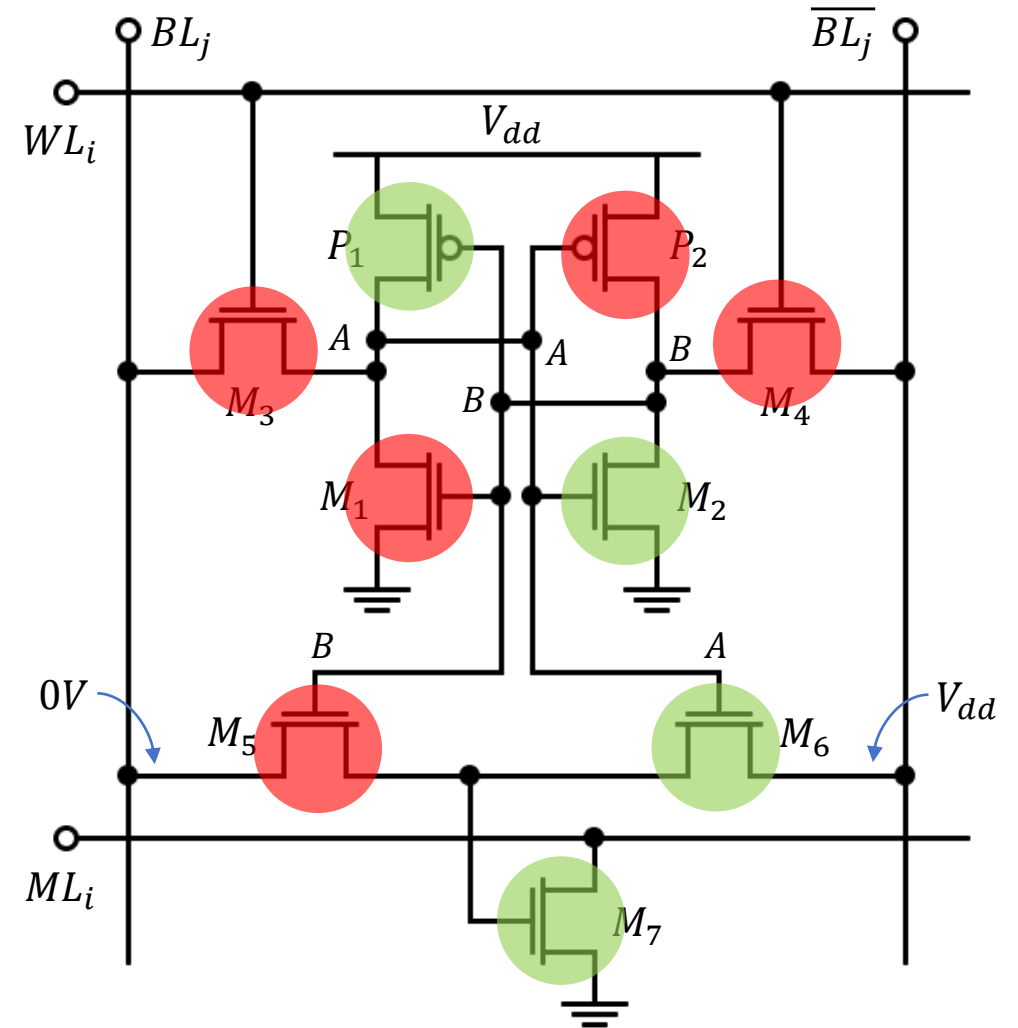
Content Addressable Memories

- Associative SRAM
 - Example 1
 - Bit cell contains logic value 1
 - M_1 OFF, M_2 ON, P_1 ON, P_2 OFF
 - $V_A = V_{dd}$, $V_B = 0V$
 - M_5 OFF, M_6 ON
 - Search for logic value 1
 - $BL_j = V_{dd}$, $\overline{BL}_j = 0V$
 - Match operation
 - M_6 is ON
 - The voltage $0V$ is driven to the gate of M_7
 - M_7 is OFF
 - The Match Line ML_i maintains the high value
 - MATCH! HIT



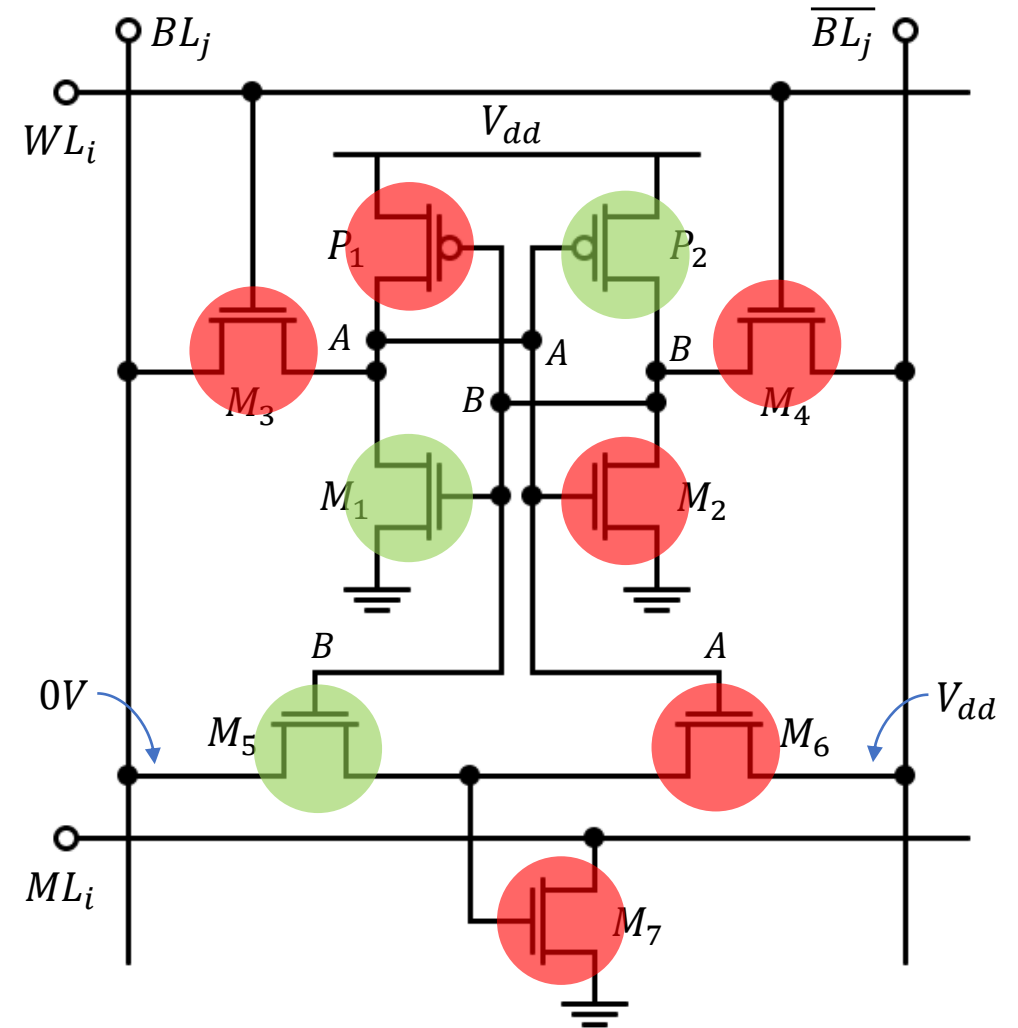
Content Addressable Memories

- Associative SRAM
 - Example 2
 - Bit cell contains logic value 1
 - M_1 OFF, M_2 ON, P_1 ON, P_2 OFF
 - $V_A = V_{dd}$, $V_B = 0V$
 - M_5 OFF, M_6 ON
 - Search for logic value 0
 - $BL_j = 0V$, $\overline{BL}_j = V_{dd}$
 - Match operation
 - M_6 is ON
 - The voltage V_{dd} is driven to the gate of M_7
 - M_7 is ON
 - The Match Line ML_i is discharged to ground
 - NO MATCH! MISS



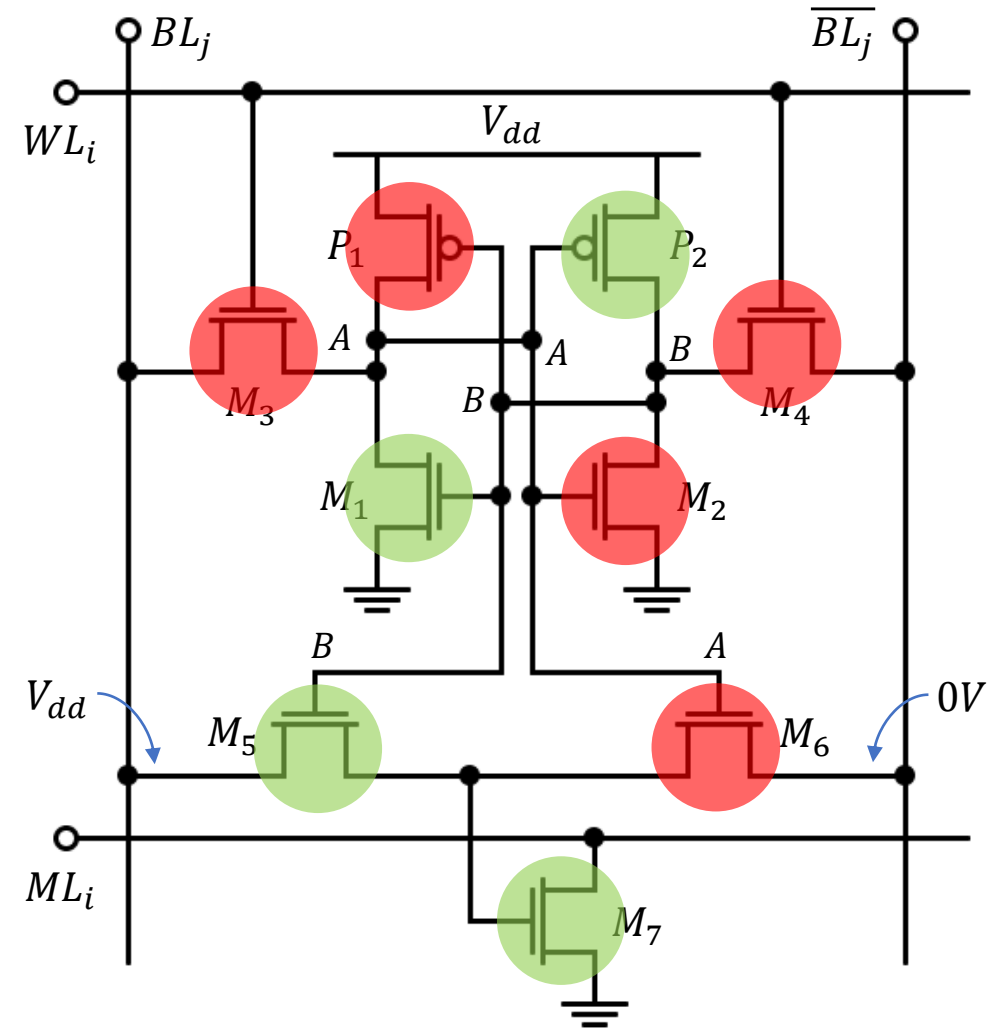
Content Addressable Memories

- Associative SRAM
 - Example 3
 - Bit cell contains logic value 0
 - M_1 ON, M_2 OFF, P_1 OFF, P_2 ON
 - $V_A = 0V$, $V_B = V_{dd}$
 - M_5 ON, M_6 OFF
 - Search for logic value 0
 - $BL_j = 0V$, $\overline{BL}_j = V_{dd}$
 - Match operation
 - M_5 is ON
 - The voltage $0V$ is driven to the gate of M_7
 - M_7 is OFF
 - The Match Line ML_i maintains the high value
 - MATCH! HIT



Content Addressable Memories

- Associative SRAM
 - Example 4
 - Bit cell contains logic value 0
 - M_1 ON, M_2 OFF, P_1 OFF, P_2 ON
 - $V_A = 0V$, $V_B = V_{dd}$
 - M_5 ON, M_6 OFF
 - Search for logic value 1
 - $BL_j = V_{dd}$, $\overline{BL}_j = 0V$
 - Match operation
 - M_5 is ON
 - The voltage V_{dd} is driven to the gate of M_7
 - M_7 is ON
 - The Match Line ML_i is discharged to ground
 - NO MATCH! MISS

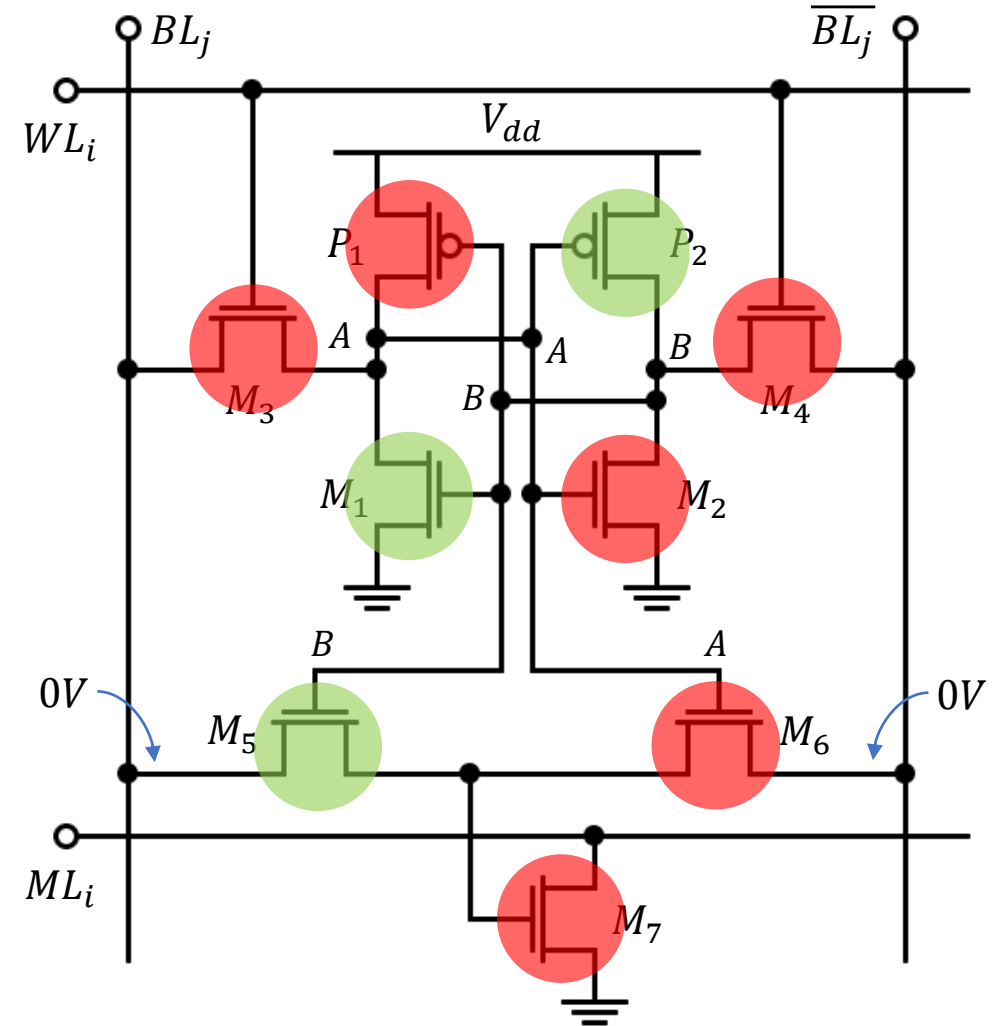


Memories

-

Content Addressable Memories

- Associative SRAM
 - Example 6
 - Bit cell contains logic value 0
 - M_1 ON, M_2 OFF, P_1 OFF, P_2 ON
 - $V_A = 0V$, $V_B = V_{dd}$
 - M_5 ON, M_6 OFF
 - Search for logic value 0 on both Bit Lines (*Don't Care*)
 - $BL_j = 0V$, $\overline{BL_j} = 0V$
 - Match operation
 - M_5 is ON
 - The voltage $0V$ is driven to the gate of M_7
 - M_7 is OFF
 - The Match Line ML_i maintains the high value
 - DON'T CARE! HIT

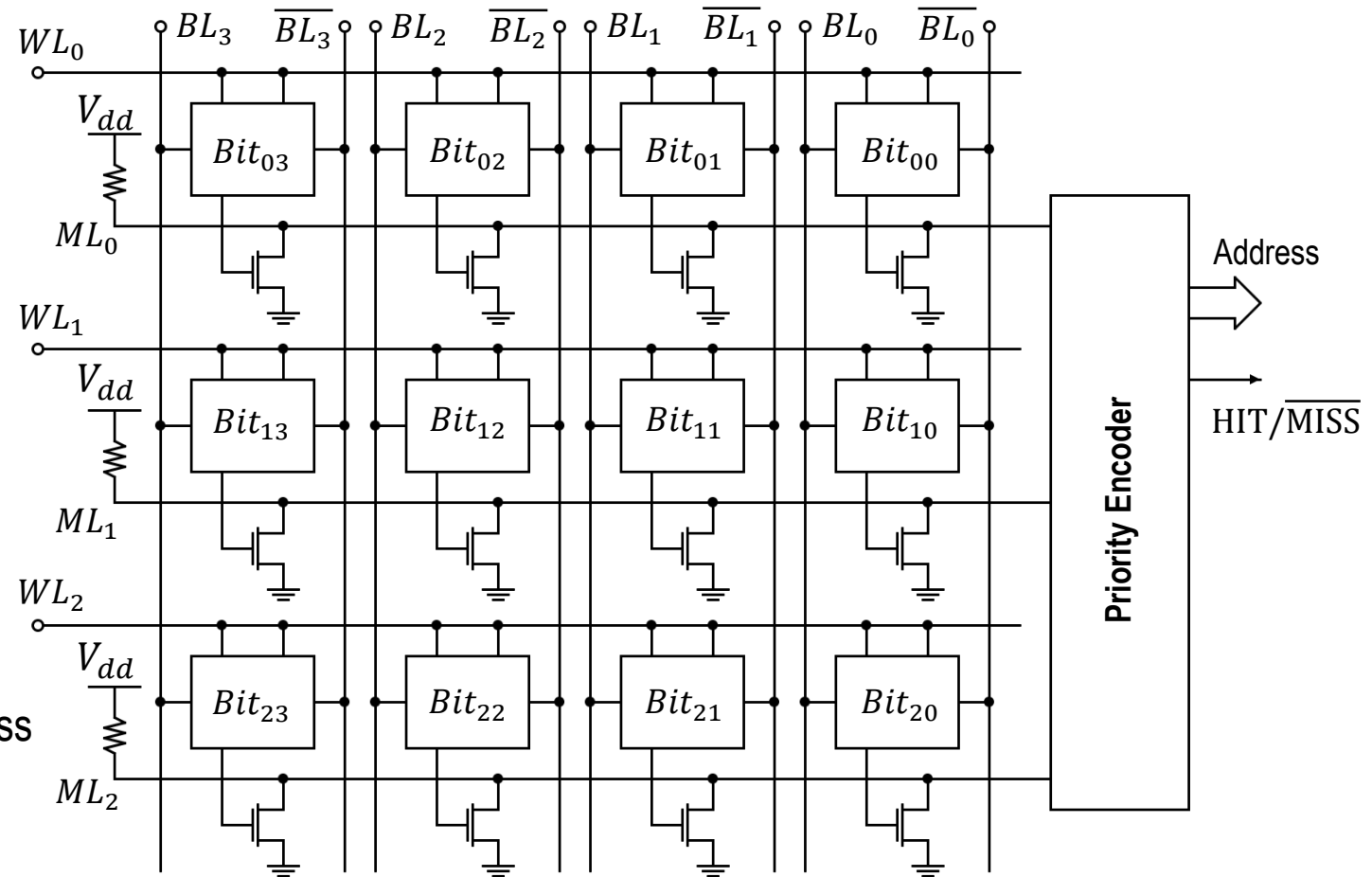


Content Addressable Memories

- Associative SRAM

- Memory structure

- Each Match Line spans an entire word
 - Not an entire row
 - When matching a word
 - If all bits in Word i match
 - The Match Line ML_i is high
 - If at least one bit in Word i does not match
 - The Match Line ML_i is low
 - If at least one Match Line is high
 - The priority encoder asserts HIT
 - It generates the corresponding address
 - If all Match Lines are low
 - The priority encoder asserts MISS

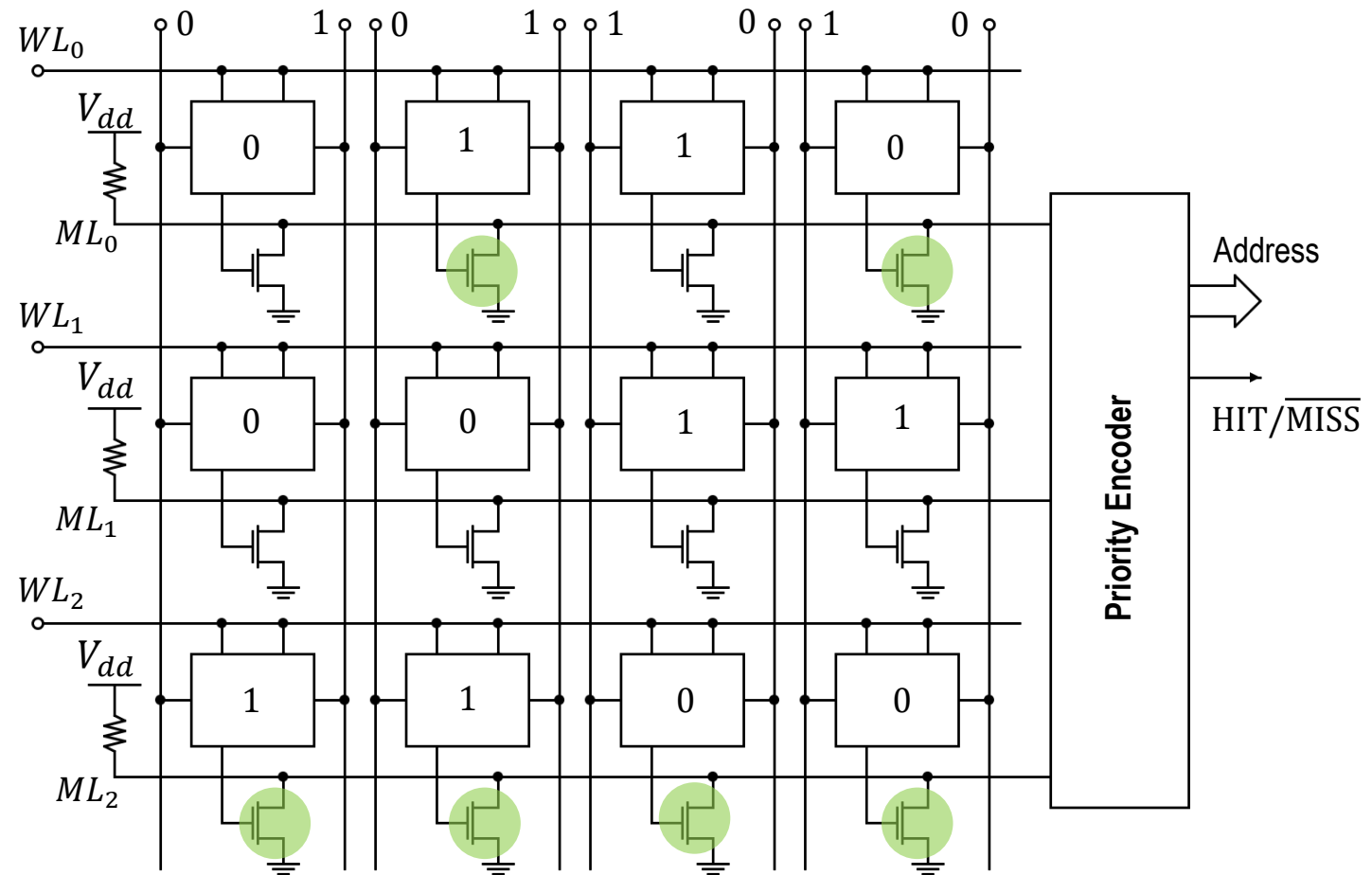


Content Addressable Memories

- Associative SRAM

- Example 1

- Match value 0011_2
 - ML_0 goes to ground
 - Two transistors are conducting
 - ML_1 remains high
 - All transistors are not conducting
 - ML_2 goes to ground
 - All transistors are conducting
 - $HIT/\overline{MISS} = 1$
 - Address = 1

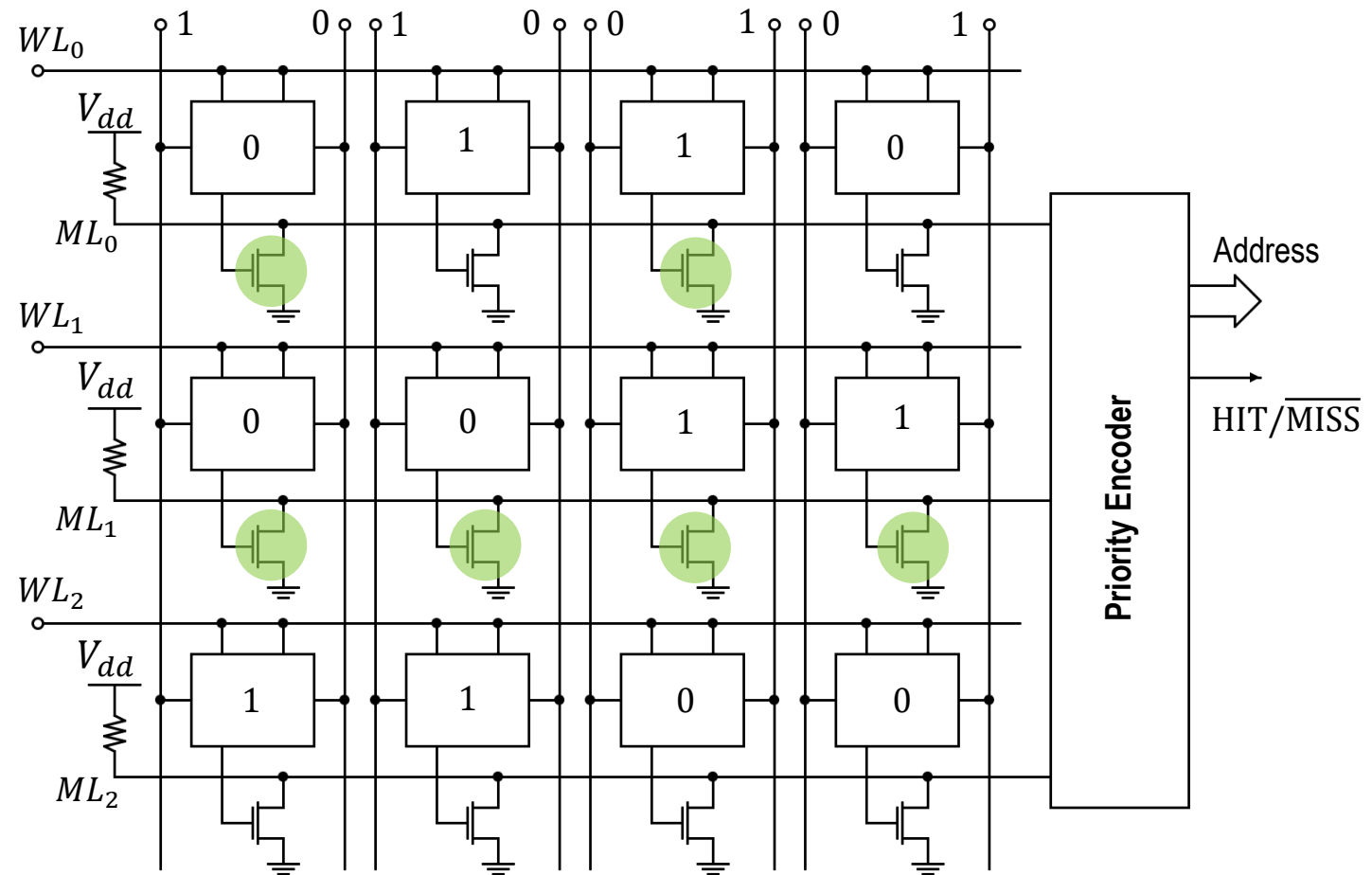


Content Addressable Memories

- Associative SRAM

- Example 2

- Match value 1100_2
 - ML_0 goes to ground
 - Two transistors are conducting
 - ML_1 goes to ground
 - All transistors are conducting
 - ML_2 remains high
 - All transistors are not conducting
 - $HIT/\overline{MISS} = 1$
 - Address = 2

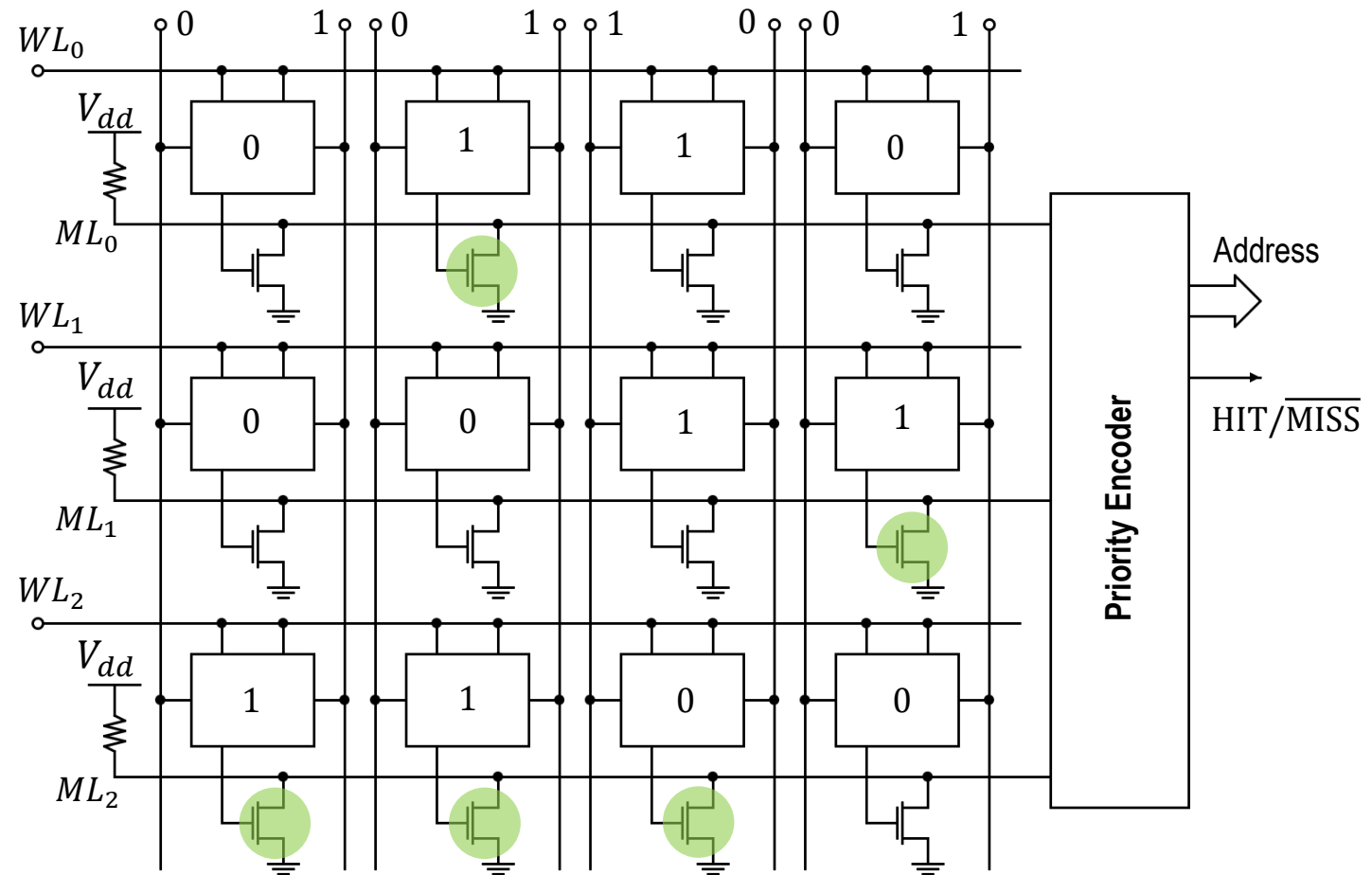


Content Addressable Memories

- Associative SRAM

- Example 3

- Match value 0010_2
 - ML_0 goes to ground
 - One transistor is conducting
 - ML_1 goes to ground
 - One transistor is conducting
 - ML_2 goes to ground
 - Three transistors are conducting
 - $HIT/\overline{MISS} = 0$
 - Address = don't care / High-Z

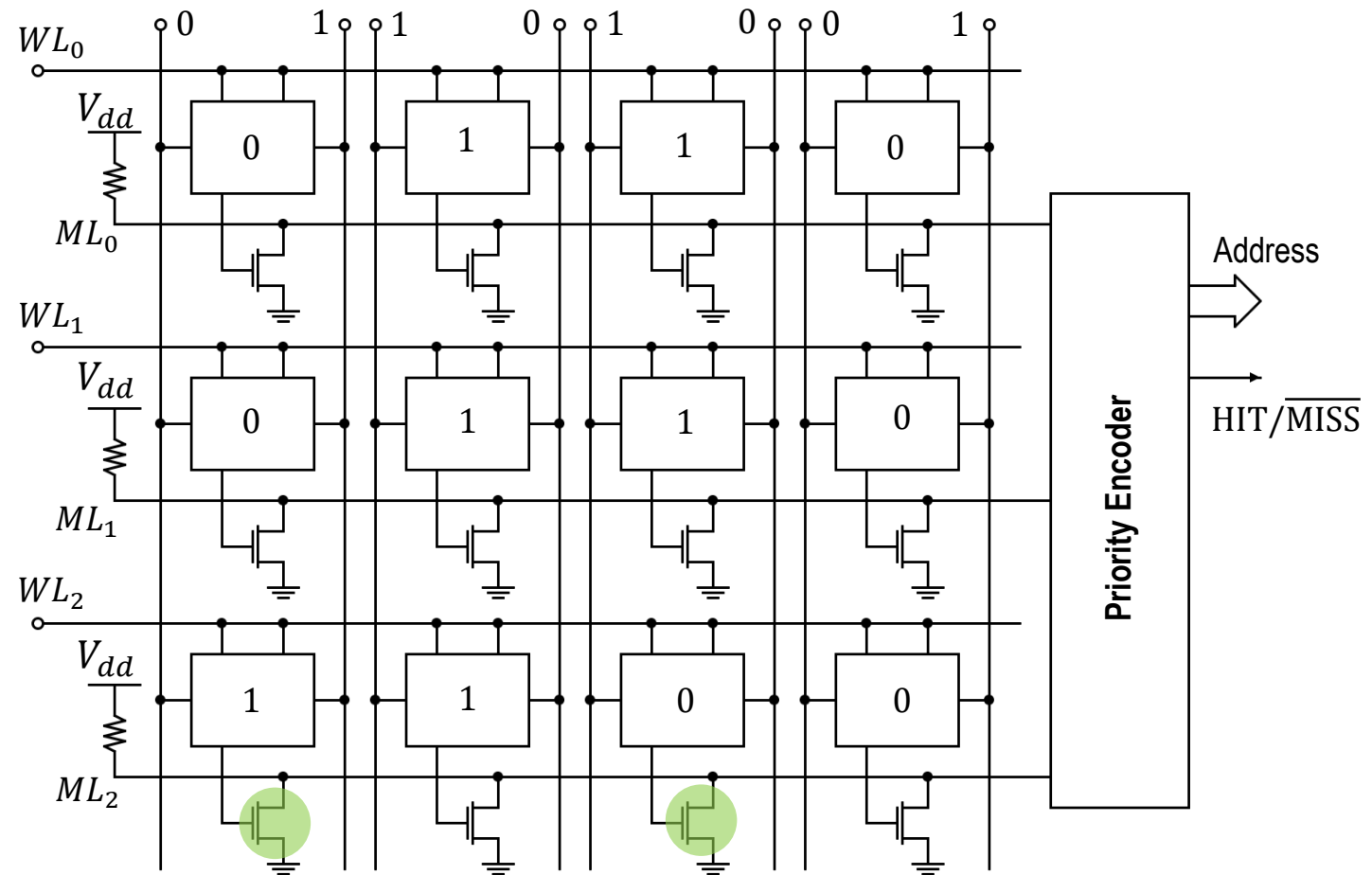


Content Addressable Memories

- Associative SRAM

- Example 4

- Match value 0110_2
- ML_0 remains high
 - All transistors are not conducting
- ML_1 remains high
 - All transistors are not conducting
- ML_2 goes to ground
 - Two transistors are conducting
- $\text{HIT}/\overline{\text{MISS}} = 1$
- Address = 0 or 1
 - Up to the encoder



Content Addressable Memories

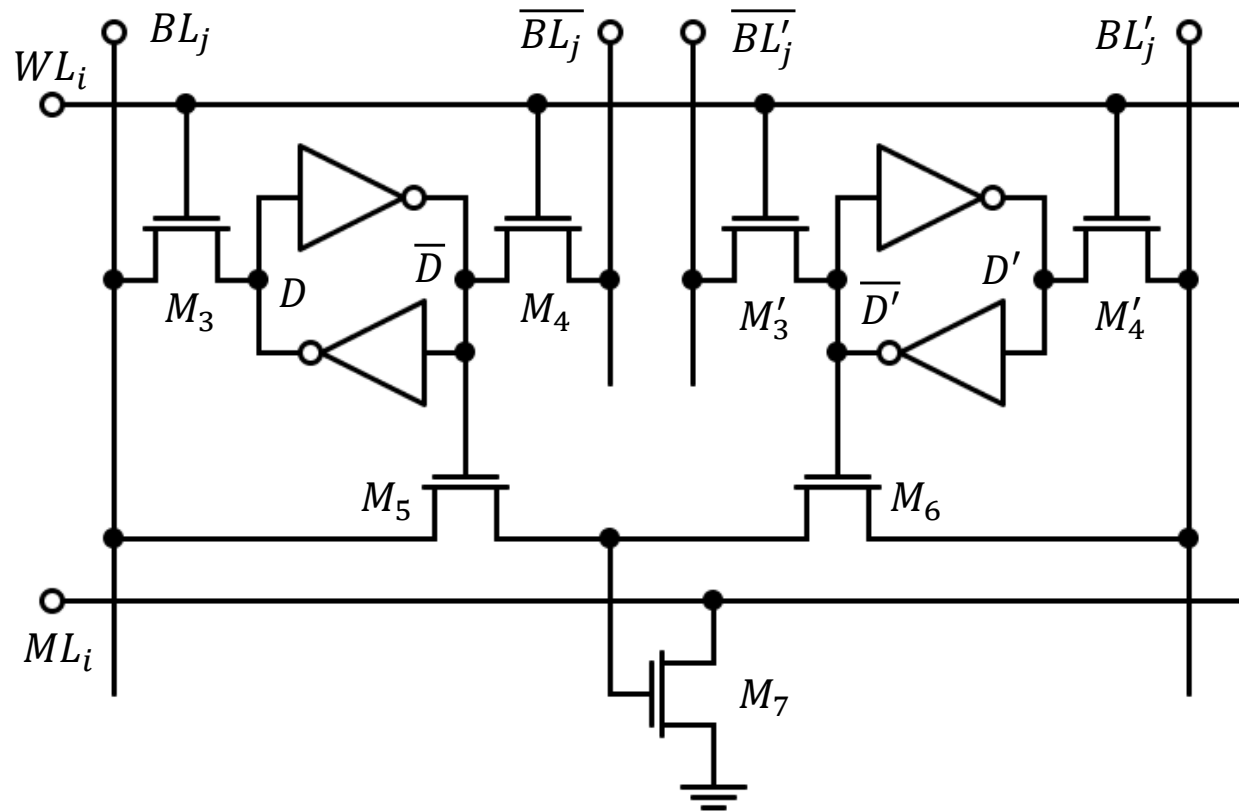
- Associative SRAM
 - When an Associative SRAM is switched ON the content is random
 - Assume some data is written, but only partially occupies the memory
 - Want to match only against the written data
 - All words in the associative memory contain an additional bit, called the *valid bit*
 - At power on also the valid bit is random
 - A special associative memory operation initializes all valid bits to 0
 - It marks all words as invalid
 - To be issued immediately after power on, or whenever you want to clear all valid bits
 - When writing a word, the corresponding valid bit is automatically set
 - When matching, all valid bits are automatically matched against logic 1
 - All invalid words do not match

Content Addressable Memories

- Associative SRAM
 - Performance
 - All matching performed in parallel
 - Much better than scanning the entire memory sequentially
 - Very quick result (tens of nanoseconds)
 - The larger the memory, the slower the matching
 - Increased parasitic capacitances
 - More expensive than an SRAM
 - 9 transistors compared to 6 transistors
 - The bit cell is larger
 - Power consumption
 - High power consumption during the match operation
 - Precharge all match lines
 - Discharge most (or all) of them

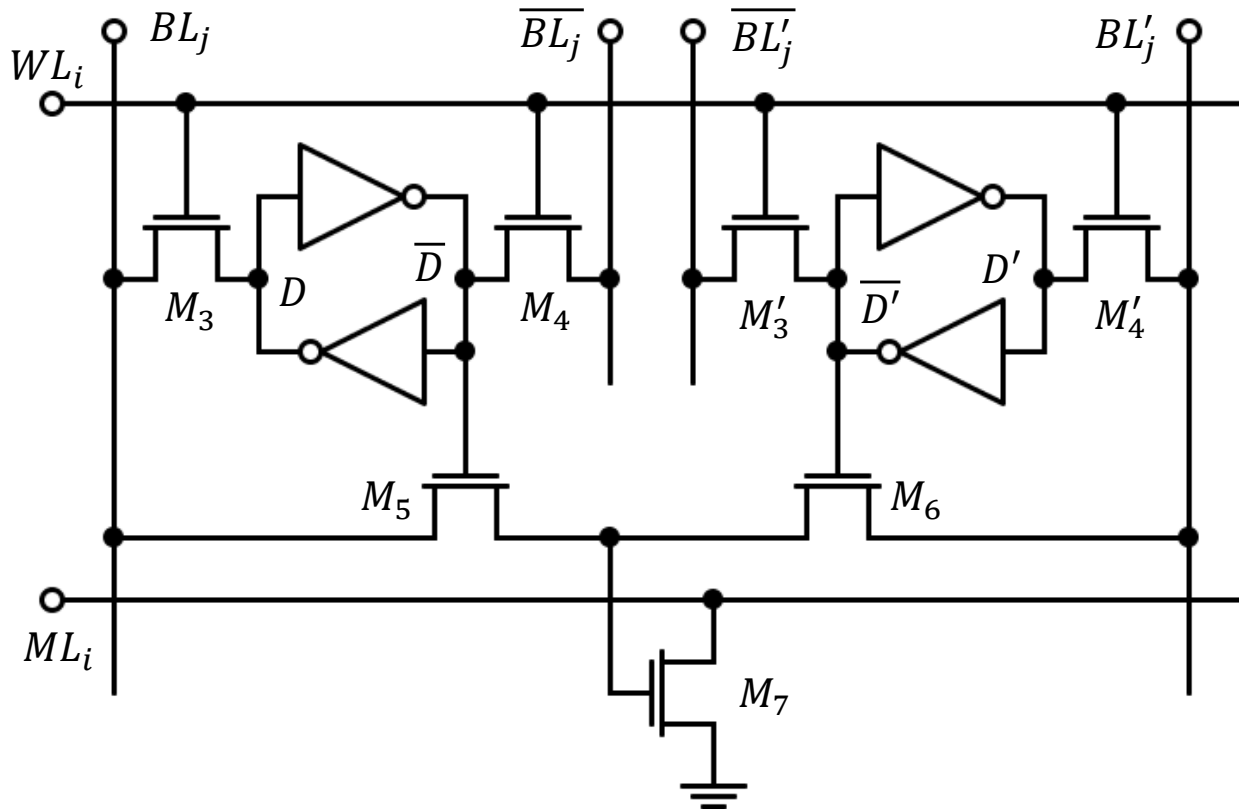
Content Addressable Memories

- Ternary CAM
 - Matching against both Bit Lines at 0 corresponds to matching a *don't care*
 - See examples 5 and 6 above
 - The content of the memory was either 0 or 1
 - Want to be able to specify a *don't care* in the content of the memory as well
 - Need 2 bit cells for each bit in the memory
 - Call them D and D'
 - There are 4 possible combinations
 - $DD' = 01$ content is 0
 - $DD' = 10$ content is 1
 - $DD' = 11$ content is *don't care*
 - $DD' = 00$ not used



Content Addressable Memories

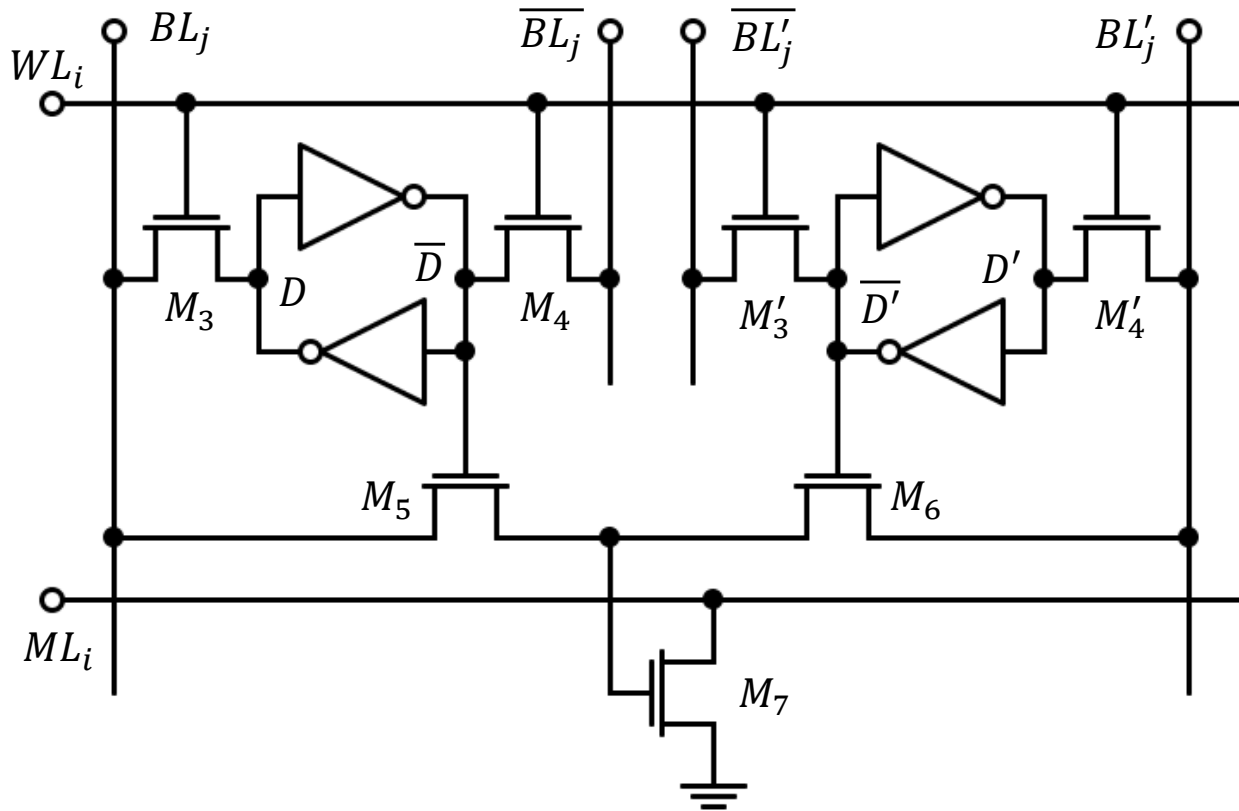
- Ternary CAM
 - When $BL_j = 0$ and $BL'_j = 0$
 - Searching for a don't care
 - M_5 or M_6 may be ON ...
 - ... but the gate of M_7 remains at $0V$, M_7 is OFF
 - $\rightarrow$ The match line ML_i is not discharged
 - When $D = 1$ and $D' = 1$
 - Matching against a don't care
 - M_5 and M_6 are both OFF ...
 - ... the gate of M_7 is floating, M_7 is OFF
 - You may want to add a gate resistance towards ground to force the gate at $0V$
 - $\rightarrow$ The match line ML_i is not discharged



Memories

- Ternary CAM

BL_j	BL'_j	D	D'	Searching	Value	ML_i
0	0	0	0	X	not used	–
0	0	0	1	X	0	1
0	0	1	0	X	1	1
0	0	1	1	X	X	1
0	1	0	0	0	not used	–
0	1	0	1	0	0	1
0	1	1	0	0	1	0
0	1	1	1	0	X	1
1	0	0	0	1	not used	–
1	0	0	1	1	0	0
1	0	1	0	1	1	1
1	0	1	1	1	X	1
1	1	0	0	not used	not used	–
1	1	0	1	not used	0	–
1	1	1	0	not used	1	–
1	1	1	1	not used	X	–



Memories

- Ternary CAM

